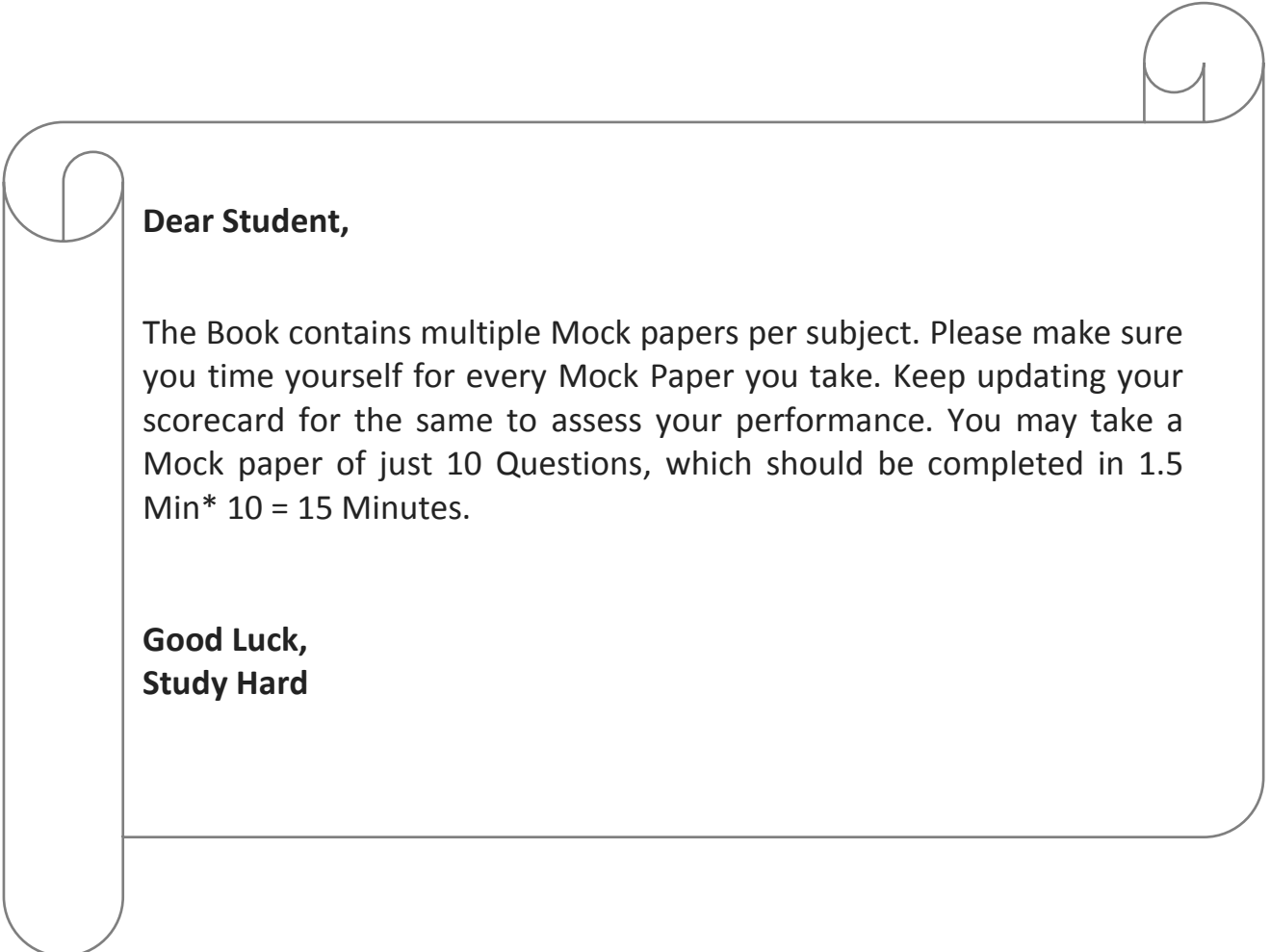


Name : _____

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Dear Student,

The Book contains multiple Mock papers per subject. Please make sure you time yourself for every Mock Paper you take. Keep updating your scorecard for the same to assess your performance. You may take a Mock paper of just 10 Questions, which should be completed in 1.5 Min* 10 = 15 Minutes.

**Good Luck,
Study Hard**

Score Card									
		A		B		C		D	
	Subject	Score	Total	Score	Total	Score	Total	Score	Total
1.	Ethical and Professional Standards		18		18		18		18
2.	Quantitative Methods		14		14		12		13
3.	Economics		12		12		12		11
4.	Financial Statement Analysis		24		24		24		24
5.	Corporate Finance		8		8		8		8
6.	Equity Investments		12		12		12		12
7.	Derivative Investments		6		6		6		6
8.	Fixed Income Investments		12		12		12		12
9.	Alternative Investments		4		4		4		4
10.	Portfolio Management		10		10		10		10

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Chapter 1

Ethical and Professional Standards

Mock A

Questions

1. A research analyst is facing a moral dilemma and decides to use an ethical decision-making framework. After looking at the facts at hand and identifying the situational influences, he still cannot make a decision on the best course of action. His least appropriate next step is to:
 - a) Determine what additional information is needed.
 - b) Ask someone else to give guidance.
 - c) decide, act, monitor, and reflect
2. The belief that one's ethical standards are above average is most likely a reflection of which of the following behavioral biases?
 - a) Overconfidence
 - b) Short-term focus
 - c) Situational influence
3. Several years ago, Leo Peek, CFA, co-founded an investment club. The club is fully invested but has not actively traded its account for at least a year and does not plan to resume active trading of the account. Peek's employer requires an annual disclosure of employee stock ownership. Peek discloses all of his personal trading accounts but does not disclose his holdings in the investment club. Peek's actions are least likely to be a violation of which of the CFA Institute Standards of Professional Conduct?
 - a) Misrepresentation
 - b) Conflicts of interest
 - c) Transaction priority
4. Verification of compliance with the GIPS standards most likely requires:
 - a) Verification for each specific composite under review.
 - b) An independent third party to carry out the verification.
 - c) an assurance that the composite presentations are accurate
5. Delaney O'Keefe, a CFA candidate, is a portfolio manager at Bahati Management Company. The company is considering investing offshore for the first time, particularly in North America, on behalf of its clientele, who are all high-net-worth individuals. O'Keefe does not have experience in offshore investments, so she hires Mark Carlson, CFA, of Carlson Consulting, on the sole basis that he is a CFA charter holder, to undertake due diligence exercises on the top 10 portfolio managers in North America, ranked by assets under management (AUM). To avoid violating any Code and Standards, O'Keefe should most likely undertake:
 - a) The due diligence exercise on the top 10 asset managers herself.
 - b) A due diligence exercise on Mark Carlson and Carlson Consulting.
 - c) A sampling of the suitability of North America for her clients.
6. Vishal Chandarana, an unemployed research analyst, recently registered for the CFA Level I exam. After two months of intense interviewing, he accepts a job with a stock brokerage company in a different region of the country. Chandarana posts on a blog how being a CFA candidate really helped him get a job. He also notes how relieved he was when his new employer did not ask him about being fired from his former employer. Which CFA Institute Standards of Professional Conduct did Chandarana least likely violate?
 - a) Misconduct
 - b) Reference to CFA Institute, the CFA Designation, and the CFA Program
 - c) Loyalty to Employers
7. Justin Blake, CFA, a retired portfolio manager, owns 20,000 shares of a small public company that he would like to sell because he is worried about the company's prospects. He posts messages on several internet bulletin

boards. The messages read, "This stock is going up once the pending patents are released, so now is the time to buy. The stock is a buy at anything below \$3. I have done some close research on these guys." According to the Standards of Practice Handbook, Blake most likely violated the Code and Standards associated with:

- a) Neither Integrity of Capital Markets nor Conflicts of Interest.
 - b) Integrity of Capital Markets, but not Conflicts of Interest.
 - c) Integrity of Capital Markets, and Conflicts of Interest.
8. In cases where applicable local laws governing calculation and presentation of investment performance conflict with the GIPS standards, firms are:
- a) Required to comply with local regulations and make full disclosure of the conflict to claim GIPS compliance.
 - b) Required to calculate and maintain two sets of performance data in order to claim GIPS compliance.
 - c) Unable to claim GIPS compliance in cases where local regulations prohibit accurate calculation.
9. The CFA Institute Code of Ethics and Standards of Professional Conduct are most likely designed to foster and reinforce a culture of:
- a) Regulatory compliance.
 - b) Service to the firm.
 - c) Responsibility and professionalism.
10. The Global Investment Performance Standards (GIPS) were developed for the benefit of
- a) Broker/dealers.
 - b) Prospective clients.
 - c) Regulators.
11. Amanda Covington, CFA, works for McJan Investment Management. McJan employees must receive prior clearance of their personal investments in accordance with McJan's compliance procedures. To obtain prior clearance, McJan employees must provide a written request identifying the security, the quantity of the security to be purchased, and the name of the broker through which the transaction will be made. Precleared transactions are approved only for that trading day. As indicated below, Covington received prior clearance.

Security	Quantity	Broker	Prior	Clearance
A	100	Easy	Trade	Yes
B	150	Easy	Trade	Yes

Two days after she received prior clearance, the price of Stock B decreased, so Covington decided to purchase 250 shares of Stock B only. In her decision to purchase 250 shares of Stock B only, did Covington violate any CFA Institute Standards of Professional Conduct?

- a) Yes, relating to her employer's compliance procedures
 - b) Yes, relating to diligence and reasonable basis
 - c) No
12. Disclosure of confidential CFA exam information will most likely be detected by the Professional Conduct staff through:
- a) Monitoring online and social media.
 - b) Annual Professional Conduct Statements.
 - c) Analysis of proctor reports.
13. From the point of view of an investor, unethical behaviour by investment professionals can most likely lead to which of the following?
- a) Increased willingness to accept risk
 - b) Rise in the demand for investments
 - c) Demand for a higher return
14. According to the CFA Institute Code of Ethics and Standards of Professional Conduct, trading on material non-public information is least likely to be prevented by establishing:
- a) Personal trading limitations.
 - b) Firewalls.

- c) Selective disclosure.
15. Which of the following best outlines the minimally acceptable behaviours expected of a member belonging to a societal group?
- a) Code of ethics
 - b) A firm's employee handbook
 - c) Standards of conduct
16. Examples of the beneficial features of using an ethical decision-making framework least likely includes analyzing:
- a) The decision maker's perspective of contemplated actions.
 - b) A broader picture from a long-term point of view.
 - c) The best course of action when alternatives are available.
17. Which of the following is not a component of the CFA Institute Code of Ethics?
- a) Practice and encourage others to practice in a professional and ethical manner that will reflect credit on themselves and the profession.
 - b) Place the integrity of the investment profession and the interests of clients above your own personal interests.
 - c) Promote financial integrity and seek to prevent and punish abuses in the financial markets.
18. Claire Jones, CFA, is an analyst following natural gas companies in the United States. At an industry energy conference, the chief financial officer of Alpine Energy states that the company is interested in making strategic acquisitions. At a separate event, Alpine's head of exploration commented that he is bullish on natural gas production prospects within north-eastern Pennsylvania. Jones is aware that Alpine currently has very little exposure to this region. She also knows another company in her universe, Pure Energy, Inc. is based in north-eastern Pennsylvania and controls significant assets in the area. Pure Energy is highly leveraged, and Jones believes it will need to raise additional capital or partner with another firm to move to the production phase with their assets. Jones attempts to contact Alpine's chief executive officer with an unrelated question and is told he is unavailable because he is on a business trip to north-eastern Pennsylvania. Jones updates her research on Pure Energy and then recommends the stock to Lisa Wong, CFA, and a portfolio manager, who purchases significant positions in client accounts. The following week, Pure Energy announces it has entered into an agreement to be purchased by Alpine for a significant premium. Has either Jones or Wong most likely violated standards with regard to the integrity of capital markets?
- a) Yes, both Jones and Wong have acted on insider information
 - b) No
 - c) Yes, Jones' recommendation is based on insider information

Answers

1. C.

The least appropriate action would be for the decision maker to go ahead and make a decision based on insufficient information. By doing so, the decision maker could cause harm and make the situation worse. The ethical decision-making framework is iterative, and users can move between phases rather than undertaking them in any one order. If a decision maker is not yet ready to make a decision, the most appropriate course of action would be to ask someone else to give guidance and determine what additional information is needed to clarify the situation.

CFA Level I

"Ethics and Trust in the Investment Profession,"

2. A.

The belief that one's ethical standards are above average illustrates an overconfidence bias. An overconfidence bias will most likely lead individuals to overestimate the morality of their own behavior and can lead to a failure to consider important inputs and variables needed to make the best ethical decisions.

CFA Level I

"Ethics and Trust in the Investment Profession,"

3. C.

There is no indication that the investment club is trading ahead of clients. See Standard VI(B).

CFA Level I

"Guidance for Standards I-VII," CFA Institute Standard I(C), Standard VI(A), Standard VI(B).

4. B.

Although a firm is responsible for its own compliance claim, it cannot perform its own verification. An independent third party must undertake the verification.

CFA Level I

"Introduction to the Global Investment Performance Standards (GIPS)"

5. B.

O'Keefe can delegate a due diligence exercise to a third party but must ensure the person or company hired to do so is competent and has the skills necessary to undertake a thorough and appropriate analysis. Although Carlson may be qualified to undertake this assignment, O'Keefe needs to take the necessary steps to ensure that he is indeed qualified. Just because a person is a CFA charterholder does not necessarily mean he or she is appropriate for the assignment.

CFA Level I

"Guidance for Standards I-VII,"

6. B.

There is no evidence Chandarana violated Standard VII(B)—Reference to CFA Institute, the CFA Designation, and the CFA Program with regard to his being a CFA candidate. Specifically, Chandarana does not overstate his competency or imply he will achieve superior performance as a result of his CFA designation. It does appear, however, Chandarana did not act with integrity when he hid information that could potentially harm his new employer's reputation, thus violating Standard I(D)—Misconduct and Standard IV(A)—Loyalty.

CFA Level I

"Guidance for Standards I-VII"

7. B

Blake violated Standard II(B) regarding the Integrity of Capital Markets by engaging in a practice that is likely to artificially inflate trading volume.

CFA Level I

"Guidance for Standards I-VII,"

8. A.

In cases where applicable local laws governing calculation and presentation of investment performance conflict with the GIPS standards, firms must comply with local regulations and fully disclose the conflict in the compliant presentation.

CFA Level I

"Global Investment Performance Standards (GIPS),"

9. C.

The CFA Institute Code of Ethics and Standards of Professional Conduct are designed to foster and reinforce a culture of responsibility and professionalism. The Code and Standards apply to all members and candidates regardless of title, position, occupation, geographic location, or specific situation, and they apply to all professional activities of investment professionals.

CFA Level I

"Ethics and Trust in the Investment Profession,"

10. B.

The GIPS standards benefit two main groups: investment management firms and prospective clients. GIPS-compliant presentations allow prospective clients to know that the track record of a GIPS-compliant fund manager is complete and fairly presented.

CFA Level I

"Introduction to the Global Investment Performance Standards (GIPS)"

11. A.

Prior-clearance processes guard against potential and actual conflicts of interest; members are required to abide by their employer's compliance procedures (Standard VI (B)).

CFA Level I

"Guidance for Standards I-VII,"

12. A.

Professional Conduct inquiries come from a number of sources, including the monitoring of online and social media to detect disclosure of confidential exam information. 2016

CFA Level 1

"Code of Ethics and Standards of Professional Conduct,"

13. C.

Unethical behavior erodes and destroys trust. Investors with low levels of trust are less willing to accept risk and, therefore, will likely demand a higher return for the use of their capital. They may also choose to invest elsewhere or to not invest at all.

CFA Level I

"Ethics and Trust in the Investment Profession,"

14. C.

Selective disclosure occurs when companies discriminate in making material nonpublic information public. Corporations that disclose information on a limited basis create the potential for insider-trading violations. See Standard II(A).

CFA Level I

"Guidance for Standards I-VII,"

15. C

Standards of conduct outline the minimally acceptable behaviors expected of a member of a societal group. The code of ethics serves as a general guide for how community members should act.

CFA Level I

"Ethics and Trust in the Investment Profession,"

16. C

An ethical decision-making framework helps a decision maker see the situation from multiple perspectives, not just from her personal perspective, and pay attention to aspects of the situation that may be less evident if a short-term, self-focused perspective is applied.

CFA Level I

"Ethics and Trust in the Investment Profession,"

17. C.

Punishing abuse in the financial markets is not one of the six components of the Code of Ethics.

CFA Level I

"Code of Ethics,"

18. B.

Jones has used the mosaic theory to combine nonmaterial, nonpublic information with material public information.

CFA Level I

"Guidance for Standards I-VII,"

Questions

1. If a firm restructures and wants to remain compliant with the GIPS standards, it should most likely:
 - a) Create all new composites with proper disclosures regarding the reorganization.
 - b) Alter the historical performance within existing composites.
 - c) Maintain the historical performance for all composites.
2. Which of the following statements concerning the Global Investment Performance Standards (GIPS) is most likely correct?
 - a) Clients or prospective clients benefit from the Standards because the historical track record of compliant firms is accurate and precise.
 - b) Compliance with the Standards enhances the credibility of investment management firms.
 - c) The Standards eliminate the need for in-depth due diligence by investors.
3. Which of the following is least likely part of the CFA Institute Standards of Professional Conduct, Standard V (B): Communication with Clients and Prospective Clients? Members and candidates must:
 - a) Disclose to clients and prospective clients significant limitations and risks associated with the investment process.
 - b) Distinguish between fact and opinion in the presentation of investment analysis and recommendations.
 - c) Make reasonable efforts to ensure that when communicating investment performance information, it is fair, accurate, and complete.
4. Zhao Xuan, CFA, is a sell-side investment analyst. While at a software industry conference, Zhao hears rumors that Green Run Software may have falsified its financial results. When she returns to her office, Zhao conducts a thorough analysis of Green Run. Based on her research, including discussions with some of Green Run's customers, Zhao is convinced that Green Run's reported 50% increase in net income during recent quarters is completely fictitious. But so far Zhao is the only analyst suspicious about Green Run's reported earnings. According to the CFA Institute Code of Ethics and Standards of Professional Conduct, the least appropriate action for Zhao is to:
 - a) Recommend her clients sell their Green Run shares immediately.
 - b) Do nothing until other analysts support her analysis.
 - c) Report her suspicions to Green Run's management.
5. Monique Greta, CFA, is a research analyst at East West Investment Bank. Previously, Greta worked at a mutual fund management company and has a long-standing client relationship with the managers of the funds and their institutional investors. Greta often provides fund managers, who work for Greta's former employer, with draft copies of her research before disseminating the information to all of the bank's clients. This practice has helped Greta avoid several errors in her reports, and she believes it is beneficial to the bank's clients, even though they are not aware of this practice. Regarding her research, Greta *least likely* violated the CFA Institute Standards of Professional Conduct because:
 - a) the long-standing client relationships are not disclosed.
 - b) this practice benefits all clients.
 - c) her report is a draft.
6. Prudence Charmaine, CFA, was recently accused in writing of cheating on a professional accounting exam. She denied cheating and successfully defended herself against the allegation. As part of her defense and as evidence of her character, Charmaine stated she is a CFA charter holder and upholds the CFA Institute Code of Ethics and Standards of Professional Conduct. On her next annual Professional Conduct Statement, Charmaine does not report this allegation to CFA Institute. Did Charmaine most likely violate the CFA Institute Code of Ethics or Standards of Professional Conduct (Code and Standards)?
 - a) Yes, she improperly used the CFA Institute Code and Standards to defend herself.
 - b) Yes, she did not report the allegation on her annual Professional Conduct Statement.
 - c) No
7. Teresa Staal, CFA, is an investment officer in a bank trust department. She manages money for celebrities and public figures, including an influential local politician. She receives a request from the politician's political party headquarters to disclose his stock holdings. The request indicates local law requires the disclosure. What steps should Staal *most likely* take to ensure she does not violate any CFA Institute Standards of

Professional Conduct?

- a) Check with her firm's compliance department to determine her legal responsibilities.
- b) Provide the information and inform her client.
- c) Send the requested documents and inform her supervisor.

8. Jan Loots, CFA, quit his job as a portfolio manager at an investment firm with whom he had a non-solicitation agreement he signed several years ago. Loots received permission to take his investment performance history with him and also took a copy of the firm's software-trading platform. Subsequently, Loots sent out messages on social media sites announcing he was looking for clients for his new investment management firm. Access to Loots's social media sites is restricted to friends, family, and former clients. Loots *least likely* violated the CFA Institute Standards of Professional Conduct concerning his:

- a) Investment performance history.
- b) Non-solicitation agreement.
- c) trading software

9. To comply with the GIPS standards, firms *most likely* must:

- a) Apply standards on a firm-wide basis.
- b) Be defined as separate legal entities.
- c) Be verified before they can claim compliance.

10. Oliver Opdyke, CFA, works for an independent research organization that does not manage any client money. In the course of his analysis of Red Ribbon Mining, he hears rumors that the president of Red Ribbon, Richard Leisberg, has recently been diagnosed with late stage Alzheimer's disease, a fact not publicly known. The final stage of Alzheimer's is when individuals lose the ability to respond to their environment, the ability to speak, and ultimately, the ability to control movement. Leisberg is the charismatic founder of Red Ribbon, and under his leadership the company grew to become one of the largest in the industry. According to the CFA Institute Code of Ethics and Standards of Professional Conduct, the most appropriate action for Opdyke is to:

- a) Immediately publish a sell recommendation for Red Ribbon Mining.
- b) Encourage Red Ribbon Mining management to disclose the president's medical condition.
- c) Confirm the president's diagnosis before publishing his research report.

11. Raymond Ortiz, CFA, provides investment advice to high-net-worth investors. Ortiz has just completed an analysis of Continental Wheat, a manufacturer of wheat-based food products. He rated the company a long-term hold for investors seeking growth and income. Ortiz's analysis included a review of the company's management team, financial data, pro forma financial positions, and dividends and dividend policy, as well as a comparison of Continental with its competitors. Although he does not tell anyone, five years ago Ortiz worked for and managed the commodities derivatives trading unit of Continental. As part of his compensation at Continental, he received stock, which he still owns. Based on his research, Ortiz recommends Continental to clients who have a moderate risk tolerance. Two weeks later, Continental announces its quarterly earnings are 30% less than a year ago. Consequently, shares of Continental drop by 50%. Ortiz *most likely* violated the CFA Institute Standards of Professional Conduct related to his stock:

- a) Recommendation.
- b) Research.
- c) Ownership.

12. Firms claiming compliance with the GIPS standards are *most likely* required to:

- a) Make negative assurance disclosures when presenting the firm's performance.
- b) Meet at least 85% of the requirements before claiming compliance.
- c) Comply with all updates, interpretations, and clarifications.

13. If you are seeking guidance from the firm's code of ethics or written policies, your actions *most likely* reflect which phase of an ethical decision-making framework?

- a) Consider
- b) Reflect
- c) Decide

14. Which of the following statements concerning why the Global Investment Performance Standards (GIPS) were created is *least likely* correct? The GIPS standards were created to:

- a) Establish a standardized, industry wide approach for investment firms to follow.
- b) Identify a set of ethical principles for firms to follow in calculating and presenting historical investment

results.

- c) Provide clients certainty in what is presented and allow them to make reasonable comparisons.

15. Which of the following is *least likely* a phase in an ethical decision-making framework?

- a) Reflection on the outcome versus what was anticipated.
- b) Multiple iterations of analysis.
- c) Consideration of situational influences, additional guidance, and alternative action

16. Robin Herring, CFA, is a government bond research analyst at an independent credit rating agency. A competitor credit rating agency just downgraded the bonds of a government Herring follows. Herring notes all of the information in the competitor's report was covered in his analysis published last week. In the past, Herring has been slow to downgrade bonds, so he starts to doubt his own analysis after seeing the competitor's report. Herring decides to reissue his credit rating of this government bond and match the competitor's downgrade. In his revised report, Herring states that new information has been made available to justify the downgrade. Herring posts the revision on the credit rating agency's website and provides it by e-mail to all clients who received the original. Herring's rating change *least likely* violated which of the following CFA Institute Standards of Professional Conduct?

- a) Communication with Clients and Prospective Clients
- b) Diligence and Reasonable Basis
- c) Fair Dealing

17. As a condition of his employment with an investment bank, Abasi Hasina, CFA, was required to sign a contract, including a non-compete clause restricting him from working for a competitor for three years after leaving the employer. After one year, Hasina quits his job for a comparable position with an investment bank in a country where non-compete clauses are illegal. Lawyers with whom he consulted prior to taking the new position determined the non-compete clause was a violation of human rights and thus illegal. Did Hasina most likely violate the CFA Institute Code of Ethics and Standards of Professional Conduct?

- a) No, because the non-compete clause is illegal in the new country of employment
- b) Yes
- c) No, because the non-compete clause violates his human rights

18. Jean-Luc Schlumberger, CFA, is an independent research analyst providing equity research on companies listed on exchanges in emerging markets. He often incorporates statistical data he obtains from the web sites of the World Bank and the central banks of various countries into the body of his research reports. Although not indicated within the reports, whenever his clients ask where he gets his information, he informs them that the information is in the public domain but he does not keep his own records. When the clients ask for the specific web site addresses, he provides the information. Which Standard has Schlumberger *least likely* violated?

- a) Performance Presentation
- b) Misrepresentation
- c) Record Retention

Answers

1. C

Changes in a firm's organization must not lead to the alteration of historical composite performance.

CFA Level I

"The GIPS Standards"

2. B

Compliance with the GIPS standards enhances the credibility of investment management.

CFA Level I

"Introduction to the Global Investment Performance Standards (GIPS),"

3. C

The statement "When communicating investment performance information, members and candidates must make reasonable efforts to ensure it is fair, accurate, and complete" is found in the CFA Institute Standards of Professional Conduct, Standard III(D): Performance Presentation. It is not part of Standard V (B): Communication with Clients and Prospective Clients.

2016 CFA Level 1

"Code of Ethics and Standards of Professional Conduct,"

4. A

The analyst has conducted thorough research that indicates the company falsified its financial results, and she should request the company address this issue publicly as recommended by Standard II (A) – Material Non public Information. If a member or candidate determines that information is material, the member or candidate should make reasonable efforts to achieve public dissemination of the information. This effort usually entails encouraging the issuer company to make the information public. If public dissemination is not possible, the member or candidate must communicate the information only to the designated supervisory and compliance personnel within the member's or candidate's firm and must not take investment action on the basis of the information.

CFA Level I

"Guidance for Standards I–VII"

5. A

The analyst does not violate any of the Standards of Professional Conduct by having long – standing client relationships and generally is not required to disclose such relationships. However, the analyst is not treating all clients fairly as required by Standard III (B)–Fair Dealing when disseminating investment recommendations; disclosure of the relationship with long – standing clients is not the issue. The analyst has advantaged some clients over others by providing advance information, and all clients do not have a fair opportunity to act on the information within the draft report. Members and candidates may differentiate their services to clients, but different levels of service must not disadvantage or negatively affect clients.

CFA Level I

"Guidance for Standards I–VII"

6. B

Chairman should have reported the cheating allegation when making her annual Professional Conduct Statement. Even though she successfully defended herself against the charges and the charges were dropped,

she has a responsibility to report the written complaint involving her integrity. The Code of Ethics requires CFA charter holders to practice and encourage others to practice in a professional and ethical manner that will reflect credit on themselves and the profession.

CFA Level I

"Code of Ethics and Standards of Professional Conduct"

7. A

In order to avoid violating Standard III (E) – Preservation of Confidentiality, Staal should determine whether applicable securities regulations required is closing the records before she provides the confidential information concerning her client's investments.

CFA Level I

"Guidance for Standards I – VII"

8. A

The portfolio manager received permission to use his investment performance history from his prior employer. The member violated his non – solicitation agreement by indicating his availability to new clients on several social media sites accessible by clients of his former employer. This action is a violation of Standard IV (A) – Loyalty because he did not act for the benefit of his former employer. In this case, the member may cause harm to his former employer if his messages result in clients moving to his new business from his former employer. The member also violated Standard IV (A) by taking his employer's property, the trading software.

CFA Level I "Guidance for Standards I – VII"

9. A

According to requirements for GIPS compliance, the GIPS standards must be applied on a firm – wide basis. They cannot be separately applied to composites.

CFA Level I

"The GIPS Standards"

10. B

Members and candidates should make reasonable efforts to achieve public dissemination of information that is material and non public, as required by Standard II (A) – Material Non public Information. This effort usually entails encouraging the issuer company to make the information public. In this case, if the diagnosis is fact and not rumor, then this information is material and should be disclosed.

CFA Level I

"Guidance for Standards I – VII"

11. C

There is a violation of Standard VI (A) – Disclosure of Conflicts; the analyst worked for Continental and still has ties to the company in the form of his stock ownership.

CFA Level I

"Guidance for Standards I – VII"

12. C

Firms must comply with all requirements of the GIPS standards, including any updates, guidance statements, interpretations, questions and answers, and clarifications published by CFA Institute and the GIPS Executive Committee.

CFA Level I - "The GIPS Standards"

13. A

If you are seeking guidance from the firm's code of ethics or written policies, you are in the Consider phase of the ethical decision making framework. This phase involves taking time to consider the situation influences as well as personal behavioral biases that could affect your thinking and decision making. During this phase, you may also seek guidance from such trusted sources as the firm's compliance department or outside counsel.

CFA Level I - "Ethics and Trust in the Investment Profession,"

14. C

The GIPS standards were created to ensure fair representation and full disclosure of investment performance, not to provide certainty in what is presented.

CFA Level I - "Introduction to the Global Investment Performance Standards (GIPS),"

15. B

Multiple iterations of analysis are not a phase in the ethical decision-making framework. The ethical decision-making process includes multiple phases, and the process of developing the framework involves multiple iterations. The iterative aspect of developing the framework is essential to the process, but it is not a phase in the ethical decision-making process.

CFA Level I - "Ethics and Trust in the Investment Profession,"

16. C

The analyst has dealt fairly with all clients by sending them an e-mail and posting his rating change on the credit rating agency's website when making material changes to his prior investment recommendation; therefore, he has not violated Standard III (B)-Fair Dealing. Clients should be treated fairly when material changes in a member's or candidate's prior investment recommendations are disseminated, which has been done.

CFA Level I - "Guidance for Standards I-VII"

17. B

By failing to adhere to the non-compete clause he agreed to abide by when signing his employment contract, Hasina shows a lack of professional integrity toward his employer. This behavior reflects poorly on the good reputation of members and is a violation of the Code of Ethics, which states *that* members and candidates must act with integrity, and Standard I (D)-Misconduct, which states that members and candidates must not engage in any professional conduct involving dishonesty, fraud, or deceit or commit any act that reflects adversely on their professional reputation, integrity, or competence. The Code of Ethics at times requires a member or candidate to uphold a higher standard than that required by law, rule, or regulation—or in this case, the strict application of the employment agreement.

CFA Level I - "Code of Ethics and Standards of Professional Conduct,"

18. A

Standard III (D)-Performance Presentation pertains to investment performance information and there is no indication any violation has occurred.

CFA Level I - "Guidance for Standards I-VII,"

Questions

1. In cases where applicable local laws governing calculation and presentation of investment performance conflict with the GIPS standards, firms are:
 - A. required to comply with local regulations and make full disclosure of the conflict to claim GIPS compliance.
 - B. unable to claim GIPS compliance in cases where local regulations prohibit accurate calculation.
 - C. required to calculate and maintain two sets of performance data in order to claim GIPS compliance.
2. Ross Nelson, CFA, manages accounts for high-net-worth clients, including his own family's account. He has no beneficial ownership in his family's account. Because Nelson is concerned about the appearance of improper behavior in managing his family's account, when his firm purchases a block of securities, Nelson allocates to his family's account only those shares that remain after his other client accounts have their orders filled. The fee for managing his family's account is based on his firm's normal fee structure. According to the *Standards of Practice Handbook*, Nelson's best course of action with regard to management of his family's account would be to:
 - A. treat the account like other employee accounts of the firm.
 - B. treat the account like other client accounts.
 - C. remove himself from any direct involvement by transferring responsibility for this account to another investment professional in the firm.
3. Alexander Newton, CFA, is the chief compliance officer for Mills Investment Limited. Newton institutes a new policy requiring the pro rata distribution of new security issues to all established discretionary accounts for which the new issues are appropriate. The policy also provides for the exclusion of newly established discretionary accounts from the distribution until they have reached their one-month anniversary date. This policy is disclosed to all existing and potential clients. Did Newton *most likely* violate any CFA Institute Standards of Professional Conduct?
 - A. No, because the policy has been adequately disclosed to all existing and potential clients
 - B. B. Yes
 - C. No, because the allocation policy is not inequitable under the standards
4. According to the GIPS standards a verification report confirms all of the following *except* whether:
 - A. specific composite presentations are accurate.
 - B. processes and procedures are designed to calculate and present compliant performance results.
 - C. a firm has complied with all firm-wide composite construction requirements.
5. Justin Blake, CFA, a retired portfolio manager, owns 20,000 shares of a small public company that he would like to sell because he is worried about the company's prospects. He posts messages on several internet bulletin boards. The messages read, "This stock is going up once the pending patents are released, so now is the time to buy. The stock is a buy at anything below \$3. I have done some close research on these guys." According to the *Standards of Practice Handbook*, Blake *most likely* violated the Code and Standards associated with:
 - A. Integrity of Capital Markets, but not Conflicts of Interest.
 - B. neither Integrity of Capital Markets nor Conflicts of Interest.
 - C. Integrity of Capital Markets, and Conflicts of Interest.
6. Miranda Grafton, CFA, purchased a large block of stock at varying prices during the trading session. The stock realized a significant gain in value before the close of the trading day, so Grafton reviewed her purchase prices to determine what prices should be assigned to each specific account. According to the *Standards of Practice Handbook*, Grafton's *least* appropriate action is to allocate the execution prices:
 - A. across the participating client accounts at the same execution price.
 - B. across the participating client accounts pro rata on the basis of account size.

- C. on a first-in, first-out basis with consideration of bundling orders for efficiency.
7. Which of the following *least likely* reflects the two primary principles of the CFA Institute Rules of Procedure for Professional Conduct?
- Fair process to the member and candidate
 - Public disclosure of disciplinary sanctions
 - Confidentiality of proceedings
8. Several years ago, Leo Peek, CFA, co-founded an investment club. The club is fully invested but has not actively traded its account for at least a year and does not plan to resume active trading of the account. Peek's employer requires an annual disclosure of employee stock ownership. Peek discloses all of his personal trading accounts but does not disclose his holdings in the investment club. Peek's actions are *least likely* to be a violation of which of the CFA Institute Standards of Professional Conduct?
- Misrepresentation
 - Transaction priority
 - Conflicts of interest
9. Madeline Smith, CFA, was recently promoted to senior portfolio manager. In her new position, Smith is required to supervise three portfolio managers. Smith asks for a copy of her firm's written supervisory policies and procedures but is advised that no such policies are required by regulatory standards in the country where Smith works. According to the *Standards of Practice Handbook*, Smith's *most* appropriate course of action would be to:
- require her firm to adopt the CFA Institute Code of Ethics and Standards of Professional Conduct.
 - decline to accept supervisory responsibility until her firm adopts procedures to allow her to adequately exercise such responsibility.
 - require the employees she supervises to adopt the CFA Institute Code of Ethics and Standards of Professional Conduct.
10. When Jefferson Piedmont, CFA, joined Branch Investing, Branch began using a quantitative stock selection model Piedmont had developed on his own personal time prior to his employment with Branch. One year later when Piedmont left Branch Investing, he found the original copy of the model he had developed in a file at his home and presented it to his new employer, which immediately began using the model. According to the *Standards of Practice Handbook*, did Piedmont *most likely* violate any CFA Institute Standards of Professional Conduct?
- No
 - Yes, because he misappropriated property now belonging to Branch
 - Yes, because he failed to inform his new employer the model was the same one used by his previous employer
11. Reiko Kimisaki, CFA, is an investment adviser for a national social security fund in a frontier market with a very limited and illiquid capital market. The labor force is young and has an investment time horizon of 25 to 30 years. Kimisaki has been asked to suggest ways to increase the investment return of the overall portfolio. After careful assessment of the fund's previous investment history, and available asset classes, she considers investment in private equity. What is Kimisaki's lowest priority to avoid any Code and Standards violations prior to making this investment recommendation?
- Assess the risk tolerance of the fund.
 - Analyze the expected returns of private equity in the market.
 - Determine whether the investment policy statement allows for alternative investments.
12. Rebecca Wong is enrolled to take the Level I CFA exam. Her friend William Leung purchased Level I study materials from a well-known CFA review program the previous year. Leung made a photocopy of the previous year's copyrighted materials and sold it to Wong to help her study. Who *most likely* violated the CFA Institute Code of Ethics or any Standards of Professional Conduct?
- Neither violated.
 - Both violated.
 - Only Leung violated.

13. According to the Global Investment Performance Standards (GIPS), firms must do all of the following *except*:
- A. provide investors with a comprehensive view of their performance only in terms of returns.
 - B. adhere to certain calculation methodologies and make specific disclosures along with their performance.
 - C. comply with all requirements of the GIPS standards, such as updates, guidance statements, and clarifications.
14. For firms to claim compliance with the GIPS standards they *most likely* must:
- A. increase the consistency and quality of the firm's compliant presentations.
 - B. take responsibility for their claim of compliance and maintaining that compliance.
 - C. hire an independent third party to test a sample of their composites.
15. While waiting in the business class lounge before boarding an airplane, Becca Msafari, CFA, an equity analyst, overhears a conversation by a group of senior managers, including members of the board, from a large publicly listed bank. The managers discuss staff changes necessary to accommodate their regional expansion plans. Msafari hears several staff names mentioned. Under what circumstances could Msafari *most likely* use this information when making an investment recommendation to her clients? She can use the information:
- A. under no circumstances.
 - B. if she does not breach the confidentiality of the names of the staff.
 - C. if the discussed changes are unlikely to affect investor perception of the bank.
16. Fundamental Asset Managers claims compliance with the CFA Institute Global Investment Performance Standards (GIPS) and manages both discretionary and non-discretionary accounts. When constructing a single composite for Fundamental, Juma Dzuya includes all discretionary, fee paying accounts with both value and growth strategies. Does the composite constructed by Dzuya *most likely* meet the criteria of the GIPS standards?
- A. No, because of non-similar investment strategies
 - B. No, because non-discretionary accounts are not included
 - C. Yes
17. Charlie Mancini, CFA, is the Managing Director for Business Development at SV Financial, (SVF), a large U.S.-based mutual fund organization. Mancini has been under pressure recently to increase revenues. In order to secure business from a large hedge fund manager based in Asia, Mancini recently approved flexible terms for the fund's client agreement. To allow for time zone differences, the agreement permits the hedge fund to trade in all of SVF's mutual funds six hours after the close of U.S. markets, which is prohibited by U.S. regulators. Did Mancini violate any CFA Institute Standards of Professional Conduct?
- A. No
 - B. Yes, with regard to Fair Dealing and Material Nonpublic Information
 - C. Yes, with regard to Fair Dealing
18. In order to achieve compliance with GIPS Standards, it is recommended that firms:
- A. define the firm by including all geographical offices operating under the same firm name.
 - B. provide existing clients a compliant presentation applicable to their portfolio, at a minimum of a bi-annual basis.
 - C. adopt the broadest, most meaningful definition of the firm.

Answers

1. Answer = A

In cases where applicable local laws governing calculation and presentation of investment performance conflict with the GIPS standards, firms must comply with local regulations and fully disclose the conflict in the compliant presentation.

CFA Level I

"Global Investment Performance Standards (GIPS)," CFA Institute Section 4.A.22

2. Answer = B

Nelson has breached his duty to his family by treating them differently from other clients. They are entitled to the same treatment as any other client of the firm. Nelson should treat his family's account like any other client account as stated in Standard III (B) related to Fair Dealing and Standard VI (B) related to Priority of Transactions.

CFA Level I

"Guidance for Standards I-VII," CFA Institute Standard III(B), Standard VI(B)

3. Answer = B

Under Standard III(B)-Fair Dealing, members and candidates should disclose to clients and prospective clients how they select accounts to participate in and how they determine the amount of securities each account will buy or sell. Trade allocation procedures must be fair and equitable, and disclosure of inequitable allocation methods does not relieve the member or candidate of this obligation. All discretionary accounts should be treated in the same manner. Treating newer accounts differently would be considered inequitable regardless of whether this policy is disclosed.

CFA Level I

"Guidance for Standards I-VII," CFA Institute Standard III(B)

4. Answer = A

GIPS verification does not ensure the accuracy of any specific composite presentations.

CFA Level I

"Introduction to the Global Investment Performance Standards (GIPS)," CFA Institute Section V-Verification

5. Answer = A

Blake violated Standard II(B) regarding the Integrity of Capital Markets by engaging in a practice that is likely to artificially inflate trading volume.

CFA Level I

"Guidance for Standards I-VII," CFA Institute Standard II(B), Standard VI(A)

6. Answer = B

According to Standard III (B) best practices include allocating pro rata on the basis of order size, not account size. All clients participating in the block trade should receive the same execution price and be charged the same commission.

CFA Level I

"Guidance for Standards I-VII," CFA Institute Standard III (B)

7. Answer = B

The two principles of the Rules of Procedure for Professional Conduct are confidentiality of proceedings and fair process to the member or candidate.

CFA Level I

"Rules of Procedure for Professional Conduct"

8. Answer = B

There is no indication that the investment club is trading ahead of clients. See Standard VI(B).

CFA Level I

"Guidance for Standards I-VII," CFA Institute

Standard I(C), Standard VI(A), Standard VI(B)

9. Answer = B

According to guidance for Standard (IV)(C), if a member cannot fulfill supervisory responsibilities because of the absence of a compliance system or because of an inadequate compliance system, the member should decline in writing to accept supervisory responsibility until the firm adopts reasonable procedures to allow the member to adequately exercise such responsibility.

CFA Level I

"Guidance for Standards I-VII," CFA Institute Standard IV(C)

10. Answer = A

Although departing employees may not take employer property when departing, as the guidance for Standard IV(A) – Loyalty outlines, the model Piedmont presented to his new employer was not Branch's property. It was created by Piedmont prior to his employment with Branch. The model was not created for Branch in the course of his employment, even though it was adopted by Branch.

CFA Level I

"Guidance for Standards I-VII," CFA Institute Standard IV(A)

11. Answer = B

Prior to undertaking analysis with regard to expected returns, an adviser must determine the suitability of an investment class, including whether it fits within the client's risk tolerance and whether it is an allowable asset class as per the client's investment policy statement. Only after these factors have been determined should she proceed, if appropriate, to analyze expected returns to determine a particular investment recommendation.

CFA Level I

"Guidance for Standards I-VII," CFA Institute Standard III(C)

12. Answer = B

Photocopying copyrighted material, regardless of the year of publication, is a violation of Standard I (A) because copyrighted materials are protected by law. Candidates and members must comply with all applicable laws, rules, and regulations and must not knowingly participate or assist in a violation of laws.

CFA Level I

"Guidance for Standards I-VII," CFA Institute Standard I (A)

13. Answer = A

Firms must provide investors with a comprehensive view of their performance in terms of risk and returns, not just returns.

CFA Level I

"The GIPS Standards," CFA Institute Section: Overview

14. Answer = B

Firms claiming compliance with the GIPS standards are responsible for their claim of compliance and for maintaining that compliance. That is, firms self-regulate their claim of compliance.

CFA Level I

"Introduction to the Global Investment Performance Standards (GIPS)," CFA Institute Section: V. Verification

15. Answer = C

To comply with the Code and Standards, a member or candidate cannot use material nonpublic information when making investment recommendations. The information overheard would not be considered material only if any public announcement of the staff removal would be unlikely to move the share price of the bank, nor would the regional expansion substantially impact the value of the bank.

CFA Level I

"Guidance for Standards I-VII," CFA Institute Standard II(A)

16. Answer = A

A composite must include all actual fee-paying, discretionary portfolios managed in accordance with the same investment mandate, objective, or strategy (Standard IV–Composites). By including both the value and growth portfolios, the composite is made up of portfolios with different investment mandates or strategies.

CFA Level I

"Introduction to the Global Investment Performance Standards (GIPS)" Composites

17 Answer = B

Clients should be treated fairly and impartially according to Standard III(B). In addition, the flexible trading terms allow the hedge fund manager to enrich itself and are a violation of Standard II(A), which concerns trading on material nonpublic information. This situation is also a conflict of interest, and thus a violation of Standard VI(A)-Disclosure of Conflicts.

CFA Level I

"Guidance for Standards I-VII," CFA Institute

Standard II(A), Standard III(B), Standard VI(A)

18 Answer = C

The Fundamentals of Compliance recommend firms should adopt the broadest, most meaningful definition of the firm.

CFA Level I

"The GIPS Standards," CFA Institute

Section: Fundamentals of Compliance – Requirements & Recommendations

Questions

1. Richard Cardinal, CFA, is the founder of Volcano Capital Research, an investment management firm whose sole activity is short selling. Cardinal seeks out companies whose stocks have had large price increases. Cardinal also pays several lobbying firms to update him immediately on any legislative or regulatory changes that may impact his target companies. Cardinal sells short those target companies he estimates are near the peak of their sales and earnings and that his sources identify as facing legal or regulatory challenges. Immediately after he sells a stock, Cardinal conducts a public relations campaign to disclose all of the negative information he has gathered on the company, even if the information is not yet public. Which of Cardinal's actions is least likely to be in violation of the CFA Institute Standards of Professional Conduct?
 - A. Selling stock short
 - B. Trading on information from lobbyists
 - C. Disclosing information about target companies
2. Beth Kozniak, a CFA candidate, is an independent licensed real estate broker and a well-known property investor. She is currently brokering the sale of a commercial property on behalf of a client in financial distress. If the client's building is not sold within 30 days, he will lose the building to the bank. A year earlier, another client of Kozniak's had expressed interest in purchasing this same property. However, she is unable to contact this client, and she has not discovered any other potential buyers. Given her distressed client's limited time frame, Kozniak purchases the property herself and forgoes any sales commission. Six months later, she sells the property for a nice profit to the client who had earlier expressed interest in the property. Has Kozniak most likely violated the CFA Institute Standards of Professional Conduct?
 - A. Yes, she did not disclose her potential conflicts of interest to either client
 - B. Yes, she profited on the real estate to the detriment of her financially stressed client
 - C. No
3. Which of the following statements concerning the Global Investment Performance Standards (GIPS) is most likely correct?
 - A. The Standards eliminate the need for in-depth due diligence by investors.
 - B. Compliance with the Standards enhances the credibility of investment management firms.
 - C. Clients or prospective clients benefit from the Standards because the historical track record of compliant firms is accurate and precise.
4. Monique Gretta, CFA, is a research analyst at East West Investment Bank. Previously, Gretta worked at a mutual fund management company and has a long-standing client relationship with the managers of the funds and their institutional investors. Gretta often provides fund managers, who work for Gretta's former employer, with draft copies of her research before disseminating the information to all of the bank's clients. This practice has helped Gretta avoid several errors in her reports, and she believes it is beneficial to the bank's clients, even though they are not aware of this practice. Regarding her research, Gretta *least likely* violated the CFA Institute Standards of Professional Conduct because:
 - A. the long-standing client relationships are not disclosed.
 - B. this practice benefits all clients.
 - C. her report is a draft.
5. Belen Zapata, CFA, is the owner of Kawah Investments. Kawah promises investors returns of up to 12% per year and claims to achieve these returns by investing in non-investment-grade bonds and other fixed-income instruments. Over the next 12 months, bond market yields reach unprecedented lows and Zapata finds it impossible to achieve the returns she expected. No investments are ever made by Kawah, and clients are completely paid back all of their original investment. Zapata *most likely* violated the CFA Institute Standards of Professional Conduct because of the:
 - A. investment mandate.
 - B. return of capital.
 - C. promised returns.

6. A central bank fines a commercial bank it supervises for not following statutory regulations regarding nonperforming loan provisions on three large loans as a result of the bank's loan provisioning policy. Louis Marie Buffet, CFA, sits on the board of directors of the commercial bank as a non-executive director, representing minority shareholders. He also chairs the bank's internal audit committee that determines the loan provisioning policy of the bank. Mercy Gatabaki, CFA, is the bank's external auditor and follows international auditing standards whereby she tests the loan portfolio by randomly selecting loans to check for compliance in all aspects of central bank regulations. Which charterholder is *most likely* in violation of the Code and Standards?
- Gatabaki
 - Buffet
 - Both
7. Sergio Morales, CFA, believes he has found evidence that his supervisor is engaged in fraudulent activity involving a client's account. When Morales confronts his supervisor, he is told the client is fully aware of the issue. Later that day, Morales contacts the client and after disclosing the fraudulent activity, he is told by the client to mind his own business. Following the requirements of local law, Morales provides all of his evidence, along with copies of the client's most recent account statements, to a government whistleblower program. Has Morales *most likely* violated the CFA Institute Standards of Professional Conduct?
- Yes, concerning Duties to Employers
 - Yes, concerning Preservation of Confidentiality
 - No
8. According the GIPS standards, for periods beginning on or after 1 January 2011, the aggregate fair value of total firm assets *most likely* includes all:
- fee- and non-fee-paying discretionary and non-discretionary accounts.
 - fee-paying discretionary accounts.
 - fee- and non-fee-paying discretionary accounts.
9. While at a bar in the financial district after work, Ellen Miffitt, CFA, overhears several employees of a competitor discuss how they will manipulate down the price of a thinly traded micro-cap stock's price over the next few days. Miffitt's clients have large positions of this stock, so when she arrives at work the next day, she immediately sells all of these holdings. Because she had determined the micro-cap stock was suitable for all of her accounts at its previously higher price, Miffitt buys back her client's original exposure at the end of the week at the new, lower price. Which CFA Institute Standards of Professional Conduct did Miffitt *least likely* violate?
- Material Nonpublic Information
 - Preservation of Confidentiality
 - Market Manipulation
10. Pia Nilsson is a sole proprietor investment adviser. An economic recession has reduced the number of clients she advises and caused revenues to decline. As a result, Nilsson has not paid her CFA Institute membership dues for the past two years. When a national financial publication recently interviewed Nilsson, she indicated that up until two years ago, she had been a CFA charterholder and a CFA Institute member in good standing. In addition, she stated the completion of the CFA Program enhanced her portfolio management skills and enabled her to achieve superior returns on behalf of her clients. Which of Nilsson's actions *most likely* violated the CFA Institute Standards of Professional Conduct?
- Nonpayment of CFA Institute membership dues
 - Indicating that being a CFA charterholder has enhanced her portfolio management skills
 - Attributing her superior returns to participation in the CFA Program

11. Molly Burnett, CFA, is a portfolio manager for a fund that only invests in environmentally friendly companies. A multinational utility company recently acquired one of the fund's best-performing investments, a wind power company. The wind power company's shareholders received utility company shares as part of the merger agreement. The utility has one of the worst environmental records in the industry, but its shares have been one of the top performers over the past 12 months. Because the utility pays a high dividend every three months, Burnett holds the utility shares until the remaining two dividends are paid for the year then sells the shares. Burnett *most likely* violated the CFA Institute Standard of Professional Conduct concerning:
- Independence and Objectivity.
 - Suitability.
 - Disclosure of Conflicts.
12. Gabrielle Gabbe, CFA has been accused of professional misconduct by one of her competitors. The allegations concern Gabbe's personal bankruptcy filing 10 years ago when she was a college student and had a large amount of medical bills she could not pay. By not disclosing the bankruptcy filing to her clients, did Gabbe *most likely* violate any CFA Institute Standards of Professional Conduct?
- No
 - Yes, related to Misrepresentation
 - Yes, related to Misconduct
13. Francesca Ndenda, CFA, and Grace Rutabingwa work in the same department for New Age Managers, with Rutabingwa reporting to Ndenda. Ndenda learns that Rutabingwa received a Notice of Enquiry from the Professional Conduct Program at CFA Institute regarding a potential cheating violation when she sat for the CFA exam in June. As Rutabingwa's supervisor, Ndenda is afraid that Rutabingwa's behavior will be seen as a violation of the Code and Standards. Does Ndenda *most likely* have cause for concern?
- Yes
 - No, because her responsibilities do not apply
 - No, not until Rutabingwa is found guilty of cheating
14. Which of the following statements concerning why the Global Investment Performance Standards (GIPS) were created is *least likely* correct? The GIPS standards were created to:
- establish a standardized, industry wide approach for investment firms to follow.
 - provide clients certainty in what is presented and allow them to make reasonable comparisons.
 - identify a set of ethical principles for firms to follow in calculating and presenting historical investment results.
15. Chan Liu, CFA, is the new research manager at the Pacific MicroCap Fund. Liu observed the following activities after she published a research report on a thinly traded micro-cap stock that included a "buy" recommendation:
- Pacific traders purchased the stock for Pacific's proprietary account and then purchased the same stock for all client accounts; and
 - Pacific marketing department employees disseminated positive, but false, information about the stock in widely read internet forums.
- Liu notes the stock's price increased more than 50% within a period of two days and was then sold for Pacific's account. Which of the following steps is most appropriate for Liu to take to avoid violating the CFA Institute Standards of Professional Conduct?
- Remove her name from the micro-cap stock research report.
 - Report the observed activities to her employer.
 - Publicly refute the false information posted on internet forums.

16. Mariam Musa, CFA, head of compliance at Dunfield Brokers, questions her colleague Omar Kassim, a CFA candidate and a research analyst, about his purchase of shares in a company for his own account immediately before he publishes a "buy" recommendation. He defends his actions by stating he has done nothing wrong because Dunfield does not have any personal trading policies in place. The CFA Institute Standards of Professional Conduct were most likely violated by:
- A. only Musa.
 - B. Only Kassim.
 - C. both Musa and Kassim.
17. James Simone, CFA, the chief financial officer of a publicly listed company, seeks to improve the quality of his company's communication with institutional fund managers. He holds an investor briefing with this group the evening before the company earnings are announced. The company's quarterly earnings are broadcast in a press release the next day before the market opens. The earnings information in the investor briefing is identical to that in the press release. Did Simone *most likely* violate the CFA Institute Standards of Professional Conduct?
- A. Yes
 - B. No, because the company releases information while the market is closed
 - C. No, because investor briefing and press release information are identical
18. Kirsten Kelso, CFA, is a research analyst at an independent research firm. Kelso is part of a team of analysts who focus on the automobile industry. Recently, Kelso disagreed with two research sell recommendations written by her team, even though she felt confident the research process was properly conducted. In a webcast open to all institutional but not retail clients, Kelso states, "Even though my name is on the sell reports, these stocks are a buy in part because sales and share prices for both auto companies will rise significantly because of strong demand for their vehicles." Kelso's actions would *least likely* violate which of the following CFA Institute Standards of Professional Conduct?
- A. Communication with Clients
 - B. Diligence and Reasonable Basis
 - C. Fair Dealing

Answers

1. Answer = A

Selling stock short is a management strategy and does not necessarily violate any aspect of the Standards of Professional Conduct.

CFA Level I

"Guidance for Standards I–VII"

Standard II(B)—Market Manipulation

2. Answer = C

Kozniak does not appear to have violated any CFA Institute Standards of Professional Conduct. Because she is known in the market for investing and brokering property and both parties have worked with Kozniak in the past, both parties would know of her interests. In addition, in both cases, she acts for her own account as a primary investor, not as a broker. She buys the property for her own portfolio and then sells the property from her own portfolio. Therefore, Kozniak did not violate Standard VI(A)—Disclosure of Conflicts. When she purchased the property for her portfolio, she saved her client from losing the building to the bank and did not charge a sales commission. Because the sale of the property to her other client did not take place until six months after her purchase, and she was unable to contact the client who had earlier expressed interest prior to her purchase, she cannot be accused of violating Standard III(A)—Loyalty, Prudence, and Care with either client.

CFA Level I

"Guidance for Standards I–VII"

Standard III(A)—Loyalty, Prudence, and Care, Standard VI(A)—Disclosure of Conflicts

3. Answer = B

Compliance with the GIPS standards enhances the credibility of investment management firms.

CFA Level I

"Introduction to the Global Investment Performance Standards (GIPS)," CFA Institute Who Benefits from Compliance?

4. Answer = A

The analyst does not violate any of the Standards of Professional Conduct by having long-standing client relationships and generally is not required to disclose such relationships. However, the analyst is not treating all clients fairly as required by Standard III(B)—Fair Dealing when disseminating investment recommendations; disclosure of the relationship with long-standing clients is not the issue. The analyst has advantaged some clients over others by providing advance information, and all clients do not have a fair opportunity to act on the information within the draft report. Members and candidates may differentiate their services to clients, but different levels of service must not disadvantage or negatively affect clients.

CFA Level I

"Guidance for Standards I–VII" Standard III(B)—Fair Dealing

5. Answer = C

The member misrepresented the returns she could realistically achieve for her clients, violating Standard I(C)—Misrepresentation, which prohibits members and candidates from guaranteeing clients any specific return on volatile investments.

CFA Level I

"Guidance for Standards I–VII"

Standard I(C)—Misrepresentation

6. Answer = B

Buffet sat on the audit committee that determined the bank's provisioning policies that were contrary to the statutory regulations of the central bank. As a result, he most likely violated Standard I—Professionalism by not abiding by regulations of a regulatory body. Gatabaki did not violate Standard I - Professionalism because it is not apparent she knowingly facilitated the incorrect provisioning policy.

CFA Level I

"Guidance for Standards I–VII," CFA Institute Standard I(A)

7. Answer = C

Because Morales believes his supervisor and potentially the client are engaged in fraudulent activity and following the requirements of local law, he has not violated Standard III(E)–Preservation of Confidentiality or Standard (V)–Duties to Employers.

CFA Level I

"Guidance for Standards I–VII"

Standard III(E) Preservation of Confidentiality, Standard (IV) Duties to Employers, Standard (V) Duties to Employers

8. Answer = A

For periods beginning on or after 1 January 2011, total firm assets must include the aggregate fair value of all discretionary and non-discretionary assets managed by the firm. This includes both fee paying and non-fee-paying portfolios.

CFA Level I

"Global Investment Performance Standards (GIPS)" GIPS Requirement 0.A.13

9. Answer = B

Miffitt has not violated Standard III (E)–Preservation of Confidentiality, which involves information about former, current, and prospective clients.

CFA Level I

"Guidance for Standards I–VII"

Standard II(A)–Material Nonpublic Information, Standard II(B)–Market Manipulation, Standard III(E)–Preservation of Confidentiality

10. Answer = C

It is a violation of Standard VII(B)–Reference to CFA Institute, the CFA Designation, and the CFA Program to claim that the CFA charter helped her to achieve superior returns.

CFA Level I

"Guidance for Standards I–VII"

Standard VII(B)–Reference to CFA Institute, the CFA Designation, and the CFA Program

11. Answer = B

The utility is not a suitable investment for a fund that only invests in companies with good environmental records. Continuing to hold this investment, therefore, was a violation of Standard III(C)–Suitability.

CFA Level I

"Guidance for Standards I–VII"

Standard I(B)–Independence and Objectivity, Standard III(C)–Suitability, Standard VI(A)–Disclosure of Conflicts

12. Answer = A

A personal bankruptcy does not necessarily constitute a violation of Standard I(C)–

Misrepresentation or Standard I(D)–Misconduct. If the circumstances of the bankruptcy involved fraudulent or deceitful business conduct, then failing to disclose it may constitute a violation of the Standards of Professional Conduct.

CFA Level I

"Guidance for Standards I–VII"

Standard I(C)–Misrepresentation, Standard I(D)–Misconduct

13. Answer = B

A supervisor's responsibilities relate to detecting and preventing violations by anyone subject to their supervision or authority regarding activities they supervise. Ndenda had no way of detecting and/or preventing Rutabingwa from cheating during the CFA exam, if in fact that is what she did, because it was an event she did not attend.

CFA Level I

"Guidance for Standards I–VII," CFA Institute Standard IV(C)

14. Answer = B

The GIPS standards were created to ensure fair representation and full disclosure of investment performance, not to provide certainty in what is presented.

CFA Level I

"Introduction to the Global Investment Performance Standards (GIPS)," CFA Institute Why Were the GIPS Standards Created?

16. Answer = B

Certain staff at Liu's employer appear to be engaged in front running, a violation of Standard VI(B)–Priority of Transactions, and market manipulation, a violation of Standard II(B)–Market

Manipulation. If Liu observes these violations without taking steps to notify her employer, she will be in violation of Standard I(A)–Knowledge of the Law. Liu should know that the conduct observed is likely a violation of applicable laws, rules, and regulations as well as a violation of the CFA Institute Standards of Professional Conduct. Her first step, therefore, should be to attempt to stop the behavior by bringing it to the attention of the employer through a supervisor or the firm's compliance department. Inaction may be construed as participation or assistance in the illegal or unethical conduct.

CFA Level I

"Guidance for Standards I–VII"

Standard I (A)–Knowledge of the Law, Standard II(B)–Market Manipulation, Standard VI(B)–Priority of Transactions

16. Answer = C

Both Musa and Kassim violated the Standards of Professional Conduct. Musa violated Standard IV(C)–Responsibilities of Supervisors by not ensuring policies were in place to prevent violations of the Standards of Professional Conduct (in this case, Standard VI(B)–Priority of Transactions) by someone subject to her supervision. As the head of compliance, Musa supervised Kassim and must meet her supervisory responsibilities outlined in the Standards of Professional Conduct. Kassim violated Standard VI(B)–Priority of Transactions because he did not give sufficient priority to Dunfield's clients before trading on his recommendation.

CFA Level I

"Guidance for Standards I–VII"

Standard IV(C)–Responsibilities of Supervisors, Standard VI(B)–Priority of Transactions

17. Answer = A

Simone violated Standard II(A)–Material Nonpublic Information by giving institutional fund managers access to material nonpublic information prior to public dissemination (i.e., the press release). By releasing earnings results to a select group of institutional fund managers prior to a public press release, Simone allows the institutional fund managers a time advantage over other investors not invited to the investor briefing.

CFA Level I

"Guidance for Standards I–VII"

Standard II(A)–Material Nonpublic Information

18. Answer = B

The recommendation is based on a reasonable and adequate research process, so the analyst could follow the research team's opinion, as required by Standard V(A)–Diligence and Reasonable Basis.

CFA Level I

"Guidance for Standards I–VII"

Standard III(B)–Fair Dealing, Standard V(A)–Diligence and Reasonable Basis, Standard V(B)–Communication with Clients and Prospective Clients

Chapter 2

Quantitative Methods

Mock A

- The arithmetic and geometric mean are calculated for the same data. If there is variability in the data, compared with the arithmetic mean, the geometric mean will most likely be:
 - Smaller.
 - Greater.
 - Equal.
- The number of permutations that are possible when choosing 4 objects from a total of 10 objects is closest to:
 - 5,040.
 - 30.
 - 210.
- The null hypothesis is most likely to be rejected when the p-value of the test statistic:
 - Falls below a specified level of significance.
 - Exceeds a specified level of significance.
 - Is negative.
- If two events, A and B, are independent and the probability of A does not equal the probability of B [i.e., $P(A) \neq P(B)$], then the probability of Event A given that Event B has occurred [i.e., $P(A|B)$] is best described as:
 - $P(B)$.
 - $P(B|A)$.
 - $P(A)$.
- A discrete uniform distribution consists of the following 12 values:

-2.5	5.3	6.7	8.8	-4.6	9.2
3.3	8.2	1.4	0.8	-5.3	6.9

On a single draw from the distribution, the probability of drawing a value between -2.0 and 2.0 from the distribution is closest to:

- 16.67%.
 - 33.33%.
 - 27.59%.
- 6.

Cumulative Probabilities for a Standard Normal Distribution $P(Z \leq x) = N(x)$ for $x \geq 0$ or $P(Z \leq z) = N(z)$ for $z \geq 0$										
x or z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.10	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.20	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.30	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
...										
1.90	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.00	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.10	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
...										
2.50	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.60	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.70	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974

A variable is normally distributed with a mean of 5.00 and a variance of 4.00. Using the excerpt above from the cumulative distribution function for the standard normal random variable table, the probability of observing a value of -0.40 or less is closest to:

- a) 8.85%.
- b) 2.44%.
- c) 0.35%.

7. Consider the investment in the following table:

Start of Year 1	One share purchased at \$100
End of Year 1	\$5.00 dividend/share paid and one additional share purchased at \$125
End of Year 2	\$5.00 dividend/share paid and both shares sold for \$140 per share

Assuming dividends are not reinvested, compared with the time-weighted return, the money-weighted return is:

- a) Lower.
 - b) Higher.
 - c) The same.
8. The probability of Event A is 40%. The probability of Event B is 60%. The joint probability of AB is 40%. The probability (P) that A or B occurs, or both occur, is closest to:
- a. 60%.
 - b. 84%.
 - c. 40%.
9. The belief that trends and patterns tend to repeat themselves and are, therefore, somewhat predictable best describes:
- a. Arbitrage pricing theory.
 - b. Technical analysis.
 - c. weak-form efficiency
10. An investor wants to maximize the possibility of earning at least 5% on her investments each year.

Portfolio	Expected Return	Standard Deviation	Roy's Safety- First Ratio
1			0.35
2			0.64
3	22%	40%	??

Using Roy's safety-first criterion, the most appropriate choice for the investor is portfolio:

- a) 3.
 - b) 2.
 - c) 1.
11. If the probability for an event Z is 14% (i.e., $P(Z) = 14\%$), the odds for Z are closest to:
- a. 0.071.
 - b. 0.163.
 - c. 0.123.
12. Based on historical returns, a portfolio has a Sharpe ratio of 2.0. If the mean return to the portfolio is 20%, and the mean return to a risk-free asset is 4%, the standard deviation of return on the portfolio is closest to:
- a. 12%.
 - b. 8%.
 - c. 10%.
13. When two mutually exclusive projects with conventional cash flows are being ranked, the net present value (NPV) and internal rate of return (IRR) decision rules are most likely to conflict when the:
- a. Projects have similar timing of cash flows.

- b. Projects' investments are of different scale.
 - c. Projects have multiple IRRs.
14. When flipping three coins simultaneously, the number of outcomes that contain at least two-heads is most likely:
- a. Three.
 - b. Eight.
 - c. Four.

Answers

1. A.

The geometric mean is always less than or equal to the arithmetic mean. The only time the two means will be equal is when there is no variability in the observations.

CFA Level I

"Statistical Concepts and Market Returns,"

2. A.

The geometric mean is always less than or equal to the arithmetic mean. The only time the two means will be equal is when there is no variability in the observations.

$$nPr = \frac{n!}{(n-r)!} \text{ IN THIS PROBLEM, } 10!/(10-4)! = 10!/6! = 5,040$$

CFA Level I

"Statistical Concepts and Market Returns,"

3. A.

If the p-value is less than the specified level of significance, the null hypothesis is rejected.

CFA Level I

"Hypothesis Testing,"

4. C.

Two events, A and B, are independent if and only if $P(A|B) = P(A)$ or, equivalently, $P(B|A) = P(B)$. The wording of the question precludes $P(A) = P(B)$; therefore, $P(B)$ and $P(B|A)$ cannot be correct.

CFA Level I

"Probability Concepts,"

5. A.

First order the values from smallest to largest.

-5.3	-4.6	-2.5	0.8	1.4	3.3
5.3	6.7	6.9	8.2	8.8	9.2

Then note that 2 of the 12 values (i.e., 0.8 and 1.4) are between -2.0 and 2.0 . Thus, the probability of a draw from the distribution being between -2.0 and 2.0 is $2/12 = 0.16667$

CFA Level I

"Common Probability Distributions,"

6. C.

First the outcome of interest, -0.40 , is standardized for the given normal distribution: $Z = (X - \mu)/\sigma = (-0.40 - 5.00)/2 = -2.70$. Then use the table to find the probability of a Z value being 2.70 standard deviations below the mean (i.e., when $z \leq 0$). The value is $1 - P(Z \leq +2.70)$. In this problem, the solution is: $1 - 0.9965 = 0.0035 = 0.35\%$.

CFA Level I

"Common Probability Distributions,"

7. A.

The following table represents cash flows of the investment:

Year	Contribution	Start of year value after Contribution	End of year Dividend	End of year value after Dividend
1	$1 \times \$100$	$1 \times \$100 = \100	$1 \times \$5 = \5	\$125
2	$1 \times \$125$	$2 \times \$125 = \250	$2 \times \$5 = \10	$(2 \times 140) + 10 = \$290$

The time-weighted rate of return (TWR) on this investment is found by taking the geometric mean of the two holding period returns (HPRs):

$TWR = [(1 + HPR_{Year\ 1}) \times (1 + HPR_{Year\ 2})]^{1/2} - 1$, where $HPR_{Year\ 1} = (\$125 - \$100 + \$5)/\$100 = 30.0\%$ $HPR_{Year\ 2} = (\$280 - \$250 + \$10)/\$250 = 16.0\%$

$TWR = [(1 + 0.30) \times (1 + 0.16)]^{1/2} - 1 = 22.80\%$

The money-weighted rate of return (MWR) is the internal rate of return (IRR) of the cash flows associated with the investment:

, where $r = MWR$.

Using the cash flow (CF) function of a financial calculator:

$CF_0 = -100$, $CF_1 = (-125 + 5)$, $CF_2 = (280 + 10)$ and solving for IRR: $MWR \text{ or } IRR = 20.55\%$.

The difference between the TWR and MWR of this investment = $22.80\% - 20.55\% = 2.25\%$, or 225 bps, with MWR being lower than TWR

CFA Level I

"Discounted Cash Flow Applications,"

8. A.

$P(A \text{ or } B) = P(A) + P(B) - P(AB) = 0.40 + 0.60 - 0.40 = 0.60$ or 60%.

CFA Level I

"Probability Concepts,"

9. A.

Technical analysts believe that trends and patterns tend to repeat themselves and are, therefore, somewhat predictable.

CFA Level I

"Technical Analysis,"

10. B.

The portfolio with the highest safety-first ratio (SFRatio) is preferred. The SFRatio is calculated by subtracting the threshold return (RL) from the expected return $[E(RP)]$ and dividing by the standard deviation (σ_P). $SFRatio = [E(RP) - RL]/\sigma_P$. For the choices given:

	Portfolio1	Portfolio2	Portfolio3
Roys's safety-first criterion	0.35	0.64	$0.425 = \frac{(22-5)}{40}$
Portfolio 2 has the highest SF RATIO, so it is the most appropriate choice			

CFA LEVEL 1

"Common Probability Distributions,"

11. B.

Odds are calculated as $P(Z)/[1 - P(Z)]$. In this problem, $0.14/0.86 = 0.16279 \sim 0.163$.

CFA Level I

"Probability Concepts,"

12. B.

The Sharpe ratio for a portfolio p , based on historical returns, is defined as

$$S_p = \frac{\bar{R}_p - \bar{R}_F}{s_p}$$

where $\bar{R}_p$ is the mean return to the portfolio, $\bar{R}_F$ is the mean return to a risk-free asset, and s_p is the standard deviation of return on the portfolio. In this instance, $2 = (20\% - 4\%)/s_p$. Solving for s_p : $s_p = (20\% - 4\%)/2 = 8\%$.

CFA Level I

"Statistical Concepts and Market Returns," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle

13. B.

Conflict between the NPV and IRR decision rules can arise when evaluating mutually exclusive projects with conventional cash flows because 1) the scale of investments may differ and/or 2) the timing of the cash flows may differ.

CFA Level I

"Discounted Cash Flow Applications,"

14. C.

The number of outcomes having at least two heads is four, as indicated in the following table.

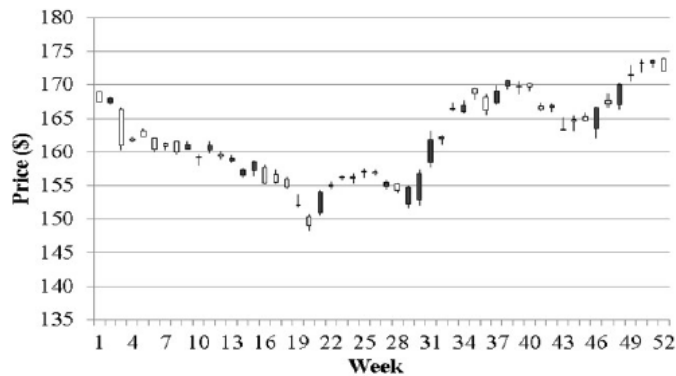
No. of Heads	Outcomes (Coin 1, Coin 2, Coin 3)	No. of Possible Outcomes
At least 2	(H,H,T), (H,T,H), (T,H,H) and (H,H,H)	4

CFA Level I

"Common Probability Distributions,"

Questions

1. The effective annual yield (EAY) for an investment is 8.0%. Its bond equivalent yield is *closest* to:
 1. 7.85%.
 2. 8.00%.
 3. 8.16%.
2. The following chart is *best* described as an example of which type of technical analysis chart?



- a) A candlestick chart
- b) A point and figure chart
- c) A bar chart
3. A price range in which selling is sufficient to stop the rise in price is *best* described as:
 - a) Change in polarity.
 - b) Support.
 - c) Resistance
4. A sample of 240 managed portfolios has a mean annual return of 0.11 and a standard deviation of returns of 0.23. The standard error of the sample mean is closest to:
 - a) 0.00710.
 - b) 0.01485.
 - c) 0.0009
5. The following is an excerpt from the cumulative distribution function for the standard normal random variable table:

Cumulative Probabilities for a Standard Normal Distribution

$P(Z \leq x) = N(x)$ for $x \geq 0$ or $P(Z \leq z) = N(z)$ for $z \geq 0$

x or z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.10	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.20	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.30	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.40	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
...										
1.10	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.883
1.20	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.30	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.40	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
...										
1.80	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.90	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.00	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.10	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857

A variable is normally distributed with a mean of 2.00 and a variance of 16.00. Using the excerpt, the probability of observing a value of 7.40 or h less is *closest* to:

- a) 63.3%.
- b) 96.8%.
- c) 91.2%.

6. Using a two-tailed test of the hypothesis that the population mean is zero, the calculated test statistic is 2.51. The sample has 23 observations. The population is normally distributed with an unknown variance.

Degrees of freedom	$p = 0.10$	$p = 0.05$	$p = 0.025$	$p = 0.01$	$p = 0.005$
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797

An analyst will *most likely* reject the null hypothesis at significance levels of:

- a) 0.10 Only.
 - b) 0.10, 0.05, and 0.01.
 - c) 0.10 And 0.05.
7. Once an investor chooses a particular course of action, the value forgone from alternative actions is *best* described as a (n):
- a) Opportunity cost.
 - b) Sunk cost.
 - c) Required return.
8. A technical analyst observes a head and shoulders pattern in a stock she has been following. She notes the following information:

Head price	\$83.50
Shoulder price	\$72.00
F	\$65.75
Current price	\$64.00

Based on this information, her estimate of the price target is *closest* to:

- a) \$48.00.
 - b) \$44.50.
 - c) \$59.50.
9. Samples of size (n₁, n₂) are drawn respectively from two populations (X₁, X₂) with associated sample means and standard deviations of (X₁, X₂) and (s₁, s₂) associated population means and standard deviations of (μ₁, μ₂) and (σ₁, σ₂) where σ₁ ≠ σ₂. In addition, $\bar{d}$ is the sample mean X₁ - X₂ of with a standard error of S $\bar{d}$ and a population mean of μ_{d0} and s_p² is a pooled estimator of the common variance.

The most appropriate test statistic to determine the equality of two population means assuming X₁ and X₂ are independent and normally distributed is:

a) $t = \frac{\bar{d} - \mu_{d0}}{s_{\bar{d}}}$

b) $t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^{0.5}}$

c) $t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\left(\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}\right)^{0.5}}$

10. If a stock's continuously compounded return is normally distributed, then the distribution of the future stock price is *best* described as being:
- a) A Student's t.
 - b) Lognormal.
 - c) normal
11. Technical analysts *most likely* study trends and patterns in security prices to forecast a company's:
- a) Earnings potential.
 - b) Intrinsic value.
 - c) Future price trends.
12. A stock's expected price movement over the next two periods is as follows:

Time = 0	Time = 1	Time = 2
$S_0 = 80$	$S_u = 88$	$S_{uu} = 96.80$
	$S_d = 72$	$S_{ud,du} = 79.20$
		$S_{dd} = 64.80$

The initial value of the stock is \$80. The probability of an up move in any given period is 75%, and the probability of a down move in any given period is 25%. Using the binomial model, the probability that the stock's price will be \$79.20 at the end of two periods is *closest* to:

- a) 18.75%.
- b) 56.25%.
- c) 37.50%.

13. The following information is available for a portfolio:

Asset Class	Asset Allocation Weight (%)	Asset Class Return (%)	Correlation with Equities Class (%)
Equities	45	16	100
Mortgages	25	12	30
Cash and equivalents	30	2	10

The return on the portfolio is *closest* to:

- a) 10.8%.
- b) 10.0%.
- c) 8.2%.

14. Which of the following, holding all else constant, will most likely increase the width of the confidence interval for a parameter estimate?

- a) Reduction in the degree of confidence
- b) Use of the t-distribution rather than the normal distribution to establish the confidence interval
- c) Increase in the sample size

Answers

1. A

$$EAY = [(1 + YTM)^{\frac{365}{T}}] - 1$$

Semiannual yield to maturity (YTM), $YTM = (1 + 0.08)^{0.5} - 1 = 0.03923 = 3.923\%$.

Bond equivalent yield = $2 \times YTM = 2 \times 3.923\% = 7.85\%$.

CFA Level I

"Discounted Cash Flow Applications,"

2. A

The chart is an example of a candlestick chart. A candlestick chart provides four prices per data point entry: the opening and closing prices and the high and low prices during the period (i.e., during a week). In a candlestick chart, a vertical line represents the range through which the security price traveled during the time period. The line is known as the wick or shadow. The body of the candle is shaded if the opening price was higher than the closing price, and the body is clear if the opening price was lower than the closing price.

CFA Level I

"Technical Analysis,"

3. C

Resistance is defined as a price range in which selling activity is sufficient to stop the rise in price.

CFA Level I

"Technical Analysis,"

4. B

For a sample, the standard error of the mean is

$$S_x$$

where s is the sample standard deviation and n is the sample size), which here is: $0.23/\sqrt{240} = 0.01485$.

CFA Level I

"Sampling and Estimation,"

5. C

First the outcome of interest, 7.40, is standardized for the given normal distribution:

$$Z = (X - \mu) / \sigma = (7.40 - 2.00) / \sqrt{16} = 1.35.$$

Then, the given table of values is used to find the probability of a Z-value being less than or equal to 1.35 standard deviations above the mean. The value is $P(Z \leq 1.35) = 0.9115 = 91.2\%$.

CFA Level I

6. C

This is a two-tailed hypothesis testing because it concerns whether the population mean is zero.

Versus With degrees of freedom (df) = $n - 1 = 23 - 1 = 22$, the rejection points are as follows:

Significance level	Rejection points for t-test
0.10	$t < -1.717$ and $t > 1.717$
0.05	$t < -2.074$ and $t > 2.074$
0.01	$t < -2.819$ and $t > 2.819$

Because the calculated test statistic is 2.51, the null hypothesis is thus rejected at the 0.05 and 0.10 levels of significance but not at 0.01.

CFA Level I

"Hypothesis Testing,"

7. A

An opportunity cost is the value that investors forgo by choosing a particular course of action.

CFA Level I

"The Time Value of Money,"

8. A

Price target = Neckline – (Head – Neckline)

In this example, Price target = \$65.75 – (\$83.50 – \$65.75) = \$65.75 – \$17.75 = \$48.00.

CFA Level I

"The Time Value of Money,"

9. B

The most appropriate test statistic for the difference between two population means (unequal and unknown population variances) is

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^{0.5}}$$

CFA Level I

"Hypothesis Testing,"

10. B

If a stock's continuously compounded return is normally distributed, then the future stock price is necessarily log normally distributed.

CFA Level I

"Common Probability Distributions,"

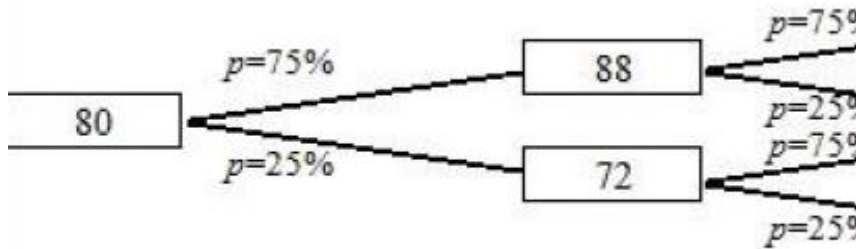
11. C

Technical analysts believe that market trends and patterns tend to repeat, so they rely on recognizing past patterns in an attempt to project future security price patterns.

CFA Level I

"Common Probability Distributions,

12. C



Across two periods, there are four possibilities:

U, U end value \$96.80,

U, D end value: \$79.20,

D, U end value: \$79.20,

D, D end value: \$64.80,

Where U is an up move and D is a down move.

The probability of an up move followed by a down move is $0.75 \times 0.25 = 0.1875$.

The probability of a down move followed by an up move is $0.25 \times 0.75 = 0.1875$.

Both of these sequences result in an end value of \$79.20.

Therefore, the probability of an end value of \$79.20 is $(0.1875 + 0.1875) = 0.375 = 37.5\%$.

Alternatively, the following formula could be used:

$$P(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x} = \frac{n!}{(n-x)!x!} p^x (1-p)^{n-x}$$

Where

$n = 2$ (number of periods)

$x = 1$ (number of up moves: ud and du)

$p = 0.75$ (probability of an up move)

$$p(1) = \binom{2}{1} 0.75^1 (1-0.75)^{2-1} = \frac{2!}{(2-1)!1!} \times 0.75^1 \times 0.25^{2-1} = 2 \times 0.75 \times 0.25 = 0.375$$

CFA Level I

13. A

The portfolio return is the weighted mean return and is calculated as follows:

$$\bar{X}_w = \sum_{i=1}^n w_i \bar{X}_i = (0.45 \times 16) + (0.25 \times 12) + (0.30 \times 2) = 10.8\%$$

CFA Level I

"Statistical Concepts and Market Returns,"

14. B

Reflecting the uncertainty of the unknown variance, confidence intervals based on the t-distribution will be larger than those using the normal distribution because $t > z$ for any sample size, n , with the exception of $n = \infty$. Larger sample sizes and reduced confidence levels, holding all else constant, both reduce the width of a confidence interval.

Questions

- The discrepancy between a statistically significant result and an economically meaningful result is *least likely* the result of:
 - transaction costs.
 - sampling errors.
 - risk tolerance.
- Which of the following *most* accurately describes a distribution that is more peaked than normal?
 - Mesokurtotic
 - Platykurtotic
 - Leptokurtotic
- When testing a hypothesis, the power of a test is *best* described as the:
 - probability of rejecting a true null hypothesis.
 - probability of correctly rejecting the null hypothesis.
 - same as the level of significance of the test.
- A company has an unsecured line of credit and needs to maintain its EBIT-to-interest coverage ratio greater than 2.0. Its EBIT is estimated to be between \$36 million and \$48 million, with all values equally likely. If the forecasted interest charge for the year is \$20 million, the probability that EBIT/interest will be more than 2.0 is *closest* to:
 - 33.3%.
 - 66.7%.
 - 61.5%.
- If the stated annual interest rate is 9% and the frequency of compounding is daily, the effective annual rate (EAR) is *closest* to:
 - 9.00%.
 - 9.86%.
 - 9.42%.
- A financial contract offers to pay €1,200 per month for five years with the first payment made immediately. Assuming an annual discount rate of 6.5%, compounded monthly the present value of the contract is *closest* to:
 - €61,663.
 - €61,330.
 - €63,731.
- Use the following values from a student's *t*-distribution to establish a 95% confidence interval for the population mean given a sample size of 10, a sample mean of 6.25, and a sample standard deviation of 12. Assume that the population from which the sample is drawn is normally distributed and the population variance is not known.

Degrees of Freedom	$p = 0.10$	$p = 0.05$	$p = 0.025$	$p = 0.01$
9	1.383	1.833	2.262	2.821
10	1.372	1.812	2.228	2.764
11	1.363	1.796	2.201	2.718

The 95% confidence interval is *closest* to a:

- lower bound of -2.20 and an upper bound of 14.70.
 - lower bound of -2.33 and an upper bound of 14.83.
 - lower bound of -0.71 and an upper bound of 13.21.
- Investors should be *most* attracted to return distributions that are:
 - normal.

- B. negatively skewed.
 - C. positively skewed.
9. The null hypothesis is *most likely* to be rejected when the p -value of the test statistic:
- A. falls below a specified level of significance.
 - B. exceeds a specified level of significance.
 - C. is negative.
10. A discrete uniform distribution consists of the following 12 values:

-2.5	5.3	6.7	8.8	-4.6	9.2
3.3	8.2	1.4	0.8	-5.3	6.9

On a single draw from the distribution, the probability of drawing a value between -2.0 and 2.0 from the distribution is *closest* to:

- A. 16.67%.
 - B. 27.59%.
 - C. 33.33%.
11. A descriptive measure of a population characteristic is *best* described as a:
- A. parameter.
 - B. frequency distribution.
 - C. sample statistic.
12. The arithmetic and geometric mean are calculated for the same data. If there is variability in the data, compared with the arithmetic mean, the geometric mean will *most likely* be:
- A. smaller.
 - B. greater.
 - C. equal.

Answers

1. Answer = B

Sampling errors will result in statistical error. A statistically significant result might not be economically meaningful after an analyst accounts for the risk, transaction costs, and applicable taxes.

CFA Level I

"Hypothesis Testing," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 2

2. Answer = C

A distribution that is more peaked than normal is called leptokurtotic.

CFA Level I

"Statistical Concepts and Market Returns," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 9

3. Answer = B

The power of a test is the probability of correctly rejecting the null hypothesis—that is, the probability of rejecting the null when it is false.

CFA Level I

"Hypothesis Testing," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 2

4. Answer = B

The EBIT-to-interest ratio is equal to 2.0 when the EBIT is \$40 million. Given that the values between

\$36 million and \$48 million are equally likely, the probability of the ratio being equal to or less than 2.0 is 33.3% ($= [\$40 \text{ million} - \$36 \text{ million}] / [\$48 \text{ million} - \$36 \text{ million}]$). Consequently, the probability of the ratio being greater than 2.0 is 66.7% (i.e., $1 - \text{Probability of the ratio being equal to or less than 2.0}$).

CFA Level I

"Common Probability Distributions," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 3.1

5. Answer = C

$\text{EAR} = (1 + \text{periodic interest rate})^m - 1 = [1 + (0.09 / 365)]^{365} - 1 = 0.094162$, rounded to 9.42%.

CFA Level I

"The Time Value of Money," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Sections 3.2, 3.3

6. Answer = A

Using a financial calculator: $N = 60$; the discount rate, $I/Y = (6.5\%/12) = 0.54166667$; $\text{PMT} = €1,200$; Future value = €0; Mode = Begin; Calculate present value (PV): $\text{PV} = €61,662.62$.

Alternatively: Treat the stream as an ordinary annuity of 59 periods and add the current value of

€1,200 to the derived answer. Using a financial calculator: $N = 59$; the discount rate, $I/Y = (6.5\%/12) = 0.54166667$; $\text{PMT} = €1,200$; Future value = €0; Mode = End; Calculate PV: $\text{PV} = €60,462.62$; Total PV = $€1,200 + €60,462.62 = €61,662.62$.

CFA Level I

"The Time Value of Money," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 6.1

7. Answer = B

With a sample size of 10, there are 9 degrees of freedom. The confidence interval concept is based on a two-tailed approach. For a 95% confidence interval, 2.5% of the distribution will be in each tail. Thus, the correct t -statistic to use is 2.262. The confidence interval is calculated as:

$$\bar{X} \pm t_{0.025} s/\sqrt{n}$$

Where $\bar{X}$ is the sample mean, s is the sample standard deviation, and n is the sample size. In this case: $6.25 \pm 2.262 \times 12/\sqrt{10} = 6.25 \pm 8.58369$ or **-2.33 to 14.83**.

CFA Level I

"Sampling and Estimation," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 4.2

8. Answer = C

Investors should be attracted by a positive skew (distribution skewed to the right) because the mean return falls above the median. Relative to the mean return, positive skew amounts to a limited, though frequent, downside compared with a somewhat unlimited, but less frequent, upside.

CFA Level I

"Statistical Concepts and Market Returns," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 8

9. Answer = A

If the p -value is less than the specified level of significance, the null hypothesis is rejected.

CFA Level I

"Hypothesis Testing," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 2

10. Answer = A

First order the values from smallest to largest.

-5.3	-4.6	-2.5	0.8	1.4	3.3
5.3	6.7	6.9	8.2	8.8	9.2

Then note that 2 of the 12 values (i.e., 0.8 and 1.4) are between -2.0 and 2.0. Thus, the probability of a draw from the distribution being between -2.0 and 2.0 is $2/12 = 0.16667$.

CFA Level I

"Common Probability Distributions," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 2.1

11. Answer = A

Any descriptive measure of a population characteristic is called a parameter.

CFA Level I

"Statistical Concepts and Market Returns," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 2.2

12. Answer = A

The geometric mean is always less than or equal to the arithmetic mean. The only time the two means will be equal is when there is no variability in the observations.

CFA Level I

"Statistical Concepts and Market Returns," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 5.4.2

Questions

- The confidence interval is *most likely* to be:
 - wider as the sample size increases.
 - wider as the point estimate increases.
 - narrower as the reliability factor decreases.
- Using a two-tailed test of the hypothesis that the population mean is zero, the calculated test statistic is 2.51. The sample has 23 observations. The population is normally distributed with an unknown variance.

Degrees of freedom	$p = 0.10$	$p = 0.05$	$p = 0.025$	$p = 0.01$	$p = 0.005$
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797

An analyst will *most likely* reject the null hypothesis at significance levels of:

- 0.10, 0.05, and 0.01.
 - 0.10 and 0.05.
 - 0.10 only.
- Independent samples drawn from normally distributed populations exhibit the following characteristics:

Sample	Size	Sample Mean	Sample Standard Deviation
A	25	200	45
B	18	185	60

Assuming that the variances of the underlying populations are equal, the pooled estimate of the common variance is 2,678.05. The t -test statistic appropriate to test the hypothesis that the two population means are equal is *closest* to:

 - 0.29.
 - 0.94.
 - 1.90.
 - Once an investor chooses a particular course of action, the value forgone from alternative actions is *best* described as a(n):
 - opportunity cost.
 - sunk cost.
 - required return.
 - The central limit theorem is *best* described as stating that the sampling distribution of the sample mean will be approximately normal for large-size samples:
 - if the population distribution is symmetrical.
 - for populations described by any probability distribution.
 - if the population distribution is normal.
 - Survivorship bias is *most likely* an example of which bias?
 - Look-ahead
 - Data mining
 - Sample selection

7. If a stock's continuously compounded return is normally distributed, then the distribution of the future stock price is *best* described as being:
- normal.
 - lognormal.
 - a Student's t .
8. The distribution of all the distinct possible values for a statistic when calculated from samples of the same size randomly drawn from the same population is *most* accurately referred to as:
- the sampling distribution of a statistic.
 - a discrete uniform distribution.
 - a multivariate normal distribution.
9. Compared with historical simulation, Monte Carlo simulation is *most* appropriate when:
- "what if" analysis is required.
 - analytical methods are required.
 - probability distributions are unavailable.
10. Consider a two-tailed test of the hypothesis that the population mean is zero. The sample has 50 observations. The population is normally distributed with a known variance.

t-Distribution			
Degrees of freedom	$p = 0.10$	$p = 0.05$	$p = 0.025$
49	1.299	1.677	2.010
50	1.299	1.676	2.009
z-Distribution	= 0.10	= 0.05	= 0.025
	1.645	1.960	2.330

At a 0.05 significance level, the rejection points are *most likely* at:

- 2.010 and 2.010.
 - 2.009 and 2.009.
 - 1.960 and 1.960.
11. A hypothesis test fails to reject a false null hypothesis. This result is *best* described as a:
- test with little power.
 - Type II error.
 - Type I error.
12. A sample of 25 observations has a mean of 8 and a standard deviation of 15. The standard error of the sample mean is *closest* to:
- 3.00.
 - 1.60.
 - 3.06.

13.

Table of the <i>F</i> -Distribution						
Panel A: Critical values for right-hand tail area equal to 0.05						
	df1					
	(read across)	1	2	3	4	5
df2						
(read down)	1	161	200	216	225	230
	2	18.5	19.0	19.2	19.2	19.3
	3	10.1	9.55	9.28	9.12	9.01
	4	7.71	6.94	6.59	6.39	6.26
	5	6.61	5.79	5.41	5.19	5.05
Panel B: Critical values for right-hand tail area equal to 0.025						
	df1	1	2	3	4	5
	(read across)					
df2						
(read down)	1	648	799	864	900	922
	2	38.51	39.00	39.17	39.25	39.30
	3	17.44	16.04	15.44	15.10	14.88
	4	12.22	10.65	9.98	9.60	9.36
	5	10.01	8.43	7.76	7.39	7.15

Which of the following statements is *most* appropriate? The critical value is:

- A. 9.60 and do not reject the null.
- B. 6.39 and reject the null.
- C. 7.15 and do not reject the null.

Answers

1. Answer = C

A confidence interval for a parameter = Point estimate $\pm$ Reliability factor $\times$ Standard error. For example, the reliability factors for confidence intervals based on the standard normal distribution are 1.65 for 90% confidence intervals and 1.96 for 95% confidence intervals. For a given point estimate and standard error, the confidence interval will be narrower with a lower reliability factor.

CFA Level I

"Sampling and Estimation," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle
Section 4.2

2. Answer = B

This is a two-tailed hypothesis testing because it concerns whether the population mean is zero.

$H_0 : \mu$ versus $H_a : \mu \neq 0$

With degrees of freedom (df) = $n - 1 = 23 - 1 = 22$, the rejection points are as follows:

Significance level	Rejection points for t -test
0.10	$t < -1.717$ and $t > 1.717$
0.05	$t < -2.074$ and $t > 2.074$
0.01	$t < -2.819$ and $t > 2.819$

Because the calculated test statistic is 2.51, the null hypothesis is thus rejected at the 0.05 and 0.10 levels of significance but not at 0.01.

CFA Level I

"Hypothesis Testing," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 3

3. Answer = B

The t -statistic for the given information (normally distributed populations, population variances assumed equal) is calculated as:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\left(\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}\right)^{0.5}}$$

In this case, we have: $s_p^2 = 2678.05$.

$$t = \frac{(200 - 185) - (0)}{\left(\frac{2678.05}{25} + \frac{2678.05}{18}\right)^{0.5}} = 0.93768 \sim \mathbf{0.94}$$

CFA Level I

"Hypothesis Testing," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 3.2

4. Answer = A

An opportunity cost is the value that investors forgo by choosing a particular course of action.

CFA Level I

"The Time Value of Money," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle
Section 2

5. Answer = B

The central limit theorem holds without regard for the distribution of the underlying population.

CFA Level I

"Sampling and Estimation," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle

Section 3.1

6. Answer = C

Sample selection bias often results when a lack of data availability leads to certain data being excluded from the analysis. Survivorship bias is an example of sample selection bias.

CFA Level I

"Sampling and Estimation," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle

Section 5.2

7. Answer = B

If a stock's continuously compounded return is normally distributed, then the future stock price is necessarily lognormally distributed.

CFA Level I

"Common Probability Distributions," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle

Section 3.4

8. Answer = A

The sampling distribution of a statistic (like a sample mean) is defined as the probability distribution of a given sample statistic when samples of the same size are randomly drawn from the same population.

CFA Level I

"Sampling and Estimation," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle

Section 2.1

9. Answer = A

Monte Carlo simulation lends itself to "what if" analysis and requires the user to provide a probability distribution or distributions. It can be a complement to analytical methods.

CFA Level I

"Common Probability Distributions," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle
Section 4

10. Answer = C

The appropriate test statistic is a z-statistic because the sample comes from a normal distributed population with known variance. A z-test does not use degrees of freedom. This test is two-sided at the 0.05 significance level, and the rejection point conditions are $z > 1.960$ and $z < -1.960$.

CFA Level I

"Hypothesis Testing," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 3

11. Answer = B

Failure to reject a false null hypothesis is a Type II error.

CFA Level I

"Hypothesis Testing," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 2

12. Answer = A

The standard error of the sample mean, when the sample standard deviation is known, is:

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}. \text{ In this case, } s_{\bar{x}} = \frac{15}{\sqrt{25}} = 3.00.$$

CFA Level I

"Sampling and Estimation," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle

Section 3.1

13. Answer = A

The test statistic makes use of the F -distribution and is the ratio of the variances, with the larger variance in the

numerator. The test statistic is $F = \frac{s_2^2}{s_1^2} = 28/4 = 7$. The degrees of freedom are 4 by 4. Because it is a two-tailed test, the correct critical value at $\alpha = 5\%$ is 9.60 (Panel B). Because the test statistic is less than the critical value (i.e., $7 < 9.60$), the null hypothesis cannot be rejected.

CFA Level I

"Hypothesis Testing," Richard A. DeFusco, Dennis W. McLeavey, Jerald E. Pinto, and David E. Runkle Section 4.2

Chapter 3

Economics

Mock A

Questions

- According to the Fisher effect, an increase in expected inflation will most likely increase:
 - The nominal interest rate.
 - Both nominal and real interest rates.
 - The real interest rate.
- Given stable aggregate supply, which of the following changes in aggregate demand is most likely to cause economic expansion?
 - An increase in foreign currency per unit of domestic currency
 - An increase in corporate income tax
 - An increase in capacity utilization
- A small country has a comparative advantage in the production of pencils. The government establishes an export subsidy for pencils to promote economic growth. Which of the following will be the most likely result of this policy?
 - The increase in the domestic producer surplus will exceed the sum of the subsidy and the decrease in the domestic consumer surplus.
 - Although domestic producers will receive a net benefit, the policy will give rise to inefficiencies that cause a deadweight loss to the national welfare.
 - As new domestic producers enter the pencils market, supply will increase and domestic prices will decline.
- Holding the working-age population constant, if the labour force participation rate declines while the number of people employed remains unchanged, the unemployment rate will most likely:
 - Decrease.
 - Remain unchanged.
 - Increase.
- An increase in both aggregate demand and supply occurs, with aggregate supply increasing more than aggregate demand. The most likely result is a decrease in the:
 - Real GDP.
 - Price level.
 - Participation rate.
- An expansionary fiscal policy is most likely associated with:
 - An increase in capital gains tax rates.
 - Crowding out of private investments.
 - An increase in government spending on social insurance and benefits.
- A New Zealand traveller returned from Singapore with SGD7, 500 (Singapore dollars). A foreign exchange dealer provided the traveller with the following quotes:

Ratio	Spot Rates
USD/SGD	1.2600
NZD/USD	0.7670
USD: US dollar NZD: New Zealand dollar	

The amount of New Zealand dollars (NZD) that the traveller would receive for his Singapore dollars is closest to:

- NZD4, 565.

- b) NZD7, 761.
- c) NZD7, 248.

8. The market demand function for item X is a function of its price, household income, and the price of item Y.

Own-price elasticity of demand for X	-0.8
Income elasticity of demand for X	1.5
Cross-price elasticity of demand for X with respect to the price of Y	0.4

Given the above elasticity coefficients for the two items, which of the following statements is most accurate?

- a) Item X is an inferior good.
 - b) X and Y are substitutes.
 - c) Demand for X is elastic.
9. According to the Solow neoclassical growth model, sustained long-term growth in potential GDP is best explained by:
- a) Growth in labour supply.
 - b) Technological change.
 - c) Capital deepening investments.
10. Three firms operate under perfect competition, producing 900 units of the same product but using different production technologies. Each company's cost structure is indicated in the table:

Company	X	Y	Z
Total Variable Costs	\$2,700	\$3,600	\$4,500
Total Fixed Costs	<u>\$2,700</u>	<u>\$1,800</u>	<u>900</u>
Total Costs	\$5,400	\$5,400	5400

Which of the following statements is most accurate? If the unit selling price is:

- a) 3.00, firm X should continue to operate in the short run, but firms Y and Z should shut down production.
 - b) \$4.50, all firms should continue to operate in the short run, but exit the market in the long run if these conditions are expected to persist.
 - c) \$6.00, all firms should exit the market in the long run.
11. The most relevant measure of income that economists use in determining household decisions to save and spend is personal:
- a) Earned income.
 - b) Taxable income.
 - c) Disposable income.
12. The elasticity of demand for a good is most likely greater when:
- a) The adjustment to a price change takes a longer time.
 - b) The good is a necessity.
 - c) A lesser proportion of income is spent on the good.

Answers

1. A.

The Fisher effect states that the nominal interest rate is the sum of the real rate of interest and the expected rate of inflation over a given time horizon. An increase in expected inflation will result in a higher nominal rate.

CFA Level I

"Monetary and Fiscal Policy,"

2. C.

An increase in capacity utilization will cause an increase in aggregate demand through higher investment and will increase GDP (economic expansion).

CFA Level I

"Aggregate Output, Prices, and Economic Growth,"

3. B.

Export subsidies interfere with the functioning of the free market and result in a deadweight loss to society. The deadweight loss arises on the producer side because the higher subsidized price causes inefficient producers to remain in the market. On the consumer side, the higher price causes those that would have purchased at the lower price to be shut out of the market.

CFA Level I

"International Trade and Capital Flows,"

4. A.

For a given working-age population, a decline in the labor force participation rate (often caused by an increase in discouraged workers) reduces the labor force. If the number of people employed remains the same while the labor force becomes smaller, the number of workers defined to be unemployed must be smaller and thus the unemployment rate lower.

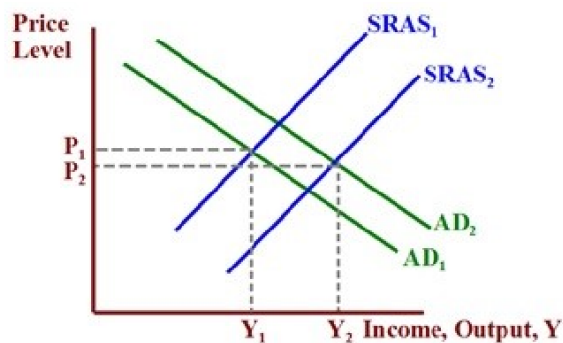
The following example illustrates the direction of change:		
	Initial Case	After Change
Working-age population	100	100
Labor force = Employed + Unemployed	$60 + 20 = 80$	$60 + 15 = 75$
Labor force participation rate	80%	75%
Unemployment rate	$20/80 = 25\%$	$15/75 = 20\%$
Labor force participation rate = Labor force/Working age population		
Unemployment rate = Unemployed/Labor force		

CFA Level I

"Understanding Business Cycles,"

5. B.

If both aggregate demand (AD) and aggregate supply (AS) increase, real GDP will increase but the impact on inflation is not clear unless we know the magnitude of the changes because an increase in AD will increase the price level, whereas an increase in AS will decrease the price level. If AD increases more than AS, the price level will increase. If AS increases more than AD, as depicted in the graph to the right, the price level will decline.



CFA Level I

"Aggregate Output, Prices, and Economic Growth." Paul R. Kutasovic and Richard G. Fritz

6. B.

Expansionary policy increases government borrowing, which may divert private sector investment from taking place (resulting in an effect known as crowding out). A rise in capital gain tax rates is a form of contractionary fiscal policy. Rises in government spending on social insurance and benefits is a form of automatic stabilizer and not due to discretionary fiscal expansion.

CFA Level I

"Monetary and Fiscal Policy,"

7. C.

The NZD/SGD cross-rate is $\text{NZD/USD} \times \text{USD/SGD} = 0.7670 \times 1.2600 = 0.9664$. The traveler will receive: NZD0.9664 per SGD; $\text{NZD}0.9664 \times \text{SGD}7,500 = \text{NZD}7,248$.

CFA Level I

"Currency Exchange Rates,"

8. B.

The cross-price elasticity is positive, indicating that as the price of Y increases, more of X is demanded, thus making X and Y substitutes.

CFA Level I

"Topics in Demand and Supply Analysis,"

9. B.

Growth in potential GDP equals growth in technology plus the weighted average growth rate of labor and capital. Because of diminishing marginal productivity (return), the only way to sustain long-term growth in potential GDP is through technological change or growth in total factor productivity.

CFA Level I

"Aggregate Output, Prices, and Economic Growth,"

10. A.

Revenue-Cost Relationship	Short-Run Decision	Long-Term Decision
$TR \geq TC$	Stay in market	Stay in market
$TR > TVC$ but $TR < TFC + TVC$	Stay in market	Exit market
$TR < TVC$	Shut down production	Exit market
where TR = Total Revenue;		
and TC = Total Costs; TVC = Total Variable Costs; TFC = Total Fixed Costs		

Hence, if the selling price is \$3.00, total revenue for all firms will be $\$3.00/\text{unit} \times 900 \text{ units} = \$2,700$. Only firm X's variable costs are covered and it should continue operating, while firms Y and Z should immediately shutdown production.

CFA Level I

"Topics in Demand and Supply Analysis,"

11. C.

Personal disposable income, which is equal to personal income minus personal taxes, is the most relevant measure of income for household spending and saving decisions.

CFA Level I

"Aggregate Output, Prices, and Economic Growth,"

12. A.

For most goods and services, the long-run demand is much more elastic than the short-run demand. For example, if gas prices rise, consumers cannot quickly change their mode of transportation, but will likely do so in the longer run.

CFA Level I

"Topics in Demand and Supply Analysis,"

Questions

1. The following data apply to a firm operating in perfect competition.

Quantity	Total Revenue	Total Cost
21	\$210	\$138
22	\$220	\$145
23	\$230	\$154
24	\$240	\$165

The firm's profit maximizing output (in units) is *most likely*:

- 23.
 - In excess of 24.
 - 21.
2. The following information applies to a hypothetical economy:

Total Population	1,100
Working age population	975
Labor Force	750
Underemployed	120
Unemployed	95
Discouraged workers	80
Frictionally unemployed	25
Voluntarily unemployed	40

The unemployment rate is closest to:

- 16.0%.
 - 9.7%.
 - 12.7%.
3. The price index that *best* resolves the substitution bias is the:
- Fisher index.
 - Laspeyres index.
 - Paasche index.
4. Cost-push inflation is *least likely* to be affected by an increase in:
- Employee wages.
 - Finished goods prices.
 - Commodity prices.
5. Which of the following is *least likely* to be a valid function/characteristic of money? Money:
- Requires a double coincidence of wants.
 - Acts as a unit of account.
 - Provides a store of wealth.
6. The two dominant supermarket chains in the area are attempting to increase their market share by moving to 24-hour service instead of closing at 9 p.m. every night. The strategic outcomes and payoff matrix that arise from their actions are depicted in the diagram (with the shaded sections representing payoffs for Chain 2).

		Chain 2	
		Close at 9 pm	Open 24 hours
Chain 1	Close at 9 pm	290 180	592 55
	Open 24 hours	75 540	140 108

According to Nash equilibrium, the best strategy is for:

- Only chain to open for 24 hours.
- Both chains to close at 9 p.m.
- Both chains to open for 24 hours.

7. The following information is available for 2011:

	New Zealand		Canada	
	Jan 1	Dec 31	Jan 1	Dec 31
Price index	1,137	1,158	117.8	119.9
Nominal exchange rate: NZD/CAD	1.2844			1.2589

The change in the real exchange rate (NZD/CAD terms) is closest to:

- 1.92
 - +1.96
 - 2.05
8. An expansionary fiscal policy is *least likely* to include an increase in:
- Government expenditures.
 - Budget deficit.
 - tax rates
9. Successful advertising and product differentiation are *most likely* to have a positive impact on the economic profits of a producer under:
- Monopolistic competition.
 - Perfect competition.
 - Monopoly.
10. An electricity producer charges lower rates to its high-volume customers and higher rates to its low-volume customers. The degree of price discrimination is best described as:
- Second.
 - Third.
 - First.
11. The statement that is most consistent with real business cycle (RBC) models is that:
- Persons are unemployed because their asking wages are too high.
 - Governments should intervene when the economy is in contraction.
 - Monetary variables have a major impact on GDP growth.
12. Which characteristic is a firm least likely to exhibit when it operates in a market with a downward sloping demand curve, many Competitors and zero economic profits in the long run?
- Differentiated product
 - Low barriers to entry
 - No pricing power

Answers

1. A.

Under perfect competition, economic profits are maximized when marginal revenue equals marginal cost—in this case, marginal cost crosses \$10 per unit. Profits are maximized at 23 units of production because marginal cost is in excess of marginal revenue at 24 units.

CFA Level I

"The Firm and Market Structures,"

2. C

$$\text{Unemployed Rate} = \frac{\text{Unemployed}}{\text{labor force}} \times 100 = \frac{95}{750} \times 100 = 12.7\%$$

CFA Level I

"Understanding Business Cycles,"

3. A

The Fisher index is a geometric mean of the Laspeyres and Paasche indexes, and it will therefore display less of a substitution bias than the other two. Both the Laspeyres index and the Paasche index ignore the substitution effect whereby people may substitute higher priced goods or services with cheaper ones. The Laspeyres index uses the historical composition of a basket of goods making it upward biased relative to the true inflation rate, while the Paasche index uses the current composition of the basket with cheaper options replacing more expensive ones, making it downward biased relative to the true inflation rate.

CFA Level I

"Understanding Business Cycles,"

4. B

Cost-push inflation arises due to increases in costs associated with production: wages and raw materials prices.

CFA Level I

"Understanding Business Cycles,"

5. A

The functions of money include being a means of payment, acting as a medium of exchange and acting as a unit of account. It does not require a double coincidence of wants, as barter does since it easily divisible and can act as a medium of exchange.

CFA Level I

"Monetary and Fiscal Policy,"

6. C

Each company will consider the other's reaction in selecting its strategy. Using the following summary, it is best for both chains to provide 24-hour service.

Chain	Consideration		Best Decision
1	It opens for 24 hours; it will see a higher payoff regardless of what Chain 2 does.		Opens for 24 hours
	Chain 2 Closes at 9 p.m.	Chain 2 Opens for 24 hours	
	Chain 1 earns 540 instead of 180	Chain 1 earns 108 instead of 55	

2	If it opens for 24 hours, it will see a higher payoff regardless of what Chain 1 does.		Opens for 24 hours
	Chain 1 Closes at 9 p.m.	Chain 2 Opens for 24 hours	
	Chain 2 earns 592 instead of 290	Chain 2 earns 140 instead of 75	

CFA Level I

"The Firm and Market Structures,"

7. C

	Formula	New Zealand	Canada
Inflation rate	$\frac{I1 - I0}{I0}$	$\frac{1,158 - 1,137}{1,137} = 1.85\%$	$\frac{119.9 - 117.8}{117.8} = 1.78\%$
Nominal Exchange Rate Change	$\frac{S1 - S0}{S0}$	$\frac{1.2589 - 1.2844}{1.2844} = -1.99\%$	
Real Exchange Rate Change	$\frac{\left(1 + \frac{\Delta SNZ/CDA}{SNZ/CDA}\right) \times \left(1 + \frac{\Delta PCANADA}{PCANADA}\right)}{\left(1 + \frac{\Delta PNZ}{PNZ}\right)} - 1$		$\frac{(1 - 0.0199) \times (1 + 0.0178)}{(1 + 0.0185)} - 1 = -2.05\%$
	Approximation:	$-1.99\% + 1.78\% - 1.85\% = -2.06\%$	

CFA Level I "Understanding Business Cycles,"

"Currency Exchange Rates,"

8. C

An expansionary fiscal policy means that the government increases its purchases of goods and services and/or cuts tax rates to increase aggregate demand. Furthermore, an increase in the budget deficit would be associated with an expansionary fiscal policy.

2014 CFA Level I

"Monetary and Fiscal Policy,"

9. A

Advertising and product differentiation are most likely to have a positive impact on the economic profits of producers under monopolistic competition. The monopoly aspect of this structure arises from the ability to differentiate its product.

CFA Level I

"The Firm and Market Structures,"

10. A

Second-degree price discrimination involves using the quantity purchased as the basis for the pricing of a particular good.

CFA Level I

"The Firm and Market Structures,"

11. A

As suggested particularly by the earliest RBC models, a person is unemployed because he or she is asking for wages that are too high

CFA Level I

“Understanding Business Cycles,”

12. C

The characteristics of monopolistic competition include a large number of competitors, low pricing power, and the production of differentiated products (through advertising and other non-price strategies), but these still result in some pricing power. The ease of entry results in zero economic profits in the long run.

CFA Level I

"The Firm and Market Structures,"

Questions

1. The following information applies to a start-up company solely owned by an entrepreneur.

	Value
Total units produced	3,550
Average revenue	\$1,110
Average variable cost	\$750
Total fixed cost	\$300,000
Total investment	\$1,550,000
Required rate of return	12.5%
Opportunity cost of owner's labor	\$125,000

The company's economic profit is *closest to*:

- A. \$659,250.
 - B. \$784,250.
 - C. \$318,750.
2. Which of the following statements concerning the Herfindahl–Hirschman Index (HHI) is *most* accurate?
- A. The HHI is a useful measure of potential barriers to entry.
 - B. The HHI is usually unaffected by mergers among the top market incumbents.
 - C. An HHI of 0.05 would be analogous to having the market shared equally by 20 firms.
3. If the quantity demanded of pears falls by 4% when the price of apples decreases by 3%, then apples and pears are *best* described as:
- A. substitutes.
 - B. inferior goods.
 - C. complements.
4. In a country with a high level of income, as domestic income rises, it is *most likely* that an increase will occur in:
- A. private saving and investment.
 - B. the fiscal balance.
 - C. the trade balance.
5. Assuming all other factors remain unchanged, which of the following changes would *most likely* cause a simultaneous increase in the participation ratio and a decrease in the unemployment rate?
- A. An increase in the number of people included in the labor force.
 - B. A decrease in the total population of working age people.
 - C. A decrease in the number of unemployed people.
6. Three firms operate under perfect competition, producing 900 units of the same product but using different production technologies. Each company's cost structure is indicated in the table:

Company	X	Y	Z
Total Variable Costs	\$2,700	\$3,600	\$4,500
Total Fixed Costs	<u>2,700</u>	<u>1,800</u>	<u>900</u>
Total Costs	\$5,400	\$5,400	\$5,400

Which of the following statements is *most* accurate? If the unit selling price is:

- A. \$4.50, all firms should continue to operate in the short run, but exit the market in the long run if these conditions are expected to persist.
 - B. \$3.00, Firm X should continue to operate in the short run, but Firms Y and Z should shut down production.
 - C. \$6.00, all firms should exit the market in the long run.
7. Which of the following statements is *most* accurate concerning the sum-of-value-added method used to determine GDP based on expenditures?
- A. The method shows a larger GDP value compared with the value-of-final-output method.

- B. The method is based on the prices consumers pay for the products and services.
- C. The method involves summing the income created at each stage of the production and distribution process.
8. A New Zealand traveler returned from Singapore with SGD7,500 (Singapore dollars). A foreign exchange dealer provided the traveler with the following quotes:

Ratio	Spot Rates
USD/SGD	1.2600
NZD/USD	0.7670
USD: US dollar NZD: New Zealand dollar	

The amount of New Zealand dollars (NZD) that the traveler would receive for his Singapore dollars is *closest* to:

- A. NZD7,248.
- B. NZD7,761.
- C. NZD4,565.
9. Which of the following statements with respect to Giffen and Veblen goods is *least* accurate?
- A. Giffen goods are "inferior," whereas Veblen goods are "high-status" goods.
- B. Both types of goods demonstrate the possibility of a positively sloping demand curve.
- C. Both types of goods violate the fundamental axioms of demand theory.
10. If a strengthening economy leads discouraged workers to return to an active employment search, at least initially, the number of unemployed people would *most likely*:
- A. increase.
- B. decrease.
- C. remain unchanged.
11. A member of the labor force quit her job last week and will begin a new job next week. During this interim period, for the purposes of calculating unemployment statistics, this person is *most likely* classified as:
- A. frictionally unemployed.
- B. hidden unemployed.
- C. voluntarily unemployed.
12. The monthly demand curve for playing tennis at a particular club is given by the following equation: $P_{Tennis\ Match} = 9 - 0.20 \times Q_{Tennis\ Match}$. The club currently charges members \$4.00 to play a match but is considering adding a membership fee. If the club continues to charge the same per play charge, the most that it will be able to charge as a membership fee is *closest* to:
- A. \$162.50.
- B. \$40.00.
- C. \$62.50.

Answers

1. Answer = A

Economic profit = Accounting profit - Total implicit opportunity costs where Accounting profit = Total revenue – Total variable costs – Total fixed costs and Total opportunity costs = opportunity cost of capital + opportunity cost of labor			
Total revenue	$3,550 \times \$1,110$	\$3,940,500	# units $\times$ average revenue
Less Total variable costs	$3,550 \times \$750$	\$2,662,500	# units $\times$ average var cost
Less Total fixed costs		<u>\$300,000</u>	given
Accounting profit		\$978,000	
Opportunity cost of capital	$\$1,550,000 \times 0.125$	\$193,750	Investment $\times$ Required return
Opportunity cost of owner's labor		\$125,000	Given
Total opportunity costs		<u>\$318,750</u>	
Economic profit		\$659,250	

CFA Level I

"The Firm and Market Structures," Richard G. Fritz, Michele Gambera Section 3.2

"Topics in Demand and Supply Analysis," Richard V. Eastin and Gary L. Arbogast Section 3.2

2. Answer = C

If there are M firms in the industry with equal market shares, the HHI equals $1/M$. With 20 firms having equal shares, the HHI = $1/20 = 0.05$.

CFA Level I

"The Firm and Market Structures," Richard G. Fritz and Michele Gambera Section 7.2

3. Answer = A

The cross elasticity of demand is defined as the percentage change in quantity demanded divided by the percentage change in the price of a substitute or complement. If the cross elasticity of demand is positive, the goods are substitutes. In this case, the 4% decline in quantity of pears is divided by the 3% decline in the price of apples, which is a positive number: $-4/-3 = +1.33$.

CFA Level I

"Topics in Demand and Supply Analysis," Richard V. Eastin and Gary L. Arbogast Section 2.4

4. Answer = A

In a country with a high level of income, as domestic income rises, private saving and investment will increase.

CFA Level I

"Aggregate Output, Prices, and Economic Growth," Paul R. Kutasovic and Richard G. Fritz Section 3.1.1

5. Answer = A

The participation ratio (or activity ratio) is the ratio of the labor force to total population of working age and the unemployment rate is the ratio of the number of unemployed to the labor force. Labor force is the numerator in the participation ratio and denominator in the unemployment rate. Therefore, assuming all else remains unchanged, an increase in the number of people included in the labor force would cause the participation ratio to increase and unemployment rate to decrease.

CFA Level I

"Understanding Business Cycles," Michele Gambera, Milton Ezrati, and Bolong Cao Section 4.1

6. Answer = B

Revenue-Cost Relationship	Short-Run Decision	Long-Term Decision
$TR \geq TC$	Stay in market	Stay in market
$TR > TVC$ but $TR < TFC + TVC$	Stay in market	Exit market
$TR < TVC$	Shut down production to zero	Exit market
where TR = Total Revenue; and TC = Total Costs; TVC = Total Variable Costs; TFC = Total Fixed Costs		
Hence, if the selling price is \$3.00, total revenue for all firms will be $\$3.00/\text{unit} \times 900 \text{ units} = \$2,700$. Only firm X's variable costs are covered and it should continue operating, while firms Y and Z should immediately shutdown production.		

CFA Level I

"Topics in Demand and Supply Analysis," Richard V. Eastin and Gary L. Arbogast Section 3.2.7

7. Answer = C

The sum-of-value-added method involves summing the value added (or income created) at each step in the production and distribution process.

CFA Level I

"Aggregate Output, Prices, and Economic Growth," Paul R. Kutasovic and Richard G. Fritz

Section 2.1

8. Answer = A

The NZD/SGD cross-rate is $\text{NZD}/\text{USD} \times \text{USD}/\text{SGD} = 0.7670 \times 1.2600 = 0.9664$.

The traveler will receive: $\text{NZD}0.9664 \text{ per SGD}; \text{NZD}0.9664 \times \text{SGD}7,500 = \text{NZD}7,248$.

CFA Level I

"Currency Exchange Rates," William A. Barker, Paul D. McNelis, and Jerry Nickelsburg Section 3.2

9. Answer = C

Veblen goods violate the fundamental axioms of demand theory, whereas Giffen goods do not.

CFA Level I

"Topics in Demand and Supply Analysis," Richard V. Eastin and Gary L. Arbogast Section 2.6

10. Answer = A

The unemployed are defined as those who are actively seeking employment but are currently without a job. Discouraged workers are without a job but have given up searching for work, so are not classified as being unemployed. A strengthening economy will lead discouraged workers to once again actively search for work; they will be reclassified as unemployed, and at least initially, the number of unemployed people will increase and the unemployment rate will rise.

CFA Level I

"Understanding Business Cycles," Michele Gambera, Milton Ezrati, and Bolong Cao Section 4.1

11. Answer = A

Frictional unemployment is short-term and transitory in nature: it includes persons who are "between jobs" as in this case and those who are not working because they are taking time to search for a job that better matches their skills, interests, and other preferences.

CFA Level I

"Understanding Business Cycles," Michele Gambera, Milton Ezrati, and Bolong Cao Section 4.1

12. Answer = C

On rearrangement, the demand function is

$$Q_{\text{Tennis Match}} = 45 - 5.0 \times P_{\text{Tennis Match}}$$

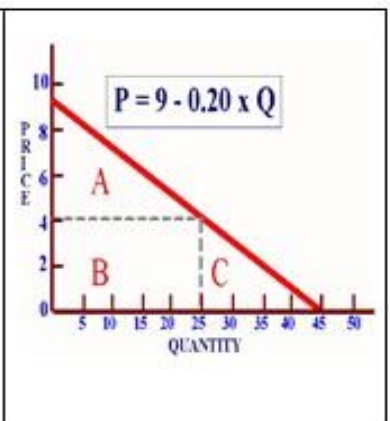
The number of matches played per month at \$4.00/match =
 $45 - 5.0 \times 4.00 = 25$

The Y-intercept of the demand curve occurs when $Q = 0$: $P = 9$

The X-intercept of the demand curve occurs when $P = 0$: $Q = 45$

In addition to the per play charge, the club will be able to charge the consumer surplus: the area under the demand curve above the current price per match to a total of 25 matches: $\frac{1}{2} \times (\$9.00 - \$4.00) \times 25 = \underline{\$62.50}$.

This is illustrated in the diagram as **triangle A**.



CFA Level I

“Topics in Demand and Supply Analysis,” Richard V. Eastin, and Gary L. Arbogast Section 2.1

Questions

- First-degree price discrimination is *best* described as pricing that allows producers to increase their economic profit while consumer surplus:
 - decreases.
 - is eliminated.
 - increases.

- The following data apply to a country in its domestic currency units:

Consumer spending on goods & services	875,060	Government spending on goods & services	305,600
Business gross fixed investment	286,400	Government gross fixed investment	84,120
Change in inventories	-68,500	Capital consumption allowance	8,540
Transfer payments	9,300	Statistical discrepancy	-2,850
Exports	219,800	Imports	250,980

Using the expenditures approach, the country's GDP is *closest* to:

- 1,466,490.
 - 1,448,650.
 - 1,451,500.
- For a given economy and a given period of time, GDP measures the:

I.	aggregate income earned by all households, all companies, and the government.
II.	total income earned by all of the country's citizens, firms, and the government.
III.	total market value produced of resalable and final goods and services.
- The *most* appropriate description of what is measured by GDP is given by:
- I and III.
 - I only.
 - I and II.
- In the short run, a firm operating in a perfectly competitive market will *most likely* avoid shutdown if it is able to earn sufficient revenue to cover which of the following costs?
 - Fixed
 - Variable
 - Marginal
 - Assume that two firms in a duopoly enter into a collusive agreement in an attempt to form a cartel and restrict output, raise prices, and increase profits. According to the Nash equilibrium, a low price is *most likely* charged by:
 - neither firm.
 - only one firm.
 - both firms.

6.

	Spot Rate	Expected Spot Rate in One Year
USD/EUR	1.3001	1.3456
USD/GBP	1.5805	1.5489

Based on the table, the appreciation of which of the following currencies is *most likely* to occur?

- A. The British pound against the US dollar by 2.00%
 - B. The euro against the US dollar by 3.50%
 - C. The US dollar against the euro by 3.38%
7. Given the inverse demand function $P_x = 13 - 3.7Q_x$, where P_x is the price per unit of good X and Q_x is the quantity demanded of good X , in units, the maximum value for Q_x is *closest* to:
- A. 9.3.
 - B. 3.5.
 - C. 13.0.
8. A country with which of the following characteristics is *least likely* to face long-term GDP growth challenges?
- A. A country with innovations in production processes
 - B. A country with large natural resources
 - C. A country with high labor quality

9.

	Spot Rate	One-Year Forward Rate
USD/EUR	1.2952	1.3001

Which of the following statements is *most* accurate based on the FX quotations in the table?

- A. The euro is trading at a forward premium of 49 points.
 - B. The forward rate is trading at a discount to the spot rate by 0.0049 points.
 - C. The US dollar is trading at a forward premium of 49 points.
10. In the demand function $Q_x^d = 4.3 - 0.7P_x + .03I - .05P_y$, where Q_x^d represents the quantity demanded of a good X , P_x is the price per unit of good X , I is consumers' income, and P_y is the price per unit of good Y , X , and Y are *best* described as:
- A. complements.
 - B. substitutes.
 - C. preferences.
11. Which of the following would be *most* useful as a leading indicator to signal the start of an economic recovery?
- A. The narrowing of the spread between the 10-year Treasury yield and the federal funds rate
 - B. An increase in aggregate real personal income (less transfer payments)
 - C. A decrease in average weekly initial claims for unemployment insurance

Answers

1. Answer = B

In first-degree price discrimination, the entire consumer surplus is captured by the producer; the consumer surplus falls to zero.

CFA Level I

"The Firm and Market Structures," Richard G. Fritz and Michele Gambera

Section 6.4

2. Answer = B

Using the expenditures approach:

GDP = Consumer spending on goods and services + Business gross fixed investment + Change in inventories + Government spending on goods and services + Government gross fixed investment + Exports – Imports + Statistical discrepancy

Consumer spending on goods and services	875,060
Business gross fixed investment	286,400
Change in inventories	(68,500)
Government spending on goods and services	305,600
Government gross fixed investment	84,120
Exports	219,800
Imports	(250,980)
Statistical discrepancy	<u>(2,850)</u>
= Gross domestic product (GDP)	1,448,650

CFA Level I

"Aggregate Output, Prices, and Economic Growth," Paul R. Kutasovic and Richard G. Fritz Sections 2.2, 2.3

3. Answer = B

Gross domestic product (GDP) can be defined in term of either output or income:

- it is the market value of all final goods and services produced within the economy in a given period of time (**output definition**) or, equivalently,
- it is the aggregate income earned by all households, all companies, and the government within the economy in a given period of time (**income definition**).

CFA Level I

"Aggregate Output, Prices, and Economic Growth," Paul R. Kutasovic, and Richard G. Fritz Section 2.1

4. Answer = B

Shutdown is defined as a situation in which the firm stops production but still confronts the payment of fixed costs in the short run. In the short run, a business can operate at a loss as long as it covers its variable costs even though it is not earning sufficient revenue to cover fixed costs. If variable costs cannot be covered in the short run, the firm will shut down operations and simply absorb the unavoidable fixed costs.

CFA Level I

"Topics in Demand and Supply Analysis," Richard V. Eastin and Gary L. Arbogast Section 3.2.7

5. Answer = A

The market outcomes for two firms in a duopoly is shown in the diagram to the right.

The lower left hand quadrant is the Nash solution when there is **no collusion**.

However, **with collusion**, if ArcCo shares at least enough of its profit in the bottom right quadrant to provide BatCo more than it would receive in the lower left, it will be the optimal solution for the pair: the **maximum joint profits** will arise where **both firms charge high prices for the product**.

Exhibit 15

Nash Equilibrium in Duopoly Market

ArcCo – Low Price 50 BatCo – Low Price 70	ArcCo – Low Price 80 BatCo – High Price 0
ArcCo – High Price 300 BatCo – Low Price 350	ArcCo – High Price 500 BatCo – High Price 300

CFA Level I

"The Firm and Market Structures," Richard G. Fritz and Michele Gambera Section 5.1

6. Answer = B

In the exchange rate quotation USD/EUR, the US dollar is the price currency and the euro is the base currency. The USD/EUR is expected to increase from 1.3001 to 1.3456. This increase represents a

3.5% appreciation of the euro against the dollar—that is, a percentage change of

$$\frac{1.3456}{1.3001} - 1 = +3.50\%$$

CFA Level I

"Currency Exchange Rates," William A. Barker, Paul D. McNelis, and Jerry Nickelsburg Section 3.1

7. Answer = B

Because price cannot be negative, the maximum value for Q_x is the value that makes $P_x = 0$. Solving $3.7Q_x = 13$, $Q_x = 3.5$.

CFA Level I

"Topics in Demand and Supply Analysis," Richard V. Eastin and Gary L. Arbogast Section 2.1

8. Answer = A

The most important factor affecting economic growth is technology because it allows an economy to overcome the limits imposed by diminishing marginal returns. A country with innovations in the production process is least likely to face long-term GDP growth challenges compared with a country that relies on input growth, such as labor or natural resources.

CFA Level I

"Aggregate Output, Prices, and Economic Growth," Paul R. Kutasovic and Richard G. Fritz Section 4.2

9. Answer = A

Forward premium = Forward rate – Spot rate = $1.3001 - 1.2952 = 0.0049$. To convert to points, scale four decimal places—that is, multiply by 10,000 = $10,000 \times 0.0049 = 49$ points. Because the forward rate exceeds the spot rate for the base currency (euro), the euro is trading at a forward premium of 49 points.

CFA Level I

"Currency Exchange Rates," William A. Barker, Paul D. McNelis, and Jerry Nickelsburg Section 3.3

10. Answer = A

The negative sign on P_y indicates that X and Y have a negative cross-price elasticity of demand and are thus complements.

CFA Level I

"Topics in Demand and Supply Analysis," Richard V. Eastin and Gary L. Arbogast Section 2.4

11. Answer = C

Average weekly initial claims for unemployment insurance is a leading indicator of economic activity. A decrease in these claims is an indicator of rehiring, which signals the start of an economic recovery.

CFA Level I

"Understanding Business Cycles," Michele Gambera, Milton Ezrati, and Bolong Cao Section 5.1

Chapter 4

Financial Reporting and Analysis

Mock A

1. An analyst gathers the following information about a company:

(\$ millions)	2014	2013
Sales	283.5	234.9
Year-end inventory (LIFO inventory method)	81.4	53.7
LIFO reserve	36.4	21.8
Cost of goods sold (LIFO)	203.9	167.3

Had the company used the first-in, first-out (FIFO) inventory method instead of last-in, first-out (LIFO), the company's 2014 gross profit margin would be closest to:

- 33.2%.
 - 15.2%.
 - 22.9%.
2. Which of the following is best described as a necessary characteristic for an effective financial reporting framework?
- Transparency to the underlying economics
 - Consistency in the measurement basis used across the balance sheet
 - Uniform treatment of transactions by different entities
3. A company acquires a license for \$6,500 with the right to use the license for four years. Management expects to derive benefits from the license for three years and uses the straight-line amortization method. Accumulated amortization at the end of Year 2 is closest to:
- \$2,167.
 - \$4,333.
 - \$3,250.
4. The following data is available on two companies that operate in the same industry:

Metric (\$ millions)	Company X	Company Y
Sales	11.2	14.5
Cost of goods sold	5.7	7.7
Administration costs	1.9	2.2
Interest expense	0.3	0.7
Research & development expenses	1.5	1.7

Which of the following statements is most appropriate? Better margin performance will be reported by:

- Y at both the gross margin and operating margin levels.
 - Y at the gross margin level and X at the operating margin level.
 - X at the gross margin level and Y at the operating margin level.
5. Which of the following best describes a component of the income statement?
- Amounts that a company owes its vendors for purchases of goods and services.
 - Obligations from past events that are expected to result in an outflow of economic benefits.
 - Outflows or depletions of assets in the course of a business's activities.
6. During its first two years of operations, a retailer sold its inventory with a constant gross margin but found that the cost of the inventory was quite variable. The following table shows the company's purchases and sales during the past two years.

Year	Inventory Purchased		Inventory Sold	
	Units	Unit Cost	Units	Selling Price
1	80,000	\$10	75,000	\$15.50
2	65,000	\$8	55,000	\$12.40

In Year 2, the company's inventory turnover ratio was 3.67, using end-of-period inventory and cost of goods sold (COGS). The method that the retailer is using to value its inventory is most likely:

- Last-in, first-out (LIFO).
 - First-in, first-out (FIFO).
 - Weighted average cost.
7. A company that prepares its financial statements in accordance with International Financial Reporting Standards (IFRS) uses the revaluation model to value land. At the end of the current year, the value of land, newly acquired this year, has increased and will be adjusted on the balance sheet. This land is the only asset in its asset class for revaluation purposes. Which of the following statements is most accurate? In the current period, the revaluation of the land will:
- Decrease the debt-to-equity ratio.
 - Increase return on assets.
 - Increase return on sales.
8. Which of the following reports is least likely to be filed with the US SEC?
- Annual report
 - Proxy statement
 - Form 10-K
9. At the time of signing a contract to build a bridge, a construction company has the following information available on the contract. Engineers estimate that the work will take three years and that 70% of the work will be completed by the end of Year 2.

	(£)
Total contract amount	500,000
Estimated cost to complete contract	300,000
Estimated expenses for Year 1	155,000
Estimated expenses for Year 2	55,000

Actual costs incurred were £150,000 and £51,000 in Years 1 and 2, respectively. The company recognizes revenue using the percentage-of-completion method based on expenses incurred. The revenue the company will recognize in Year 2 is closest to:

- £166,667.
 - £85,000.
 - £91,667.
10. According to the International Financial Reporting Standards (IFRS), which of the following conditions should be satisfied to report revenue from the sale of goods on the income statement?
- Goods have been delivered to the customer.
 - Payment has been received.
 - Costs can be reliably measured.
11. The common shareholders' equity reported on a company's balance sheet is seldom an appropriate measure of the market or intrinsic value of the company's common shares. The most likely reason for this fact is that the balance sheet:
- Fails to include all aspects of a company's ability to generate future cash flow.
 - Evaluates a company's financial position spanning a period of time.
 - Recognizes items only when future economic benefits are reasonably certain.
12. Providing information about the performance of a company, its financial position, and changes in financial position that is useful to a wide range of users is most accurately described as the role of:
- The audit report.

- b. Financial statement analysis.
 - c. Financial reporting.
13. Holding all else constant, a company that develops intangible assets internally rather than purchasing them is most likely to report:
- a. Higher investing cash outflows.
 - b. Lower amounts of assets.
 - c. Lower operating cash outflows.
14. The notes to the financial statements of a company reporting under US GAAP contains the following information for the year 2014:

Note 11. Property and Equipment (all figures in \$ thousands)
Depreciation expense for 2014 is \$362. This amount includes capitalized interest of \$143. interest is allocated and capitalized to construction in progress by applying the firm's cost of borrowing rate to qualifying assets. Interest capitalized in 2014 is \$170.

Ignoring the effects of income taxes, the expensing of previously capitalized interest most likely causes the company's cash flow from operations to be:

- a) Lower.
 - b) Higher.
 - c) Unaffected
15. According to the International Accounting Standards Board's (IASB) Conceptual Framework for Financial Reporting, the two fundamental qualitative characteristics that make financial information useful are best described as:
- a. Timeliness and accrual accounting.
 - b. Relevance and faithful representation.
 - c. Understand ability and verifiability.
16. If all of the assets and liabilities are listed on the balance sheet broadly in order of how easily they can be converted into cash, the presentation format is best described as:
- a. Classified.
 - b. Current/non c current.
 - c. Liquidity based.
17. For which of the following assets is it most appropriate to test for impairment at least annually?
- a. A trademark with an indefinite expected life
 - b. A patent with a legal life of 20 years
 - c. Land
18. An analyst gathers the following information about a company:

LIFO reserve as of 31 December 2013	\$420,000
LIFO reserve as of 31 December 2014	\$450,000
Marginal tax rate	30%

If the company had used the first-in, first-out (FIFO) method instead of last-in, first-out (LIFO), its 2014 net income would most likely have been:

- a) \$9,000 higher.
 - b) \$30,000 lower.
 - c) \$21,000 higher.
19. What is the most likely effect on the accounting equation when a company purchases office equipment with cash?
- a. Assets decrease and owners' equity decreases
 - b. Assets increase and liabilities increase
 - c. No effect on the accounting equation
20. The following excerpt was taken from the notes of a company's financial statements that were prepared in accordance with International Financial Reporting Standards. All figures are in thousands of Australian dollars.

Note 12: Broadcast Licenses

During 2014, the company successfully disposed of broadcast licenses that were held for sale for A\$37,900 (net book value of A\$23,500). Based on the successful completion of that sale, the impairment losses taken in 2012 on other licenses have been reversed, restoring those intangible assets to their amortized historical cost. Broadcast licenses are amortized over a period of 15–25 years.

The note leads an analyst to believe that the rapid reversal of the impairment loss related to the broadcast licenses arose as an attempt by management to manage earnings.

If the analyst's belief is correct, her analysis of the original 2013 financial statements would most likely have shown that, compared with the economic reality in 2013, the company had:

- a) Understated fixed asset turnover.
- b) Understated ROA.
- c) Overstated net profit margin.

21. For which of the following inventory valuation methods is the gross profit margin least likely to be the same under both a perpetual inventory system and a periodic inventory system?

- a. LIFO
- b. Specific identification
- c. FIFO

22. The following information is available for a manufacturing company:

	\$ Millions
Cost of ending inventory computed using FIFO	4.3
Net realizable value	4.1
Current replacement cost	3.8

If the company is using International Financial Reporting Standards (IFRS) instead of US GAAP, its cost of goods sold (in millions) is most likely:

- a) \$0.3 lower.
- b) \$0.3 higher.
- c) The same.

23. A regional jet manufacturer delivers 20 regional jets to an airline under long-term leases. The lease terms are for 15 years with annual payments of \$5 million per plane; the first payment is due on delivery. The company classifies the leases as finance leases and prepares its financial statements according to US GAAP. The company usually sells these jets for \$45 million each, with production cost averaging \$40 million per jet. In the year in which the leases are signed, if an interest rate of 7% is used to determine the present value of the lease payments on the deal, the gross profit on this transaction will be closest to:

- a. \$100 million.
- b. \$111 million.
- c. \$175 million

24. Selected information from a company that uses the FIFO inventory method is provided:

Events	Units	\$/Unit	Total (\$)
Opening Inventory	1,000	7.50	7,500
First Purchase	250	7.60	1,900
Sales	550	12.00	6,600
Second Purchase	300	7.70	2,310
Sales	600	12.00	7,200
Ending Inventory	400		

If the company used a perpetual system versus a periodic inventory system, the gross margin would most likely be:

- a) Lower.
- b) The same.
- c) Higher.

Answers

1. A.

The FIFO cost of goods sold (COGS) is determined from the LIFO COGS less the change in LIFO reserve

Change in LIFO Reserve	2014 LIFO reserve – 2013 LIFO reserve	= 36.4 – 21.8 = 14.6
FIFO COGS =	LIFO COGS – Change in LIFO reserve	= 203.9 – 14.6 = 189.3
FIFO gross profit	Sales – FIFO COGS = 283.5 – 189.3 = 94.2	
FIFO gross profit	margin Gross profit/Sales= 94.2/283.5 = 33.2%	

CFA Level I

"Inventories,"

2. A.

An effective framework should enhance the transparency of the underlying economics through the financial statements; transparency arises through full disclosure and fair presentation.

CFA Level I

"Financial Reporting Standards,"

3. B.

Accumulated amortization for the intangible asset at the end of Year 2 is closest to \$4,333. At the end of the second year, amortization taken = 2 years × (6,500/3) = \$4,333.

CFA Level I

"Long-Lived Assets,"

4. C.

Common size statements offer a convenient way to compare companies of different magnitudes. Company X reports better (higher) gross margin performance. Company Y reports better (higher) operating margin performance.

Metric (common size)	Company X	Company Y	Comparison
Sales	100%	100%	
Cost of goods sold	51	53	
Gross margin (GM)	49	47	X's GM is higher
Administrative costs	17	15	
Research & development expenses	13	12	
Operating margin (OM)	19	20	Y's OM is higher

CFA Level I

"Understanding Income Statements,"

5. C.

Expenses are a component of the income statement and are defined as outflows, asset depletions, and liabilities incurred in the course of a business's activities.

CFA Level I

"Understanding Income Statements,"

6. C.

At the end of Year 1, 5,000 units will remain ($0 + 80,000 - 75,000$) at a unit cost of \$10. All units in the following tables are in thousands.

Under Weighted Average Cost		
Year	1	
Average cost	\$10	$5 \times \$10 + 65 \times \8 $= \$8.14/\text{unit } 70 \text{ units}$
Inventory	$5 \times \$10 = \50	$15 \text{ units} \times \$8.14/\text{unit} = \$122$
COGS	$75 \times \$10 = \750	$55 \times \$8.14 = \448
Inventory turnover		$\text{COGS}/\text{Inv} = \$448/\$122 = 3.67$
Because the costs assigned to COGS and inventory are the same, only the units going to each need to be tracked.		

Under LIFO		
Year	1	2
Inventory	$5 \times \$10 = \50	$5 \times \$10 + 10 \times \$8 = \$130$
COGS	$75 \times \$10 = \750	$55 \times \$8 = \440
Inventory turnover		$\text{COGS}/\text{Inv} = \$440/\$130 = 3.38$
In Year 2, all of the most recent purchases at \$8 will go to COGS. Inventory will consist of the 5 units remaining from Year 1 and the 10 units remaining from Year 2 purchases.		
Under FIFO		
Year	1	2
Inventory	$5 \times \$10 = \50	$15 \times \$8 = \120
COGS	$75 \times \$10 = \750	$5 \times \$10 + 50 \times \$8 = \$450$
Inventory turnover		$\text{COGS}/\text{Inv} = \$450/\$120 = 3.75$
In Year 2, 5 units of the remaining Year 1 inventory and 50 units of Year 2 purchases will go to COGS. Inventory will consist of the remaining 15 units from Year 2 purchases.		

CFA Level I

"Inventories,"

7. A.

The increase in the value of the land bypasses the income statement and goes directly to a revaluation surplus account in equity, assuming no previous decreases in value in the asset class for revaluation purposes. Equity increases, thereby decreasing the debt-to-equity ratio.

CFA Level I

"Long-Lived Assets,"

8. A.

The annual report is not a requirement of the SEC.

CFA Level I

"Financial Reporting Standards,"

9. B.

The company will recognize revenue as follows:

Year	Costs Incurred (£)	Percentage of Costs Incurred to Date	Revenue Recognized per Year
1	150,000	$\text{£150,000/£300,000} = 50\%$	$50\% \times \text{£500,000} = \text{£250,000}$
2	51,000	$\text{£201,000/£300,000} = 67\%$ Amount recognized is 67% less amounts previously recognized (50%)	$67\% \times \text{£500,000} - \text{£250,000} = \text{£85,000}$

CFA Level I

"Understanding Income Statements,"

10. C.

The IFRS conditions that should be met to recognize revenue from the sale of goods include that the costs incurred can be reliably measured, that the economic benefits will flow to the entity, and that the significant risks and rewards of ownership have been transferred, which is normally when the goods have been delivered but not always. The actual receipt of any payment is not a condition.

CFA Level I

"Understanding Income Statements,"

11. A.

A company's value is a function of many factors, including expected future cash flows and current market conditions. Important aspects of a company's ability to generate future cash flows—for example, its reputation and management skills—are absent from the balance sheet.

CFA Level I

"Understanding Balance Sheets,"

12. C.

The role of financial reporting is to provide information about the performance of a company, its financial position, and changes in financial position that is useful to a wide range of users in making economic decisions.

CFA Level I

"Financial Statement Analysis: An Introduction,"

13. B.

Costs associated with internally developing intangible assets are usually expensed; thus, a company that has internally developed intangible assets through expenditures on R&D will recognize a lower amount of assets than a company that has obtained intangible assets through external purchase.

CFA Level I

"Long-Lived Assets,"

14. C.

The expensing of the previously capitalized interest is a non-cash amount and does not affect cash flow from operations. Under US GAAP, cash flow from operations is higher as a result of the initial capitalizing of interest but not its subsequent expensing. If the interest had not been capitalized, interest expense would have been greater and net income and cash from operations lower.

CFA Level I

"Long-Lived Assets,"

15. B.

Relevance and faithful representation are the two fundamental qualitative characteristics that make financial information useful, according to the IASB Conceptual Framework.

CFA Level I

"Financial Reporting Standards,"

16. C.

A liquidity-based presentation format presents all of the assets and liabilities on the balance sheet broadly in order of liquidity. With respect to a particular asset or liability, liquidity refers to its "nearness to cash."

CFA Level I

"Understanding Balance Sheets,"

17. A.

Intangible assets with indefinite lives need to be tested for impairment at least annually. Property, plant, and equipment (including land) and intangibles with finite lives are only tested if there has been a significant change or other indication of impairment.

CFA Level I "Understanding Balance Sheets,"

18. C.

Change in LIFO reserve (\$ thousands)	2014 LIFO reserve – 2013 LIFO reserve	= \$450 – \$420 = \$30
FIFO cost of goods sold (COGS) =	LIFO COGS – Change in LIFO reserve	LIFO COGS – \$30
If an increase in the LIFO reserve occurs, LIFO COGS will be higher than FIFO by the amount of the increase.		
With lower COGS under FIFO, pre-tax income will be higher by \$30,000.		
With a lower COGS under FIFO, after-tax income will be higher by $\\$30,000 \times (1 - 0.30) = \\$21,000$.		

CFA Level I

Inventories,

19. C.

There would be no effect on the accounting equation because the company has exchanged one asset for another. Cash has decreased and office equipment, a capital asset, has increased.

CFA Level I

Financial Reporting Mechanics,

20. C.

The broadcast licenses were written down in 2012, but the write-down was reversed in 2014. Therefore, during 2013 the intangible assets were understated, which would have understated amortisation expense for the year and increased profit. Thus, in 2013 net profit margin was overstated.

CFA Level I

Long-lived Assets,

21. A. The periodic and perpetual systems result in the same inventory and cost of goods sold values (and thus gross profit margin) using both FIFO and specific identification valuation methods but not always under LIFO.

CFA Level I

“Inventories,”

22. A.

Under IFRS, the inventory would be written down to its net realizable value (\$4.1 million), whereas under US GAAP, market value is defined as current replacement cost and thus would be written down to its current replacement cost (\$3.8 million). The smaller write-down under IFRS will reduce the amount charged to the cost of goods sold compared with US GAAP and result in a lower cost of goods sold of \$0.3 million.

CFA Level I

“Inventories,”

23. C.

The company classifies the long-term leases as financing leases. The present value of the lease payments, using the stated interest rate, is more than the company's production cost (carrying value) of the asset (\$40 million). Therefore, the leases would be classified as sales-type leases under US GAAP.

In sales-type leases, the gross margin is recognised in the year the lease is signed and is the difference between the present value of the lease payments and the cost to produce the aircraft.

On a per-plane basis (\$ millions):		
The first lease payment is due on delivery; hence the leases are annuities due.		
The present values (PV) of the lease payments on a per-plane basis is:		
	PV of an annuity due, 15 annual payments of \$5 million at 7%	48.727
	Minus: Production cost (per plane)	40.000
	Gross profit per plane	8.727
For the overall transaction (\$ millions):		
	Gross profit on transaction: 20 planes × 8.727 per plane	174.5

CFA Level I

“Long-lived Assets”

“Non-Current (Long-Term) Liabilities,”

24. B.

When using the FIFO inventory method, the ending inventory, the cost of goods sold, and the gross margin are the same under either the perpetual or periodic methods. The use of a perpetual or periodic system makes a difference under weighted average and LIFO.

CFA Level I

“Inventories,”

Questions

1. A company owns the following three assets:

- A customer list expected to provide benefits for two years. The company expects to sell products to customers from this list for

The foreseeable future.

- A license that expires in two years but can be renewed at little or no cost and relates to a service the company plans to market for the foreseeable future.

- A trademark that expires in two years but can be renewed and relates to a product the company plans to sell for the foreseeable future.

Which of the listed intangible assets is most likely to be amortized?

- Customer list
- Trademark
- License

2. Selected information from a company's comparative income statement and balance sheet is presented below:

Selected Income Statement Data for the Year Ended 31 August (\$ thousands)		
	2012	2012
Sales Revenue	100,000	95,000
Cost of goods sold	47,000	47,500
Depreciation expense	4,000	3,500
Net Income	11,122	4,556

Selected Balance Sheet Data as of 31 August (\$ thousands)		
	2013	2012
Current Assets		
Cash and investments	21,122	25,000
Accounts Receivables	25,000	13,500
Inventories	13,000	8,500
Total Current Assets	59,122	47,000
Current Liabilities		
Accounts Payable	15,000	15,000
Other Current Liabilities	7,000	9,000
Total Current Liabilities	22,000	24,000

The cash collected from customers in 2013 is closest to:

- \$88,500.
 - \$111,500.
 - \$96,100.
3. A company acquired a customer list for \$300,000 and a trademark for \$5,000,000. Management expects the customer list to be useful for three years, and it expects to use the trademark for the foreseeable future. The trademark must be renewed every 10 years with the Patent and Trademark office for a nominal amount; otherwise it expires.

If the company uses straight-line depreciation for all its intangible assets, the annual amortization expense for these two assets will be closest to:

- \$600,000.
- \$0.

- c) \$100,000.
4. Which of the following suggestions is best aligned with CFA Institute's advocacy for financial reporting that reflects economic reality?
- Detailed cost information for long-lived assets
 - Cash flow statements prepared using the direct format for cash flow from operations (CFO)
 - Conservatism in revenue recognition policies
5. Amounts recorded as deferred revenue are most likely included in income when they are:
- Paid.
 - Invoiced.
 - Earned.
6. The following information for a company was taken from its financial statements and the accompanying notes:

Income Statement (\$ thousands)		
For periods ending 31 December	2014	2013
Net Sales	11,159	8,895
Cost of goods sold (COGS)	9,898	7,901

Note 5. Inventories

Inventories are reported on a last-in, first-out (LIFO) basis. The LIFO reserve was \$867 thousand and \$547 thousand at the end of 2014 and 2013, respectively. During 2014, the company liquidated certain LIFO inventories that had been carried at lower costs in prior years, and the effect of the liquidation was to decrease COGS by \$263 thousand. No LIFO liquidation occurred in 2013.

After adjusting for the LIFO liquidation in 2014, the change in gross profit margin compared with 2013 is most likely:

- Higher by 2.5%.
 - Lower by 2.3%.
 - Essentially unchanged.
7. Financial statement disclosures relating to inventory are least likely to include which of the following?
- Information about the:
- amount of inventories pledged as security for liabilities
 - types of inventory
 - inventory valuation method used
8. An analyst is evaluating the amortization of a product patent acquired by a company. She gathers the following information:

Information about Patent and Production using the Patent	
Acquisition Cost	£ 2,650,000
Acquisition Date	1 January 2014
Patent expiration date	31 December 2018
Plant capacity	50,000 units per year
Production of product in fiscal year ended 31 December 2014	30,000 units
Expected production of product during life of patent	200,000 units

If the analyst uses the units-of-production method, the amortization expense for fiscal year 2014 is closest to:

- £397,500.
 - £1,590,000.
 - £662,500.
9. If a company has a deferred tax asset reported on its statement of financial position and the tax authorities reduce the tax rate, which of the following statements is most accurate concerning the effect of the change?
- The existing deferred tax asset will:
- Decrease in value.
 - Not be affected.
 - Increase in value.

10. An analyst is comparing the inventory turnover ratio, based on the year's average inventory, of two competitors. Company a uses first-in, first-out (FIFO) for its inventory costing and has an inventory turnover ratio of 3.50 for the current year. Company B uses last-in, first-out (LIFO). The analyst has the following information available on Company B:

(\$ thousands)	Current Year	Prior Year
Sales	2,841	2,668
Gross Profit	839	798
Inventory	534	526
LIFO reserve	48	20

The best conclusion the analyst can make is that compared with Company A, Company B's inventory turnover ratio, on a FIFO

Basis is:

- Lower.
 - The same.
 - Higher.
11. A company purchased equipment in 2013 for £25,000. The year-end values of the equipment for accounting purposes and tax purposes are as follows:

Year ending:	2014	2013
Carrying amount for accounting purposes	£20,000	£22,500
Tax base for tax purposes	£16,000	£20,000
Tax Rate	25%	30%

Which of the following statements best describes the effect of the change in the tax rate on the company's 2014 financial statements? The deferred tax liability:

- Decreases by £200.
 - Decreases by £800.
 - Increases by £250.
12. The objective of general purpose financial reporting is best described as:
- Providing information about financial performance to a wide range of users.
 - Reporting an entity's economic resources and claims, and changes therein, to shareholders.
 - Facilitating resource allocation decisions by current and potential investors and creditors.
13. In a period of rising prices and stable inventory levels, which inventory valuation method will most likely result in the highest?
- Inventory turnover ratio, all else being equal?
- Weighted average cost
 - First-in, first-out (FIFO)
 - Last-in, first-out (LIFO)
14. An e-commerce company sells hotel room nights on its website under agreement from a large number of major hotel chains. The hotel chains grant the company flexibility for the rooms they supply to the company's website and for the prices charged. These major chains bear the responsibility for providing all services once a customer books a room from the website. During the current year, the company received \$5 million in payments from the sale of hotel rooms. The cost of these rooms was \$4.5 million, which does not include \$250,000 in direct selling costs.

Under US GAAP, the e-commerce company's cost of sales is closest to:

- \$4,750,000.
 - \$4,500,000.
 - \$250,000.
15. The following items are from a company's cash flow statement.

Classification of Cash Flow	Description	Amount (£ thousands)
Operating activities	Cash received from customers	55,000
Investing activities	Interest and dividends received	10,000
Financial activities	Net repayment of revolving credit loan	12,000

Which of the following standards and formats did the company most likely use in the preparation of its financial statements?

- a) IFRS, indirect format
- b) Either IFRS or US GAAP, direct format
- c) IFRS, direct format

16. The following information is available about a company for its current fiscal year:

Accounting profit (earnings before taxes)	\$250,000
Taxable income	\$215,000
Tax rate	30%
Income taxed paid in year	\$61,200
Deferred tax liability, start of year	\$82,400
Deferred tax liability, end of year	\$90,650

The income tax expense reported on the current year's statement of earnings is closest to:

- a) \$69,450.
- b) \$72,750.
- c) \$64,500.

17. Which of the following most likely indicates effective inventory management?

- a) Current year's days of inventory on hand exceeds the prior year's days of inventory on hand
- b) Finished goods growth rate exceeds growth rate of inventories other than finished goods
- c) Sales growth rate exceeds finished goods inventory growth rate

18. For a company that prepares its financial statements under IFRS, for which of the following assets is it most likely that it could report using the fair value model?

- a) A building owned by the company and leased out to tenants
- b) A building the company owns and uses to house its administrative activities
- c) Houses built by the company for sale to customers

19. To evaluate the potential effect of an innovative and unique type of business transaction on financial statements, an analyst's best approach is to:

- a) Gain an understanding of the transaction's economic purpose.
- b) Consider the approach taken for "new" transactions that arose in the past.
- c) Monitor the actions of standard setters and regulators.

20. A company has located a building that it expects to need for its operations during the next five years. It can arrange for using the building in one of two ways:

- Lease option: Enter into a five-year operating lease.
- Purchase option: Finance the purchase of the building with a mortgage, and then sell the building after five years.

The purchase option would most likely be preferred under which of the following debt covenant constraints?

- a) Interest coverage
- b) Minimum EBITDA
- c) Debt-to-equity

21. The best description of a classified statement of financial position is one that:

- a) Is supported by note disclosures relevant to understanding its components.
- b) Has not been audited.
- c) Distinguishes between current and non-current assets and liabilities.

22. Along with relevance, the most critical qualitative characteristic of financial information is:

- a) Faithful representation.
- b) Understandability.
- c) comparability

23. The following selected balance sheet and ratio data are available for a company:

Metric	Current Year	Previous Year
Cash and cash Equivalents	98.0	
Marketable Securities	389.2	
Accounts Receivables	12.0	
Other current assets	<u>120.1</u>	
Total current assets	619.3	
Deferred Revenues	85.0	
Other current Liabilities	92.3	
Total Current Liabilities	177.3	
Cash Ratio		2.37
Quick Ratio		2.97
Current Ratio		3.27

Which of the following ratios most likely decreased this year?

- a) Current
- b) Quick
- c) Cash

24. Because of a problem with the production process, a manufacturer produced a batch of defective finished goods with a total cost of \$18,000. The sales value of this batch in its current condition is \$6,000. With \$3,000 of additional processing, however, the batch could be sold for \$11,000. The value of the unsold inventory, on the balance sheet of a company using International Financial Reporting Standards (IFRS), is closest to:

- a) \$11,000.
- b) \$9,000.
- c) \$8,000.

Answers

1. A

An acquired customer list that has an expected useful life of two years is an example of an intangible asset with a finite useful life, even if the company plans to continue selling products or services to customers on the list into the foreseeable future. Assets with finite useful lives are amortized. Although the other two assets appear to have finite lives, they can be renewed at little or no cost and, therefore, would be considered to have indefinite lives.

CFA Level I

"Long-Lived Assets,"

2. A

Cash collected from customers = Revenues - Increase in accounts receivable = \$100 - (25 - 13.5) = \$88.5 thousand.

CFA Level I

"Understanding Cash Flow Statements,"

3. C

The trademark can be renewed at a minimal cost, therefore it is considered to have an indefinite life and amortization expense is not required.

Annual amortization expense on the customer list = \$300,000 ÷ 3 years = \$100,000.

CFA Level I

"Long-Lived Assets,"

4. B

CFA Institute's 2007 position paper, "A Comprehensive Business Reporting Model: Financial Reporting for Investors," advocated for financial reporting that reflects economic reality. It specifically identified a preference for using the direct format for CFO in cash flow statements.

CFA Level I

"Financial Reporting Standards,"

5. C

Deferred revenue is a liability account that arises when money has been collected for goods or services that have not been delivered.

Revenue is recognized (included in income) as it is earned, and the deferred revenue liability will decrease accordingly.

CFA Level I

"Understanding Balance Sheets,"

6. B

$$\text{Gross profit} = \frac{\text{Sales} - \text{COGS}}{\text{Sales}} \times 100$$

Gross profit under LIFO in 2014 (\$ thousands) = 11,159 - 9,898 = 1,261

This figure arose in part from the LIFO liquidation, which decreased COGS by \$263,000 and hence

Increased gross profit.

Adjusting the gross profit downward by this amount gives adjusted gross profit (\$ thousands) of $1,261 - 263 = 998$.

Adjusted gross profit margin in 2014 = $\frac{998}{11159} \times 100 = 8.9\%$

Gross profit margin in 2013 = $\frac{8895 - 7901}{8895} \times 100 = 11.2\%$

After adjusting for the LIFO liquidation, gross profit margin is lower by $(11.2\% - 8.9\%) = 2.3\%$.

CFA Level I

"Inventories,"

7. B

Neither IFRS nor US GAAP requires disclosure of types of inventory.

CFA Level I

"Inventories,"

8. A

Using the units-of-production method, the amortization expense is closest to £397,500 and is calculated as follows: First, determine the amortization rate per unit, and then determine the cost for 2014 based on the number of units produced that period.

	Calculation	Amount
Step 1	Cost per unit = Acquisition cost/Production capacity = $\text{£}2,650,000 / 200,000 \text{ units} =$	13.25 per unit
Step 2	Number of units produced $\times$ Cost per unit = $30,000 \text{ units} \times \text{£}13.25 =$	£397,500

CFA Level I

"Inventories,"

9. A

A decrease in the tax rate will result in a decrease in the previously reported amounts of deferred tax assets. That is, the value of the future tax assets, based on the new lower rate, is reduced for offsetting future tax payments.

CFA Level I

"Income Taxes,"

10. B

Convert Company B's results to a FIFO basis as follows:

(\$ thousands)	Company B
Current year sales	-839
Less: Increase in LIFO reserve	-28

(48 - 20)	
FIFO Cost of goods sold	1,974
Opening FIFO inventory (Inventory + LIFO reserve)	526 + 20 = 546
Ending FIFO inventory (Inventory + LIFO reserve)	534 + 48 = 582
Average FIFO inventory (Opening + Ending)/2	564
Inventory Turnover = Cost of goods sold/Average inventory	1,974/564 = 3.50

Company B's FIFO-based inventory turnover ratio is the same as Company A's.

CFA Level I

"Inventories,"

11. A

Deferred tax liability = Taxable temporary difference $\times$ Tax rate.

In 2014, if the rates had not changed, the deferred tax liability would be:	0.30 $\times$ £4,000 =	£1,200.
But with the lower tax rate, the deferred tax liability will be:	0.25 $\times$ £4,000 =	<u>£1,000</u>
Effect of the change in rate thus is a decrease in the liability		-£200
Alternative calculation = Change in rate $\times$ Taxable difference	-0.05 $\times$ £4,000 =	-£200

CFA Level I

"Income Taxes,"

12. C

According to the Conceptual Framework for Financial Reporting 2010 within the International Financial Reporting Standards, as well as Concept Statement 8 under US GAAP, "the objective of general purpose financial reporting is to provide financial information about the reporting entity that is useful to existing and potential investors, lenders, and other creditors in making decisions about providing resources to the entity."

CFA Level I

"Financial Reporting Standards,"

13. C

In a period of rising prices, the most recently purchased units of inventory carry the highest cost. Under the LIFO approach, it is these high-cost units (those that are "last in") that are transferred to the income statement ("first out") as cost of goods sold. The lowest-cost units remain on the balance sheet as inventory. With a high cost of goods sold value (numerator) and a low inventory value (denominator), the inventory turnover ratio is highest under LIFO.

CFA Level I

"Inventories,"

14. C

Cost of sales is reported on the same basis as revenue. To report revenue under gross reporting, the e-commerce company must meet four criteria:

Criteria	Met/Not Met
The e-commerce company must be the primary obligor under	
Be the primary obligor under the contract.	Not met
Bear the inventory risk and credit risk.	Not met
Be able to choose its supplier.	Met
Also have reasonable latitude to establish pricing.	Met

The first criterion is not met. The major hotel chains have the obligation of fulfilling the room contract once it is entered into. The second criterion is not met either because the e-commerce company did not incur costs for vacant rooms. The major chains bear the inventory risk. Because all criteria are not met, the e-commerce company must use net reporting for which revenue is \$500,000 and cost of sales is \$250,000.

CFA Level I

"Understanding Income Statements,"

15. A

The direct method of cash flow statement presentation shows the specific cash inflows and outflows that result in reported cash flow from operating activities (e.g., cash from customers and cash to suppliers). Companies using IFRS can decide to report interest and dividend receipts as either an investing or operating activity, whereas under US GAAP, they must report such income as an operating activity. The listed operating and investment activities indicate that the company reports under IFRS using the direct method.

CFA Level I

"Understanding Cash Flow Statements,"

16. B

Income tax expense equals income tax payable (the tax rate multiplied by the taxable income) plus the increase in the deferred tax liabilities.

$$(0.30 \times \$215,000) + (\$90,650 - \$82,400) = \$64,500 + \$8,250 = \$72,750.$$

CFA Level I

"Income Taxes,"

17. C.

When the sales growth rate exceeds the finished goods inventory growth rate, the company is managing to service its increased sales level with a relatively lower level of inventory, indicating effective inventory management.

CFA Level I

"Inventories,"

18. A.

Under IFRS a building owned for the purpose of earning rentals or capital appreciation - in this case the one owned by the company and leased out to tenants - is an investment property and can be reported under either the cost model or fair value model.

CFA Level I

"Long-Lived Assets,"

19. A.

By understanding the economic purpose of a transaction and applying the conceptual framework, an analyst may be able to evaluate the potential effect on financial statements, even in the absence of specific standards.

CFA Level I

"Financial Reporting Standards,"

20. B.

The purchase option would be preferred under a minimum EBITDA constraint. The company's EBITDA calculation would be unaffected under this option because the calculation would exclude both the mortgage interest expense and the depreciation expense associated with it. Under the lease option, however, EBITDA would be reduced by the annual rent expense amount.

CFA Level I

"Long-Lived Assets,"

21. C.

Classified statements of financial position distinguish between current and non-current assets and liabilities. Classified statements are required under International Financial Reporting Standards unless a liquidity-based presentation provides more relevant and reliable information.

CFA Level I

"Financial Reporting Standards,"

22. A.

According to the conceptual frameworks adopted under both International Financial Reporting Standards and US GAAP, faithful representation and relevance are the two fundamental qualitative characteristics that make financial information useful.

CFA Level I

"Financial Reporting Standards,"

23. B.

Metric	Current year	Previous year	Conclusion
Cash ratio = (Cash + Marketable securities)/Current liabilities	$(98 + 389.2)/177.3 = 2.75$	2.37	Increase
Quick ratio = (Cash + Marketable securities + Receivables)/Current liabilities	$(98 + 389.2 + 12)/177.3 = 2.82$	2.97	Decrease

Current ratio = Current assets/Current liabilities	619.3/177.3 = 3.49	3.27	Increase
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CFA Level I

"Understanding Balance Sheets,"

24. C.

Under IFRS, inventory is carried at the lower of cost and net realizable value (NRV). The company would logically choose to sell the batch at its highest realizable value.

Cost	Given	\$18,000
NRV		
Estimated selling price	\$11,000	
Less: Costs to sell	\$0	
Less: Costs to prepare inventory for sale	<u>\$3,000</u>	\$8,000
Lower of cost or NRV		\$8,000

CFA Level I

"Inventories,"

Questions

- According to the International Financial Reporting Standards (IFRS), which of the following conditions *should* be satisfied to report revenue from the sale of goods on the income statement?
 - Goods have been delivered to the customer.
 - Payment has been received.
 - Costs can be reliably measured.

- At the start of the year a company that uses U.S. GAAP entered a contract to design and build a bridge with the following terms:

Contract length	3 years
Fixed contract price	\$40 million
Estimated contract cost	\$32 million
Costs incurred in first year	\$12 million

The company was initially quite certain about its cost estimates and intended to recognize revenue based on them. However, unexpected problems during the first year have caused engineers to suggest that a more expensive design may be required, costing up to \$8 million more. If the appropriate design cannot be determined before the company's financial statements are issued, the difference in the amount of revenue the company would recognize is *closest* to:

- \$15 million.
 - \$ 0.
 - \$ 3 million.
- Previously, a manufacturer of high-quality industrial electrical generators only sold its units to customers, but it has just introduced a leasing program. The generators have expected useful lives of about 25 years, and the company anticipates that the leases will have a term of 20 years or more. The company reports under International Financial Reporting Standards. Which of the following statements about the first year of the new leasing program is *most* accurate?
 - If the lease is classified as an operating lease, the company's profits should be higher for a given leased asset than they would be under a finance lease.
 - Regardless of how the company classifies the lease, its total cash flow
 - If the lease is classified as a finance lease, it will decrease the company's liquidity position compared with when the company was only selling its generators.
 - In early January 2015, an analyst sees a news release that a company he follows (which reports under US GAAP) will be forced to reduce output from one of its major product lines at its highly specialized ceramics plant in response to a new technology introduced by its major competitor. The table summarizes information and estimates that the analyst has gathered from various sources about the plant and its future prospects.

Selected Information Related to the Ceramics Production Plant End of 2014 (\$ thousands)	
Carrying amount of plant	1,604
Undiscounted expected future net cash flows	1,350
Present value of expected future net cash flows	1,050
Fair value of plant	1,225
Revised estimate of useful life	4 years
Depreciation method	Straight line
Revised estimate of residual value	\$200

If the above information and estimates prove accurate, the depreciation expense that should be reported for 2015 related to the plant will be *closest* to:

- \$213 thousand.
 - \$306 thousand.
 - \$256 thousand.
- A company incurs the following costs related to its inventory during the year:

Cost	¥ millions
Purchase price	100,000

Trade discounts	5,000
Import duties	20,000
Shipping of raw materials to manufacturing facility	10,000
Manufacturing conversion costs	50,000
Abnormal costs as a result of waste material	8,000
Storage cost of finished goods prior to shipping to customers	2,000

The amount charged to inventory cost (in millions) is *closest* to:

- A. ¥177,000.
 - B. ¥185,000.
 - C. ¥175,000.
6. The following is selected balance sheet data for a company along with information about its financial and operating lease obligations.

	As of 31 December 2014 (€ millions)
Long-term debt	1,347
Total shareholder's equity	11,268
Total assets	20,097

Note 18. Financial and Operating Leases

A. Financial Leases

The implicit interest rate on finance leases for 2014 was 6.0%.

B. Operating Lease Commitments as of 31 December 2014 (€ Millions)

*After 2019, all lease payments are assumed to be the same as in 2019.

Due 1 January 2015	130
Due 1 January 2016	130
Due 1 January 2017	130
Due 1 January 2018	130
Due 1 January 2019	80
Total of future lease payments thereafter*	<u>240</u>
Total commitments	840

If the company were to capitalize its long-term leases, its adjusted long-term debt-to-assets ratio as of the end of December 2014 would be *closest* to:

- A. 9.9%.
 - B. 10.2%.
 - C. 10.4%.
7. Which of the following is *least likely* to be a general feature underlying the preparation of financial statements within the International Financial Reporting Standards (IFRS) Conceptual Framework?
- A. Matching
 - B. Materiality
 - C. Accrual basis
8. At the beginning of the year, a company purchased a fixed asset for \$500,000 with no expected residual value. The company depreciates similar assets on a straight line basis over 10 years, whereas the tax authorities allow declining balance depreciation at the rate of 15% per year. In both cases, the company takes a full year's depreciation in the first year and the tax rate is 40%.
- Which of the following statements concerning this asset at the end of the year is *most* accurate?
- A. The deferred tax asset is \$10,000.
 - B. The temporary difference is \$25,000.
 - C. The tax base is \$500,000.
9. The *least likely* reason that a security analyst needs to understand the accounting process is to:
- A. aid in the assessment of management's judgment in accruals and valuations.

- B. prevent earnings manipulation by management.
- C. make adjustments to reflect items not reported in the financial statements.
10. Under International Financial Reporting Standards (IFRS), which of the following is *most likely* one of the general features underlying the preparation of financial statements?
- A. Timeliness
- B. Consistency
- C. Understandability

11. A firm that prepares its financial statements according to US GAAP and uses a periodic inventory system had the following transactions during the year:

Date	Activity	Tons (thousands)	\$ per Ton
	Beginning inventory	1	600
February	Purchase	5	650
May	Sale	2	700
August	Purchase	3	680
November	Sale	4	750

The cost of sales (in thousands) is *closest* to:

- A. \$5,890 using weighted average.
- B. \$4,080 using LIFO.
- C. \$3,850 using FIFO.
12. The International Financial Reporting Standards (IFRS) Conceptual Framework identifies fundamental qualitative characteristics that make financial information useful. Which of the following is *least likely* to be one of these characteristics?
- A. Relevance
- B. Materiality
- C. Faithful representation
13. At the start of the year, a company acquired new equipment at a cost of €50,000, estimated to have a three-year life and a residual value of €5,000. If the company depreciates the asset using the double declining balance method, the depreciation expense that the company will report for the third year is *closest* to:
- A. €3,328.
- B. €3,705.
- C. €555.
14. During the year, a retailer purchases 1,000 units of inventory at £20.20 per unit. In addition, the following items relate to inventory acquisition and handling during the year.

Item description	£ '000s
Volume rebate received	404
Import and sales taxes	2,970
Transport and transport insurance costs	325
Storage costs of finished goods	1,250
Warehouse administrative costs	3,300

The total costs (in thousands) that will be included in inventory are *closest* to:

- A. £23,091.
- B. £22,766.
- C. £24,341.
15. The convergence of global accounting standards has advanced to a degree that the Securities & Exchange Commission in the United States now mandates that foreign private issuers who use IFRS may report under:
- A. U.S. GAAP or under IFRS.

- B. U.S. GAAP with voluntary supplemental reporting under IFRS.
- C. U.S. GAAP or under IFRS with a reconciliation to U.S. GAAP.

16. An analyst has compiled the following information on a company:

	£ thousands
Beginning of the year values	
Share capital	2,000
Retained earnings	8,850
During the year	
Revenues	12,000
Total expenses	10,150
Proceeds from shares issued	500
End of year values	
Total current assets	9,200
Total non-current assets	12,750
Investments	350
Total liabilities	9,400

The amount of dividends declared (£ thousands) during the year is *closest* to:

- A. 300.
 - B. 150.
 - C. 650.
17. The following information is available on a company for the current year.

Net income	\$1,000,000
Average number of common shares outstanding	100,000
Details of convertible securities outstanding:	
Convertible preferred shares outstanding	2,000
o Dividend/share	\$10
o Each preferred share is convertible into five shares of common stock	
Convertible bonds, \$100 face value per bond	\$80,000
o 8% coupon	
o Each bond is convertible into 25 shares of common stock	
Corporate tax rate	40%

The company's diluted EPS is *closest* to:

- A. \$7.72.
 - B. \$7.57.
 - C. \$7.69.
18. Which of the following is *best* described as a necessary characteristic for an effective financial reporting framework?
- A. Consistency in the measurement basis used across the balance sheet
 - B. Transparency to the underlying economics
 - C. Uniform treatment of transactions by different entities
19. For which of the following inventory valuation methods is the gross profit margin *least likely* to be the same under both a perpetual inventory system and a periodic inventory system?
- A. Specific identification
 - B. FIFO
 - C. LIFO
20. According to the International Accounting Standards Board's (IASB) Conceptual Framework for Financial Reporting, the two fundamental qualitative characteristics that make financial information useful are *best* described as:
- A. relevance and faithful representation.

- B. understandability and verifiability.
- C. timeliness and accrual accounting.

21. In 2015, a company undertook the following two transactions:

1.	Borrowed money from an insurance company and pledged some of its production facilities as collateral for the loan.
2.	Entered into an agreement with a local construction company to build a new research facility at a fixed price. Construction is to begin by 1 January 2016 and be completed by 31 December 2018.

With respect to required disclosures in the company's financial statements, which of the following is *most* accurate? If the company reports under:

- A. International Financial Reporting Standards (IFRS), neither transaction must be disclosed.
 - B. US GAAP, only the pledged borrowing must be disclosed.
 - C. US GAAP, neither transaction must be disclosed.
22. For which of the following assets is it *most* appropriate to test for impairment at least annually?
- A. A trademark with an indefinite expected life
 - B. A patent with a legal life of 20 years
 - C. Land
23. In a period of rising prices, when compared with a company that uses weighted average cost for inventory, a company using FIFO will *most likely* report higher values for its:
- A. debt-to-equity ratio.
 - B. return on sales.
 - C. inventory turnover.
24. The following excerpt was taken from the notes of a company's financial statements that were prepared in accordance with International Financial Reporting Standards. All figures are in thousands of Australian dollars.
- Note 12: Broadcast Licenses
- During 2014, the company successfully disposed of broadcast licenses that were held for sale for A\$37,900 (net book value of A\$23,500). Based on the successful completion of that sale, the impairment losses taken in 2012 on other licenses have been reversed, restoring those intangible assets to their amortized historical cost. Broadcast licenses are amortized over a period of 15–25 years.
- The note leads an analyst to believe that the rapid reversal of the impairment loss related to the broadcast licenses arose as an attempt by management to manage earnings.
- If the analyst's belief is correct, her analysis of the original 2013 financial statements would *most likely* have shown that, compared with the economic reality in 2013, the company had:
- A. understated ROA.
 - B. understated fixed asset turnover.
 - C. overstated net profit margin.

Answers

1. Answer = C

The IFRS conditions that should be met to recognize revenue from the sale of goods include that the costs incurred can be reliably measured, that the economic benefits will flow to the entity, and that the significant risks and rewards of ownership have been transferred, which is normally when the goods have been delivered but not always. The actual receipt of any payment is not a condition.

CFA Level I

"Understanding Income Statements," Elaine Henry and Thomas R. Robinson Section 3.1

2. Answer = A

U.S. GAAP requires that long term contracts whose outcomes can be reliably measured should be accounted for using the percentage-of-completion method, based on the stage of completion.

Under the original assumptions, the company would have recognized \$15 million of revenue.

Calculations Under the Percentage-of-Completion Method	
Costs incurred to date	\$12 million
Estimated total costs	\$32 million
% total costs incurred to date	37.5%
Total contract revenue	\$40 million
% revenue to be recognized	37.5%
Current year revenue	\$15 million

Now that the company is unclear on the appropriate design and thus the cost, the outcome cannot be reliably measured. The completed contract method is used. Under this approach, no revenue (\$ 0) is recognized until the contract is substantially complete. The difference in reported revenue under the two methods is: \$15 million - \$0 = \$15 million.

CFA Level I

"Understanding Income Statements," Elaine Henry and Thomas R. Robinson Section 3.2.1

3. Answer = C

Whether the company sells or leases the asset, inventory will be reduced. For sales, the company would report an accounts receivable classified as a current asset (assuming sales terms are not in question). If the leases qualify as finance leases, then the company will report a lease receivable, which is primarily long term. Therefore, compared with selling units outright, the company's current assets are lower under leasing and its liquidity position will decrease.

CFA Level I

"Long-Lived Assets," Elaine Henry and Elizabeth A. Gordon

Section 9.2.2

"Non-Current (Long-Term) Liabilities," Elizabeth A. Gordon and Elaine Henry Section 3.2.2

4. Answer = C

At the end of 2014, a test of impairment is required because "events or changes in circumstances indicate that its carrying amount may not be recoverable." (All amounts \$ thousands)	
US GAAP Impairment Test:	
Step 1: Assess recoverability: Compare carrying amount with undiscounted future net cash flows.	
Carrying amount = 1,604 > 1,350 (expected future net cash flows):	The recoverability test is not satisfied, so an impairment loss is required.
Step 2: Determine impairment loss:	Carrying amount - Fair value = 1,604 - 1,225 = 379 New carrying value: 1,225
Estimated depreciation in 2015	$\frac{\text{New carrying value} - \text{revised residual value}}{\text{revised useful life}} = \frac{\$1,225 - \$200}{4 \text{ year}} = 256$

CFA Level I

"Long-Lived Assets," Elaine Henry and Elizabeth A. Gordon Sections 3.1 and 5.1

5. Answer = C

The costs to include in inventories are all costs of purchase, costs of conversion, and other costs incurred in bringing the inventories to their present location and condition. It does not include abnormal waste costs or storage of finished product.

Cost	¥ Millions
Purchase price	100,000
Minus trade discounts	-5,000
Import duties	20,000
Shipping of raw materials to manufacturing facility	10,000
Manufacturing conversion costs	
Total inventory costs	<u>50,000</u> 175,000

CFA Level I

"Inventories," Michael A. Broihahn Section 2

6. Answer = A

If the leases were capitalized, both total assets and liabilities would increase by the present value of the lease payments, as shown in the following table.

Present Value of Operating Lease Payments (€ Millions)	
The lease commitments after 2019 are assumed to be the same as in 2019, so there are estimated to be 240/80 = 3 additional payments.	
The present value of the operating lease payments can be calculated as the sum of the present values of two annuities-in-advance (PVA^{ADV}): a four-year annuity starting immediately (beginning of 2015) and another four-year annuity starting in four years (2019)	

Years	Cash Flow × Annuity-in-Advance Factor	Discount by	Present Value at Start of 2015
2015 to 2018	$130 \times PVA^{ADV} (4 \text{ years}, 6\%) = 477.5$	Not required	477.5
2019 and beyond	$80 \times PVA^{ADV} (4 \text{ years}, 6\%) = 293.8 \text{ at } 2019$	$\frac{1}{(1.06)^4}$	<u>232.7</u>
		Total	710.2

$PVA^{ADV} (4 \text{ years}, 6\%)$ by financial calculator: $N = 4$; $I = 6$; $PMT = 1$; Mode = BGN; Compute PV

Adjusted Long-Term Debt/Asset Ratio Calculation

Adjusted long-term debt	$1,347 + 710 =$	2,057
Adjusted total assets	$20,097 + 710 =$	20,807
Adjusted long-term debt/asset ratio	$2,057/20,807 =$	9.9%

Alternatively, the individual cash flows can be separately discounted.

Present Value of Operating Lease Payments (€ millions)

Year	Cash Flow	Cash Flow × PV Factor	PV
0	130	$130 \times PV(0y, 6.0\%)$	130.0
1	130	$130 \times PV(1y, 6.0\%)$	122.6
2	130	$130 \times PV(2y, 6.0\%)$	115.7
3	130	$130 \times PV(3y, 6.0\%)$	109.1
4	80	$80 \times PV(4y, 6.0\%)$	63.3
Beyond 4	240/80 per year = 3 years	$80 \times PVA(3y, 6.0\%) \times PV(4y, 6.0\%)$	<u>169.4</u>
		Total	710.1

$PVA (3 \text{ years}, 6\%)$ by financial calculator: $N = 3$; $I = 6$; $PMT = 1$; Mode = END; Compute PV

CFA Level I

"Long-Lived Assets," Elaine Henry and Elizabeth A. Gordon

Section 9.2.1

"Non-Current (Long-Term) Liabilities," Elizabeth A. Gordon and Elaine Henry Section 3.2.1

7. Answer = A

The IFRS Conceptual Framework specifies a number of general features underlying the preparation of financial statements, including materiality and accrual basis. Matching is not one of those general features; it is a general principle of expense recognition.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning, and Thomas R. Robinson Sections 5.5, 5.5.2

"Understanding Income Statements," Elaine Henry and Thomas R. Robinson Section 4.1

8. Answer = B

The temporary difference is the difference between the net book value (NBV) of the asset for accounting purposes and the NBV for taxes

NBV accounting	[500,000 – (500,000/10)]	\$450,000
NBV taxes	[500,000– 0.15 × (500,000)]	<u>\$425,000</u>
Temporary difference		\$25,000

"Income Taxes," ElbieLouw and Michael A. Broihahn

Sections 2.2, 4.1, 4.3

"Long-Lived Assets," Elaine Henry and Elizabeth A. Gordon Section 3.1

9. Answer = B

Understanding the accounting process may assist an analyst in identifying earnings manipulation, but it will not prevent the manipulation of earnings by management. It is important for an analyst to understand the accounting process so that they can make adjustments for items not reported and aid in the assessment of management's judgment of accruals and valuations.

CFA Level I

"Financial Reporting Mechanics," Thomas R. Robinson, Jan Hendrik van Greuning, Karen O'Connor Rubsam, Elaine Henry, and Michael A. Broihahn Section 7

10. Answer = B

Consistency is one of the general features underlying the preparation of financial statements based on IFRS.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning, and Thomas R. Robinson Section 5.5.2

11. Answer = C

Under FIFO, the oldest units are sold first, thus for the six units sold FIFO, cost of sales is \$3,850, as follows: 1 unit at \$600 + 5 units at 650 = \$3,850.

CFA Level I

"Inventories," Michael A. Broihahn Sections 3.2, 3.5, 3.6

12. Answer = B

The two fundamental qualitative characteristics that make financial information useful are relevance and faithful representation. Materiality relates to the level of detail of the information needed to achieve relevance.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning, and Thomas R. Robinson Section 5.2

13. Answer = C

Under the double declining balance method, the depreciation rate is $2 \times$ Straight line rate. The straight line rate is 33.3% (i.e., $1/3$ years), so the double declining rate is 66.6%, or two-thirds depreciation rate per year. But the asset should not be depreciated below its assumed residual value in any year.

Double Declining Balance Method of Depreciation			
Year	Net Book Value at Start of Year	Depreciation	Net Book Value at End of Year
1	€50,000	€33,333	€16,667
2	16,667	11,111	5,555
3	5,555*	555**	5,000
*	Alternative calculation for start of Year 3 net book value: $€50,000 \times (1 - 0.667) \times (1 - 0.667) = €5,555$.		
**	Depreciation cannot be $2/3 \times €5,555 = €3,705$ because that would reduce book value to less than the estimated €5,000.		

CFA Level I

"Understanding Income Statements," Elaine Henry and Thomas R. Robinson Section 4.2.3

"Long-Lived Assets," Elaine Henry and Elizabeth A. Gordon Section 3.1

14. Answer = A

Inventory costs include all direct costs of acquisition including import taxes, transportation costs and transportation insurance costs, but not storage costs of finished goods or warehouse administrative costs. Volume rebates, and similar items reduce the price paid and the costs of purchase.

Cost determination	£ '000s
Purchase price ($1,000 \times £20.20$)	20,200
Volume rebate	(404)
Import and sales taxes	2,970
Transport and transport insurance	<u>325</u>
Total costs to be inventoried	£23,091

CFA Level I

"Inventories," Michael A. Broihahn Section 2

15. Answer = A

Historically, the Securities & Exchange Commission required reconciliation for foreign private issuers that did not prepare financial statements in accordance with U.S. GAAP. However the reconciliation requirement was eliminated as of 2008 for companies that prepared their financial statements under IFRS.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning and Thomas R. Robinson Sections 4, 7

16. Answer = C

Total assets = Current assets + Non-current assets	
= 9,200 + 12,750	Total assets = £21,950 thousand
Assets = Liabilities + Equity	
21,950 = 9,400 + Equity	Equity = £12,550 thousand
Equity = Share capital + Retained earnings	
12,550 = (2,000 + 500) + Retained earnings	Retained earnings = £10,050 thousand
Retained earnings = Beginning retained earnings + Net income – Dividends	
10,050 = 8,850 + (12,000 – 10,150) – Dividends	Dividends = £650 thousand

CFA Level I

"Financial Reporting Mechanics," Thomas R. Robinson, Jan Hendrik van Greuning, Karen O'Connor Rubsam, Elaine Henry, and Michael A. Broihahn Sections 3.2, 4.2

17. Answer = A

Because both the preferred shares and the bonds are dilutive, they should both be converted to calculate the diluted EPS. Diluted EPS is the lowest possible value.

	Basic EPS	Diluted EPS: Bond Converted	Diluted EPS: Preferred Converted	Diluted EPS: Both Converted
Net income	\$1,000,000	\$1,000,000	\$1,000,000	\$1,000,000
Preferred dividends	-\$20,000	-\$20,000	0	0
After-tax cost of interest				
$8\% \times \$80,000 \times (1 - 0.40)$		\$3,840		\$3,840
Numerator	\$980,000	\$983,840	\$1,000,000	\$1,003,840
Average common shares outstanding	100,000	100,000	100,000	100,000

Preferred converted			10,000	10,000
Bond converted		20,000		20,000
Denominator	100,000	120,000	110,000	130,000
EPS	\$9.80	\$8.20	\$9.09	\$7.72

CFA Level I

"Understanding Income Statements," Elaine Henry and Thomas R. Robinson Sections 6.2, 6.3

18. Answer = B

An effective framework should enhance the transparency of the underlying economics through the financial statements; transparency arises through full disclosure and fair presentation.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning and Thomas R. Robinson Section 6.1

19. Answer = C

The periodic and perpetual systems result in the same inventory and cost of goods sold values (and thus gross profit margin) using both FIFO and specific identification valuation methods but not always under LIFO.

CFA Level I

"Inventories," Michael A. Broihahn

Section 3.6

20. Answer = A

Relevance and faithful representation are the two fundamental qualitative characteristics that make financial information useful, according to the IASB Conceptual Framework.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning, and Thomas R. Robinson Section 5.2

21. Answer = C

Under US GAAP, neither transaction must be disclosed, but disclosure of both transactions is required under IFRS.

CFA Level I

"Long-Lived Assets," Elaine Henry and Elizabeth A. Gordon Section 7

22. Answer = A

Intangible assets with indefinite lives need to be tested for impairment at least annually. Property, plant, and equipment (including land) and intangibles with finite lives are only tested if there has been a significant change or other indication of impairment.

CFA Level I

"Understanding Balance Sheets," Elaine Henry and Thomas R. Robinson Sections 4.1, 4.3

"Long-Lived Assets," Elaine Henry and Elizabeth A. Gordon Sections 5.1, 5.2, 5.3

23. Answer = B

In periods of rising prices, FIFO results in a higher inventory value and a lower cost of goods sold and thus a higher net income. The higher net income increases return on sales. The higher reported net income also increases retained earnings and thus results in a lower debt-to-equity ratio, not a higher one. The combination of higher inventory and lower cost of goods sold (CGS) decreases inventory turnover (CGS/Inventory).

CFA Level I

"Inventories," Michael A. Broihahn

Sections 3.2, 3.3, 3.5, 3.7

24. Answer = C

The broadcast licenses were written down in 2012, but the write-down was reversed in 2014. Therefore, during 2013 the intangible assets were understated, which would have understated amortization expense for the year and increased profit. Thus, in 2013 net profit margin was overstated.

CFA Level I

"Long-Lived Assets," Elaine Henry and Elizabeth A. Gordon

Sections 5.2 and 5.5

"Financial Reporting Quality," Jack Ciesielski, Elaine Henry, and Thomas I. Selling Sections 4.2.1 and 4.2.3

Questions

1. A company values its ending inventory using the prices of its most recent purchases. The inventory valuation method that the company is most likely using is:
 - A. LIFO.
 - B. FIFO.
 - C. Weighted average cost.
2. If a company capitalizes an expenditure related to capital assets instead of expensing it, ignoring taxes, the company will most likely report:
 - A. the same free cash flow to the firm (FCFF) in that period.
 - B. a lower cash flow per share in that period.
 - C. a higher earnings per share in future periods.
3. To evaluate the potential effect of an innovative and unique type of business transaction on financial statements, an analyst's best approach is to:
 - A. gain an understanding of the transaction's economic purpose.
 - B. consider the approach taken for "new" transactions that arose in the past.
 - C. monitor the actions of standard setters and regulators.
4. Operating segments are most likely reportable if they constitute 10% or more of the total for all operating segments of which financial metrics?
 - A. Capital expenditures, liabilities, or profit/loss
 - B. Amortization expense, assets, or revenue
 - C. Assets, profit/loss, or revenue
5. Which of the following is least likely to be an acceptable approach for accounting standard setting bodies to use when developing accounting standards?
 - A. Objectives-oriented
 - B. Rules-based
 - C. Revenue/expense-based
6. Net revenue most likely refers to revenue minus:
 - A. volume discounts and estimated returns.
 - B. revenues attributable to non-controlling interests.
 - C. estimates of warranty expense.
7. Analysts can best address the challenges of comparing financial statements prepared under US GAAP with those prepared under International Financial Reporting Standards (IFRS) by:
 - A. monitoring changes in both sets of standards and interpreting cautiously.
 - B. assuming differences are minor given US GAAP and IFRS convergence.
 - C. referring to the reconciliation from IFRS to US GAAP provided in the notes.
8. Compared with the management discussion and analysis (MD&A), notes to the financial statements are the most appropriate source for:
 - A. aspects of accounting policy choices most important to understanding the financial statements.
 - B. information on capital expenditures and how they support the entity's strategic direction.
 - C. a comprehensive description of all of the entity's accounting policy choices.

9. Under IFRS it is most appropriate to include which of the following pension costs of a defined benefit plan in other comprehensive income?
- Actuarial gains or losses
 - Employees service cost
 - Net interest expense accrued on the beginning net pension liability
10. The following relates to a company's common equity over the course of the year:

Outstanding shares, at start of the year	2,000,000
Stock options outstanding, at start and end of the year (Exercise price: \$5)	100,000
Shares issued on 1 April	300,000
Shares repurchased (treasury shares) on 1 July	100,000
Average market price of common shares for the year	\$20/share

If the company's net income for the year is \$5,000,000, its diluted EPS is *closest* to:

- \$2.22.
 - \$2.20.
 - \$2.17.
- 11.
- | (£ millions) | 2014 | 2013 |
|---------------------------------|-------|-------|
| Accounts receivable, gross | 6,620 | 4,840 |
| Allowance for doubtful accounts | 92 | 56 |
| Write-offs during the year | 84 | 42 |
- Based on the presented information about a company's trade receivables, the bad debt expense (in £ millions) for 2014 is *closest* to:
- 84.
 - 120.
 - 36.
12. For a company that prepares its financial statements under IFRS, for which of the following assets is it most likely that it could report using the fair value model?
- A building owned by the company and leased out to tenants
 - Houses built by the company for sale to customers
 - A building the company owns and uses to house its administrative activities
13. The following common-size income statement data and tax rates are available on a company.

Financial Item	Current Year (%)
Revenues	100
Cost of goods sold	38.6
Interest expense	3.1
Research expenses	4.4
Selling and general expenses	32.9
Income tax rate	22%
Prior Year's Profitability Ratios	
Gross profit margin	60.5%
Operating profit margin	23.3%
Net profit margin	15.8%

The profitability ratio that had the *largest* absolute increase in value in the current year is the:

- operating profit margin.
- gross profit margin.
- net profit margin.

14. Which of the following best describes a responsibility of the SEC?
- Promoting the adoption of global financial reporting standards
 - verseeing the Public Companies Accounting Oversight Board (PCAOB)
 - Prosecuting analysts who disseminate conclusions based on non-material non-public information
15. Selected information from a company's comparative income statement and balance sheet is presented below:

Selected Income Statement Data for the Year Ended 31 August (\$ thousands)		
	2013	2012
Sales revenue	100,000	95,000
Cost of goods sold	47,000	47,500
Depreciation expense	4,000	3,500
Net Income	11,122	4,556

Selected Balance Sheet Data as of 31 August		(\$ thousands)	
		2013	2012
Current Assets			
Cash and investments		21,122	25,000
Accounts receivable		25,000	13,500
Inventories		<u>13,000</u>	<u>8,500</u>
Total current assets		59,122	47,000
Current liabilities			
Accounts payable		15,000	15,000
Other current liabilities		<u>7,000</u>	<u>9,000</u>
Total current liabilities		22,000	24,000

The cash collected from customers in 2013 is *closest* to:

- \$111,500.
 - \$96,100.
 - \$88,500.
16. Along with relevance, the most critical qualitative characteristic of financial information is:
- faithful representation.
 - comparability.
 - understandability.
17. One of the notable differences between IFRS and US GAAP when dealing with income tax is best illustrated by the fundamental treatment of:
- temporary differences between the carrying amount and tax base of assets and liabilities.
 - non-deductible goodwill.
 - the revaluation of property, plant and equipment.
18. The best description of a classified statement of financial position is one that:
- distinguishes between current and non-current assets and liabilities.
 - is supported by note disclosures relevant to understanding its components.
 - has not been audited.
19. The objective of general purpose financial reporting is best described as:
- facilitating resource allocation decisions by current and potential investors and creditors.
 - reporting an entity's economic resources and claims, and changes therein, to shareholders.
 - providing information about financial performance to a wide range of users.

20. A company acquires some new depreciable assets. It uses straight-line depreciation for all of its assets. Which of the following combinations of estimated residual values and useful lives is *most likely* to produce the highest net profit margin? Estimated residual values should be:
- A. high with long average lives.
 - B. high with short average lives.
 - C. low with long average lives.

21. Income statements for two companies (A and B) and the common-size income statement for the industry are provided in the following table:

(\$ thousands)	Company A	Company B	Industry
Sales	\$10,500	\$8,250	100.0%
Cost of goods sold	6,353	5,239	62.8%
Selling, general, and administrative expenses			
	2,625	2,021	24.8%
Interest expense	840	536	7.0%
Pretax earnings	683	454	5.4%
Taxes	<u>205</u>	<u>145</u>	<u>1.7%</u>
Net earnings	\$478	\$309	3.7%

The *best* conclusion an analyst can make is that:

- A. Company A earns a higher gross margin than both Company B and the industry.
 - B. both companies' tax rates are higher than the industry average.
 - C. Company B's interest rate is lower than the industry average.
22. The role of the International Organization of Securities Commissions (IOSCO) is best described as:
- A. promoting the use of International Financial Reporting Standards and the convergence of national accounting standards.
 - B. enforcing financial reporting requirements for entities participating in capital markets.
 - C. promoting cross-border cooperation and uniformity in securities regulation.
23. The SEC's approach to addressing the significant differences in financial reporting under International Financial Reporting Standards (IFRS) and US GAAP is *best* described as:
- A. mandating that non-US issuers provide a reconciliation to US GAAP.
 - B. requiring issuers to provide disclosures describing key differences.
 - C. publicly advocating for global accounting standards and convergence.
24. The method a high-end custom-built motorcycle manufacturer uses to value its inventory results in the matching of the physical flow of the particular items sold, and the items remaining in inventory, to their actual cost. Which of the following inventory valuation methods is the manufacturer *most likely* using?
- A. FIFO
 - B. Specific identification
 - C. Weighted average cost

Answers

1. Answer = B

FIFO values ending inventory using the most recent costs of goods purchased.

CFA Level I

"Inventories," Michael A. Broihahn Sections 3.2, 3.3, 3.4

2. Answer = A

The FCFF [Cash flow from operations (CFO) + Interest $\times$ (1 – t) – Capital expenditures] would be the same. CFO and capital expenditures would both increase by the same amount (ignoring taxes). Therefore, net effect on FCFF would be zero.

Example	Capitalizing delivery cost as opposed to expensing it	
	Ignoring taxes	
FCFF	CFO + Interest $\times$ (1 – t) – Capital expenditures	
	Capital expenditures	If capitalized, the amount capitalized increases capital expenditures and is recorded as a cash outflow from investing activities
	CFO	The CFO will be higher by amount capitalized (i.e., the amount not expensed)
Because capital expenditures and CFO increase by the same amount, ignoring taxes, FCFF is unchanged.		

CFA Level I

"Understanding Cash Flow Statements," Elaine Henry, Thomas R. Robinson, Jan Hendrik van Greuning, and Michael A. Broihahn Section 4.3

"Long-Lived Assets," Elaine Henry and Elizabeth A. Gordon Section 2.1

3. Answer = A

By understanding the economic purpose of a transaction and applying the conceptual framework, an analyst may be able to evaluate the potential effect on financial statements, even in the absence of specific standards.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning, and Thomas R. Robinson Section 8.1

4. Answer = C

A company must disclose separate information about any operating segment that constitutes 10% or more of the combined operating segments' revenue, assets, or operating profit/loss.

CFA Level I

"Financial Analysis Techniques," Elaine Henry, Thomas R. Robinson, and Jan Hendrik van Greuning Section 7.1

5. Answer = C

The revenue/expense-based approach is a measurement approach, not a standard setting approach.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning, and Thomas R. Robinson Sections 2, 6.2

6. Answer = A

Net revenue means that the revenue number is reported after adjustments for cash or volume discounts or for estimated returns.

CFA Level I

"Understanding Income Statements," Elaine Henry and Thomas R. Robinson Section 2

7. Answer = A

Significant differences still exist between IFRS and US GAAP, and in most cases, analysts will lack the information necessary to make specific adjustments to address these differences. As such, comparisons must be interpreted cautiously.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning, and Thomas R. Robinson Section 7

8. Answer = C

The notes provide a comprehensive description of all of the entity's accounting policies, irrespective of whether judgment was required or whether the policies are important in understanding the financial statements.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning, and Thomas R. Robinson Section 8.3.1

9. Answer = A

Under IFRS only actuarial gains or losses can be recognized in other comprehensive income.

CFA Level I

"Non-Current (Long-Term) Liabilities," Elizabeth A. Gordon and Elaine Henry Sections 4

10. Answer = A

First, determine the incremental shares issued from stock option exercise (treasury stock method):

Shares issued at exercise price 100,000 share x \$5 = \$500,000	100,000 shares
Minus share purchased with cash received at average market price: 500,000/20	-25,000 shares
Incremental shares issued	75,000 shares

Weighted average shares outstanding		
Original shares	2,000,000	2,000,000 shares x (12 months/ 12 months)
Incremental shares issued, assuming options were exercised	75,000	75,000 shares x (12 months/ 12 months)
Shares issued 1 April	225,000	300,000 shares x (9 months/ 12 months)
Shares repurchased 1 July	-50,000	100,000 shares x (6 months/ 12 months)
Weighted average shares outstanding	<u>2,250,000</u>	

Diluted EPS	
<i>Net income – Preference dividends</i>	$\frac{\$5,000,000 - 0}{2,250,000 \text{ shares}} = 2.22/\text{share.}$
<i>Weighted average number of shares</i>	

CFA Level I

"Understanding Income Statements," Elaine Henry and Thomas R. Robinson Sections 6.2, 6.3.3

11. Answer = B

The allowance for doubtful accounts increases by the bad debt expense recognized for the year and decreases by the amounts written off during the year.

Beginning balance allowance for doubtful accounts	£56 million
Plus bad debt expense	?
Minus write-offs	<u>-£84 million</u>
Ending balance allowance for doubtful accounts	£92 million
Solve for bad debt expense = £120 million.	

CFA Level I

"Understanding Balance Sheets," Elaine Henry and Thomas R. Robinson Section 3.1.3

12. Answer = A

Under IFRS a building owned for the purpose of earning rentals or capital appreciation – in this case the one owned by the company and leased out to tenants - is an investment property and can be reported under either the cost model or fair value model.

CFA Level I

"Long-Lived Assets," Elaine Henry, and Elizabeth A. Gordon

Section 8

"Inventories," Michael A. Broihahn

Section 6

13. Answer = B

The gross profit margin increased the most in the current year:

	Current Year (%)	Prior Year (%)	Increase
Revenues	100		
Cost of goods sold	38.6		
Gross profit margin	61.4	60.5	+0.9
Research expenses	4.4		
Selling and general expenses	32.9		
Operating margin	24.1	23.3	+0.8
Interest expense	3.1		
Earnings before tax	21.0		
Minus income tax expense	22% × 21 = 4.6		
Net profit margin	16.4	15.8	+0.6

CFA Level I

"Understanding Income Statements," Elaine Henry and Thomas R. Robinson Sections 5.5, 7.2

"Financial Analysis Techniques," Elaine Henry, Thomas R. Robinson, and Jan Hendrik van Greuning Section 4.5

14. Answer = B

The SEC is responsible for overseeing the PCAOB under the Sarbanes–Oxley Act of 2002.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning, and Thomas R. Robinson, Section 3.2.2

15. Answer = C

Cash collected from customers = Revenues – Increase in accounts receivable = \$100 – (25 – 13.5) = \$88.5 thousand.

CFA Level I

"Understanding Cash Flow Statements," Elaine Henry, Thomas R. Robinson, Jan Hendrik van Greuning, and Michael A. Broihahn

Section 3.2.1.1

16. Answer = A

According to the conceptual frameworks adopted under both International Financial Reporting Standards and US GAAP, faithful representation and relevance are the two fundamental qualitative characteristics that make financial information useful.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning, and Thomas R. Robinson Section 5.2

17. Answer = C

US GAAP prohibits the revaluation of PPE. Therefore, this is a source of an important difference between US GAAP and IFRS with respect to reporting of income taxes.

CFA Level I

"Income Taxes," ElbieAntonites and Michael A. Broihahn Section 8

18. Answer = A

Classified statements of financial position distinguish between current and non-current assets and liabilities. Classified statements are required under International Financial Reporting Standards unless a liquidity-based presentation provides more relevant and reliable information.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning, and Thomas R. Robinson Section 5.5.3

"Understanding Balance Sheets," Elaine Henry and Thomas R. Robinson Section 2.2

19. Answer = A

According to the *Conceptual Framework for Financial Reporting 2010* within the International Financial Reporting Standards, as well as Concept Statement 8 under US GAAP, "the objective of general purpose financial reporting is to provide financial information about the reporting entity that is useful to existing and potential investors, lenders, and other creditors in making decisions about providing resources to the entity."

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning, and Thomas R. Robinson Section 2

20. Answer = A

A high residual value estimate reduces the depreciable base and thus depreciation expense. Long average lives reduce the annual depreciation expense for any given depreciable base. The combination of the two would result in the lowest depreciation expense, which would lead to the highest net income and profit margins.

CFA Level I

"Long-Lived Assets," Elaine Henry and Elizabeth A. Gordon Section 3.1

21. Answer = A

Common-sized analysis of the income statements shows that Company A has a lower percentage cost of goods sold and thus a higher gross margin than the industry and Company B.

	Company A	Company B	Industry	Company A	Company B
Sales	\$10,500	\$8,250	100.0%	100%	100%
Cost of goods sold	6,353	5,239	<u>62.8%</u>	<u>60.5%</u>	<u>63.5%</u>
Gross margin			37.2%	39.5%	36.5%
Company A earns a higher gross margin than both Company B and the industry.					
Pretax earnings	\$683	\$454	5.4%	6.5%	5.5%
Taxes	205	145	1.7%	2.0%	1.8%
Tax rate = Taxes/Pretax earnings			32%	30%	32%
The tax rates for the companies are not higher than the industry.					

The tax rates for the companies are not higher than the industry. The interest rate is not a function of sales and cannot be analyzed on a common-size income statement. Tax rates are determined based on Taxes/Pretax earnings, not as a percentage of sales (as shown in common-size analysis).

CFA Level I

"Understanding Income Statements," Elaine Henry and Thomas R. Robinson Section 7

"Financial Analysis Techniques," Elaine Henry, Thomas R. Robinson, and Jan Hendrik van Greuning Sections 3.1, 3.2.2

22. Answer = C

IOSCO provides a forum for regulators from different jurisdictions to work together toward fair, efficient, and transparent markets, promoting cross-border cooperation and uniformity in securities regulation.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning, and Thomas R. Robinson Section 3.2.1

23. Answer = C

The SEC now advocates for global accounting standards through public announcements, such as its

"Statement in Support of Convergence and Global Accounting Standards" (2010). In the past, the SEC had required reconciliations between IFRS and US GAAP, but these requirements were withdrawn in 2008. The SEC now imposes no requirements on its issuers.

CFA Level I

"Financial Reporting Standards," Elaine Henry, Jan Hendrik van Greuning, and Thomas R. Robinson Section 4

24. Answer = B

Specific identification is the inventory method that results in the matching of the physical flow of the particular items sold and would be most suitable for high-end custom-built motorcycles that are not ordinarily considered interchangeable.

CFA Level I - "Inventories," Michael A. Broihahn - Section 3.1

Chapter 5

Corporate Finance

Mock A

Questions

1. When computing the weighted average cost of capital (WACC) and assuming a fixed-rate non-callable bond is currently selling above par value, the before-tax cost of debt is closest to the:
 - a. Current yield.
 - b. Yield to maturity.
 - c. coupon rate
2. A company has 100 million shares outstanding. The share price of a company's stock is £15 just prior to announcing a £100 million expansionary investment in a new plant and estimates that the present value of future after-tax cash flows will be £150 million. Analysts, however, estimate the new plant's profitability will be lower than the company's expectations. The company's stock price will most likely:
 - a. Drop below £15 per share due to the cannibalization of revenue from the new plant.
 - b. Increase by less than £0.50 per share.
 - c. Increase by the new plant's net present value per share.
3. Which of the following scenarios can best be described as offering superior protection of shareholder interests?
 - a. When CEO duality is common
 - b. When common law is practiced
 - c. When stakeholder theory prevails
4. A company's \$100 par value preferred stock with a dividend rate of 9.5% per year is currently priced at \$103.26 per share. The company's earnings are expected to grow at an annual rate of 5% for the foreseeable future. The cost of the company's preferred stock is closest to:
 - a. 9.2%.
 - b. 9.7%.
 - c. 9.5%.
5. Which of the following sources of short-term financing is most likely used by smaller companies?
 - a. Commercial paper
 - b. Uncommitted lines
 - c. Collateralized loans
6. A 20-year \$1,000 fixed-rate non-callable bond with 8% annual coupons currently sells for \$1,105.94. Assuming a 30% marginal tax rate and an additional risk premium for equity relative to debt of 5%, the cost of equity using the bond-yield-plus-risk-premium approach is closest to:
 - a. 13.0%
 - b. 9.9%
 - c. 12.0%
7. Which of the following transactions is most likely to affect a company's financial leverage ratio?
 - a. Completion of a previously announced 1-for-20 reverse stock split
 - b. An increase in cash dividends paid
 - c. Payment of a 9% stock dividend
8. Which of the following is most likely considered an example of matrix pricing when determining the cost of debt?
 - a. Debt-rating approach only.
 - b. Both the yield-to-maturity and the debt-rating approaches.
 - c. Yield-to-maturity approach only

Answers

1. B.

With a fixed-rate non-callable bond, the before-tax cost of debt is the bonds yield to maturity.

CFA Level I

“Cost of Capital,”

2. B.

The value of a company is the value of its existing investments plus the net present values of all of its future investments. The NP of this new plant is £150 million - £100 million = £50 million. The price per share should increase by NP per share or £50 million / 100 million shares = £0.50 per share. As the new plants profitability is less than expectations, the NP per share (and hence the increase in the stock price) should therefore be slightly below £0.50 per share.

CFA Level I

“Capital Budgeting”,

3. B.

Unlike civil law systems, common law systems provide judges with the ability to create law by setting precedents that are followed in subsequent cases. Shareholders are viewed as better protected under common law because judges may rule against management actions in situations that are not specifically addressed by statutes.

CFA Level I

“Corporate Governance and ESG: An Introduction”,

4. A.

Question not answered $r_p = D_p/P_p$ (or Dividend/Price) = $(\$100 \times 0.095)/\$103.26 = 9.2\%$.

CFA Level I “Cost of Capital,”

5. C.

Smaller companies use collateralised loans, factoring, or loans from non-bank companies as their sources of short-term financing. Larger companies can take advantage of commercial paper, banker’s acceptances, uncommitted lines, and revolving credit agreements.

CFA Level I

“Working Capital Management”,

6. C.

First, determine the yield to maturity, which is the discount rate that sets the bond price to \$1,105.94 and is equal to 7%. This calculation can be done with a financial calculator:

$F = \$1,000$, $P = \$1,105.94$, $N = 20$, $PMT = \$80$, solve for i , which will equal 7%.

The bond-yield-plus-risk-premium approach is calculated by adding a risk premium to the cost of debt (i.e., the yield to maturity for the debt), making the cost of equity 12.00% (= 7% + 5%).

CFA Level I “Cost of Capital,”

7. B.

Cash dividends affect a company’s capital structure and financial leverage ratios by reducing assets and shareholders’ equity.

CFA Level I

“Dividends and Share Repurchases: Basics”,

8. A.

The debt-rating approach is an example of matrix pricing.

CFA Level I

“Cost of Capital,”

Questions

- The per unit contribution margin for a product is \$12. Assuming fixed costs of \$12,000, interest costs of \$3,000, and taxes of \$2,000, the operating breakeven point (in units) is closest to:
 - 1,250.
 - 1,417.
 - 1,000.
- Financial risk is least likely affected by:
 - Long-term leases.
 - Dividends.
 - Debentures.
-

Income Statements	Millions (\$)
Revenues	10.2
Variable operating costs	4.6
Fixed operating costs	2.0
Operating Income	3.6
Interest	1.2
Taxable Income	2.4
Tax	1.0
Net Income	1.4

The degree of financial leverage (DFL) is closest to:

- 2.6.
 - 1.5.
 - 1.7.
- A company decides to repurchase 5 million of its outstanding 20 million shares with debt funding. After the repurchase, the company's after-tax earnings decline by 20%. The new earnings per share (EPS) is most likely:
 - Less than the pre-repurchase EPS.
 - Greater than the pre-repurchase EPS.
 - Equal to the pre-repurchase EPS.
 - Recent trends in corporate governance most likely include:
 - Expanding the scope to consider the interests of employees, customers, and suppliers.
 - Increasing the diversity of corporate governance systems tailored to specific jurisdictions.
 - Focusing on the corporate governance system's responsibility to maximize shareholder value.
 - A company's optimal capital budget most likely occurs at the intersection of the:
 - Marginal cost of capital and investment opportunity schedule.
 - Marginal cost of capital and net present value profiles.
 - Net present value and internal rate of return profiles.
 - The market price of a company's stock is \$5 per share with 50 million shares outstanding. The company decides to use its cash reserves to undertake a \$10 million share buyback. Just prior to the buyback, the company reports total assets of \$650 million and total liabilities of \$450 million. The company's book value per share after the share buyback is closest to:
 - \$3.96.
 - \$4.17.
 - \$3.80.

8. A company has decided to switch to using accelerated depreciation from straight-line depreciation. Holding other factors constant, the degree of total leverage (DTL) will most likely:
- a) Decrease.
 - b) Not change.
 - c) Increase.

Answers

1. C.
2. B.

By taking on fixed obligations, such as debt (including debentures) and long-term leases, a company increases its financial risk. Dividends will not increase financial risk.

CFA Level I

Measures of Leverage,”

3. B.
4. B.

The pre-repurchase EPS is Net income (NI)/20 million. The EPS after the repurchase is

$[NI \times (1 - 20\%)/15 \text{ million}]$. To connect the two values algebraically:

$$(NI/20 \text{ million}) \times X = [NI \times (1 - 20\%)/15 \text{ million}]$$

$$X = (1 - 20\%) \times (20 \text{ million}/15 \text{ million}) = 1.067$$

Because X is greater than 1, the EPS has increased after the repurchase.

CFA Level I

“Dividends and Share Repurchases: Basics,”

5. A.

A significant majority of OECD member countries have ratified the influential "Principles of Corporate Governance." Most recently updated in 2015, the principles call for an expanded scope of stakeholders to be considered as part of a prudent corporate governance system. Regulators and practitioners have responded by moving toward a more effective balance of stakeholder interests.

CFA Level I

"Corporate Governance and ESG: An Introduction,"

6. A.

The point at which the marginal cost of capital intersects the investment opportunity schedule is the optimal capital budget.

CFA Level I

“Capital Budgeting,”

“Cost of Capital,”

7. A.

No. of shares purchased	\$10 million ÷ \$5 per share =	\$10 million ÷ \$5 per share =
Remaining no. of shares after share buyback	50 million - 2 million =	48 million shares
Book value of company after buyback:	Total assets less cash used minus total liabilities: \$650 million - \$10 million - \$450 million =	\$190 million
BVPS after buyback	(\$200 million - \$10 million) ÷ 48 million =	\$3.96 per share

CFA Level I

"Dividends and Share Repurchases: Basics,"

8. C.

Based on the following equation:

$$DTL = \frac{Quantity \times (Price - Variable\ cost)}{[Quantity \times (Price - Variable\ cost) - Fixed\ costs - Financing\ costs]}$$

The change to accelerated depreciation increases the fixed costs, thus making DTL increase (i.e., the numerator does not change and the denominator decreases).

CFA Level I

“Measures of Leverage,”

Questions

1. A company's asset beta is 1.2 based on a debt-to-equity ratio (D/E) of 50%. If the company's tax rate increases, the associated equity beta will *most likely*:
 - A. remain unchanged.
 - B. decrease.
 - C. increase.
2. The following information is available for a company:
 - Bonds are priced at par and have an annual coupon rate of 9.2%.
 - Preferred stock is priced at \$8.18 and pays an annual dividend of \$1.35.
 - Common equity has a beta of 1.3.
 - The risk-free rate is 4% and the market premium is 11%.
 - Capital structure: Debt = 30%; Preferred stock = 15%; Common equity = 55%.
 - The tax rate is 35%.The weighted average cost of capital (WACC) for the company is *closest* to:
 - A. 13.4%.
 - B. 14.3%.
 - C. 11.5%.
3. Using the debt-rating approach to find the cost of debt is *most* appropriate when market prices for a company's debt are:
 - A. stable.
 - B. unreliable.
 - C. below par value.
4. Given the following information about a firm:
 - debt-to-equity ratio (D/E) of 50%,
 - tax rate of 40%,
 - cost of debt of 8%, and
 - cost of equity of 13%,the firm's weighted average cost of capital (WACC) is *closest* to:
 - A. 7.5%.
 - B. 8.9%.
 - C. 10.3%.

5. A 20-year \$1,000 fixed-rate non-callable bond with 8% annual coupons currently sells for \$1,105.94. Assuming a 30% marginal tax rate and an additional risk premium for equity relative to debt of 5%, the cost of equity using the bond-yield-plus-risk-premium approach is *closest* to:
- 9.9%
 - 12.0%
 - 13.0%
6. The following data apply to two comparable companies that are in direct competition.

	Company A	Company B
Times interest earned ratio	2.50	2.50
Return on equity (ROE)	10.13%	16.88%
Return on assets (ROA)	6.75%	11.25%
Asset turnover	1.50	2.50

Which of the following statements is *most* accurate?

- Company A has a higher degree of financial leverage than Company B.
 - Company A has a lower net profit margin.
 - Both companies have the same amount of interest expense.
7. When computing the cash flows for a capital project, which of the following is *least likely* to be included?
- Financing costs
 - Opportunity costs
 - Tax effects
8. The acceptance of which of the following capital budgeting projects is *most likely* to expose a company to the highest level of uncertainty?
- Replacement of worn out equipment
 - Newly launched product or services
 - Expansion projects

Answers

1. Answer = B

$$\beta_{equity} = \beta_{asset} \times \left[1 + \left((1 - \text{Tax rate}) \times \frac{D}{E} \right) \right]$$

If the tax rate increases, then the bracketed term $(1 - \text{Tax rate})$ decreases, thus making the equity beta decrease because the asset beta is unchanged.

CFA Level I

“Cost of Capital,” Yves Courtois, Gene C. Lai, and Pamela Peterson Drake Section 4.1

2. Answer = B

The yield to maturity on a par value bond is the coupon rate of the bond

$$\begin{aligned} r_d &= 9.2\% \\ r_p &= \frac{D_p}{P_p} = \frac{\$1.35}{\$8.18} = 16.5\% \\ r_e &= R_F + \beta[E(R_M) - R_F] = 4\% + 1.3[11\%] = 18.3\% \end{aligned}$$

$$\begin{aligned} \text{WACC} &= w_d r_d (1 - t) + w_p r_p + w_e r_e \\ \text{WACC} &= 30\% \times 9.2\% \times (1 - 35\%) + 15\% \times 16.5\% + 55\% \times 18.3\% = 14.33\% \end{aligned}$$

CFA Level I

“Cost of Capital,” Yves Courtois, Gene C. Lai, and Pamela Peterson Drake Section 2, 2.1, 3.2, 3.3

“Portfolio Risk and Return: Part II,” Vijay Singal Section 3.2.6

3. Answer = B

The debt-rating approach is used when the market prices for debt are unreliable or nonexistent.

CFA Level I

“Cost of Capital,” Yves Courtois, Gene C. Lai, and Pamela Peterson Drake Section 3.1.2

4. Answer = C

$$\text{Convert D/E to the weight for debt: } \frac{1}{3} = \frac{D/E}{(1+D/E)} = \frac{0.50}{(1+0.50)}$$

$$\text{The weight for equity is one minus the weight of debt: } \frac{2}{3} = 1 - \frac{1}{3}$$

$$\begin{aligned} \text{WACC} &= \text{Weight of debt} \times \text{Cost of debt} \times (1 - \text{Tax rate}) + \text{Weight of equity} \times \text{Cost of equity} \\ \frac{1}{3} \times 0.08 \times (1 - 0.40) + \frac{2}{3} \times 0.13 &= 0.1026 = 10.3\% \end{aligned}$$

CFA Level I

“Cost of Capital,” Yves Courtois, Gene C. Lai, and Pamela Peterson Drake Sections 2, 2.1, 2.2

5. Answer = B

First, determine the yield to maturity, which is the discount rate that sets the bond price to \$1,105.94 and is equal to 7%. This calculation can be done with a financial calculator:

FV = −\$1,000, PV = \$1,105.94, N = 20, PMT = −\$80, solve for i, which will equal 7%.

The bond-yield-plus-risk-premium approach is calculated by adding a risk premium to the cost of debt (i.e., the yield to maturity for the debt), making the cost of equity 12.00% (= 7% + 5%).

CFA Level I

“Cost of Capital,” Yves Courtois, Gene C. Lai, and Pamela Peterson Drake Sections 3.1.1, 3.3.3

6. Answer = C

		Company A	Company B	Comparison
Net profit margin	$\frac{\text{ROA}}{\text{Asset turnover}}$	$6.75 \div 1.50 = 4.50\%$	$11.25 \div 2.50 = 4.50\%$	Same
Financial leverage	$\frac{\text{ROE}}{\text{ROA}}$	$10.13 \div 6.75 = 1.50$	$16.88 \div 11.25 = 1.50$	Same

In this instance, times interest earned can be found as the correct answer by process of eliminating the other choices as potential correct answers. Keep in mind, however, that even when companies have equal times interest earned ratios, it does not mean that the amount of interest expense is the same for both because the companies may not be of equal size.

CFA Level I

"Financial Analysis Techniques," by Elaine Henry, Thomas R. Robinson, and Jan Hendrik van Greuning

Sections 4.2–4.3

"Measures of Leverage," by Pamela Peterson Drake, Raj Aggarwal, Cynthia Harrington, and Adam Kobor Section 3.4

7. Answer = A

Financing costs are not included in a cash flow calculation but are considered in the calculation of the discount rate.

CFA Level I

"Capital Budgeting," John D. Stowe and Jacques R. Gagné Section 3

8. Answer = B

Investments related to new products or services expose the company to even more uncertainties than expansion projects. These decisions are more complex and will involve more people in the decision-making process.

CFA Level I

"Capital Budgeting," John D. Stowe, and Jacques R. Gagne Section 2

Questions

- Which of the following dates in the dividend chronology can fall on a weekend? The
 - payment date.
 - ex-date.
 - record date.

- The following information is available for a firm:

Sales price per unit	€85
Variable cost per unit	€65
Fixed operating costs	€50 million
Fixed financial costs	€30 million

The firm's breakeven quantity of sales (in million units) is *closest* to:

- 1.0.
 - 2.5.
 - 4.0.
- Financial risk is least likely affected by:
 - long-term leases.
 - dividends.
 - debentures.
 - A company's optimal capital budget most likely occurs at the intersection of the:
 - marginal cost of capital and investment opportunity schedule.
 - marginal cost of capital and net present value profiles.
 - net present value and internal rate of return profiles.
 - A company has decided to switch to using accelerated depreciation from straight-line depreciation. Holding other factors constant, the degree of total leverage (DTL) will *most likely*:
 - increase.
 - not change.
 - decrease.

- The following information is available for a company's bank account:

Total deposits (millions)	\$16.0
Average daily float (millions)	\$2.5
Number of days	15

The float factor for the company is *closest* to:

- 6.4.
 - 2.3.
 - 0.4.
- Which method of calculating the firm's cost of equity is most likely to incorporate the long-run return **relationship** between the firm's stock and the market portfolio?
 - Capital asset pricing model
 - Dividend discount model
 - Bond yield plus risk premium approach
 - A company decides to repurchase 5 million of its outstanding 20 million shares with debt funding. After the repurchase, the company's after-tax earnings decline by 20%. The new earnings per share (EPS) is *most likely*:
 - less than the pre-repurchase EPS.
 - greater than the pre-repurchase EPS.
 - equal to the pre-repurchase EPS.

Answers

1. Answer = A

The payment date can occur on a weekend or holiday unlike other pertinent dates, such as the exdate and record date which occur only on business days.

CFA Level I

"Dividends and Share Repurchases: Basics," George H. Troughton, and Gregory Noronha Section 3

2. Answer = C

Breakeven quantity of sales,

$$Q_{BE} = \frac{F + C}{P - V}$$

= (€50 million + €30 million) ÷ (€85 - €65) = **4.0** million units.

CFA Level I

"Measures of Leverage," Pamela Peterson Drake, Raj Aggarwal, Cynthia Harrington, and Adam Kobor

Section 3.6

3. Answer = B

By taking on fixed obligations, such as debt (including debentures) and long-term leases, a company increases its financial risk. Dividends will not increase financial risk.

CFA Level I

"Measures of Leverage," Pamela Peterson Drake, Raj Aggarwal, Cynthia Harrington, and Adam Kobor

Section 3.4

4. Answer = A

The point at which the marginal cost of capital intersects the investment opportunity schedule is the optimal capital budget.

CFA Level I

"Capital Budgeting," John D. Stowe and Jacques R. Gagné

Section 4.7

"Cost of Capital," Yves Courtois, Gene C. Lai, and Pamela Peterson Drake Section 2.3

5. Answer = A

Based on the following equation:

$$DTL = \frac{\text{Quantity} \times (\text{Price} - \text{Variable cost})}{[\text{Quantity} \times (\text{Price} - \text{Variable cost}) - \text{Fixed costs} - \text{Financing costs}]}$$

the change to accelerated depreciation increases the fixed costs, thus making DTL increase (i.e., the numerator does not change and the denominator decreases).

CFA Level I

"Measures of Leverage," Pamela Peterson Drake, Raj Aggarwal, Cynthia Harrington, and Adam Kobor

Section 3.5

6. Answer = B

Float factor = Average daily float/Average daily deposit

= \$2.5 million/(\$16 million/15) = **2.3**

CFA Level I

"Working Capital Management," Edgar A. Norton, Jr., Kenneth L. Parkinson, and Pamela Peterson Drake

Section 5.2, Example 4

7. Answer = A

The capital asset pricing model uses the firm's equity beta, which is computed from a market model regression of the company's stock returns against market returns.

CFA Level I

"Cost of Capital," Yves Courtois, Gene C. Lai, and Pamela Peterson Drake Section 3.3

8. Answer = B

The pre-repurchase EPS is Net income (NI)/20 million. The EPS after the repurchase is $[NI \times (1 - 20\%)/15 \text{ million}]$. To connect the two values algebraically:

$$(NI/20 \text{ million}) \times X = [NI \times (1 - 20\%)/15 \text{ million}]$$

$$X = (1 - 20\%) \times (20 \text{ million}/15 \text{ million}) = 1.067$$

Because X is greater than 1, the EPS has increased after the repurchase.

CFA Level I

"Dividends and Share Repurchases: Basics," George H. Troughton and Gregory Noronha Section 4.2.1

Chapter 6

Equity Investments

Mock A

Questions

- An observation that stocks with above average price-to-earnings ratios have consistently underperformed those with below average price-to-earnings ratios least likely contradicts which form of market efficiency?
 - Semi-strong form
 - Strong form
 - Weak form
- Which of the following financial intermediaries is most likely to provide liquidity service to its clients?
 - Dealers
 - Exchanges
 - Brokers
- An investor gathers the following information for an index:

Value of the index as of 31 December 2013	1,000
Interest income over the year 2013	\$23.50
Dividend income over the year 2013	\$21.50
Total return of the index over the year 2013	-4.50%

The value of the index as of 1 January 2013 is closest to:

- 1,070.
 - 1,094.
 - 1,047.
- Accounting standards and reporting requirements that produce meaningful and timely financial disclosures are most critical for achieving which of the following efficiencies associated with a well-functioning financial system?
 - Informational
 - Operational
 - Allocation
 - A closed-end fund is trading at a premium to its net asset value. This scenario most likely reflects:
 - Concerns about the quality of management.
 - Excess demand for redemption of the shares.
 - A belief that the portfolio securities are undervalued.
 - Returns from a depository receipt are least likely affected by which of the following factors?
 - Exchange rate movements
 - Number of depository receipts
 - Analysts' recommendations
 - Dark pools are best described as:
 - Operated by investment dealers that specialize in high-risk securities.
 - Certain groups of similar assets that issue securities representing shared ownership.
 - Trading venues that exercise little regulatory authority over their subscribers.
 - A long contract requires that the investor
 - sell securities in the future.
 - buy securities in the future.
 - hedge in the future
 - For portfolio managers of passive funds, market indices are least useful as:
 - Proxies to measure systematic risk.

- b. Tools to develop exchange-traded funds for non-accessible markets.
 - c. benchmarks for portfolio performance attribution
10. When parties exchange fixed cash payments for payments that depend on the returns to a stock or a stock index, they are purchasing a(n):
- a. Equity swap.
 - b. Index fund.
 - c. stock option
11. A market index contains the following two securities:

Stock	Shares in Index	Start of period Price (\$)	End of Period Price (\$)	Dividend per share (\$)
A	600	40	37	2.00
B	500	50	52	1.50

The total return on an equal-weighted basis is closest to:

- a) 2.78%.
 - b) 2.25%.
 - c) -1.75%.
12. The behavioural bias in which investors tend to avoid realizing losses but rather seek to realize gains is best described as:
- a. The gambler's fallacy.
 - b. Mental accounting.

Answers

1. C.

The observation that stocks with high above average price-to-earnings ratios has consistently underperformed those with below average price-to earnings ratios is a cross-sectional anomaly. It is a contradiction to the semi-strong form of market efficiency and strong form market efficiency because all the information used to categorize stocks by their price-to-earnings ratios is publicly available. It is not a contradiction to weak form market efficiency.

CFA Level I

"Market Efficiency,"

2. A.

The service that dealers provide is liquidity. Liquidity is the ability to buy or sell with low transaction costs when investors want to trade. By allowing their clients to trade when they want to trade, dealers provide liquidity to them.

CFA Level I

"Market Organization and Structure,"

3. B.

The total return of an index is the price appreciation, or change in the value of the price return index, plus income (dividends and/or interest) over the period, expressed as a percentage of the beginning value of the price return index. $TRI = (V_{PRI1} - V_{PRI0} + Inc_i) / V_{PRI0}$

Where,

TR_i = the total return of the index portfolio (as a decimal number)

V_{PRI1} = the value of the price return index at the end of the period

V_{PRI0} = the value of the price return index at the beginning of the period

Inc_i = the total income (dividends and/or interest) from all securities in the index held over the period

$$-4.5\% = (1000 - V_{PRI0} + 23.5 + 21.5) / V_{PRI0};$$

$$V_{PRI0} = (1000 + 23.5 + 21.5) / (1 - 0.045) = 1,094.$$

CFA Level I

"Security Market Indices,"

4. A.

Accounting standards and reporting requirements that allow meaningful and timely financial disclosures reduce the costs of obtaining fundamental information and thereby allow analysts to form more accurate estimates of fundamental values. They support informationally efficient markets.

CFA Level I

"Market Organization and Structure,"

5. C.

Closed-end funds may trade at a premium (discount) to net asset value when investors believe that the portfolio securities are undervalued (overvalued).

CFA Level I

"Market Organization and Structure,"

6. B.

The price of each depository receipt (and, in turn, returns) will be affected by factors that affect the price of the underlying shares—such as company fundamentals, market conditions, analysts' recommendations, and exchange rate movements. The number of depository receipts issued affects their price but does not affect the returns.

CFA Level I

"Overview of Equity Securities,"

7. C.

Dark pools are trading venues that function like exchanges but do not exercise regulatory authority over their subscribers except with respect to the conduct of their trading in those venues.

CFA Level I

"Market Organization and Structure,"

8. B

A long contract requires that the investor buy securities in the future.

CFA Level 1

9. C.

Market indices are used as benchmarks for actively managed portfolios which is not relevant to passively managed funds.

CFA Level I

"Security Market Indices,"

10. A.

Equity swaps consist of parties exchanging fixed cash payments for payments that depend on the returns to a stock or a stock index.

CFA Level I

"Market Organization and Structure,"

11. B.

Stock	Shares on Index		Start- of - period price (\$)	End-of- Period Price (\$)	Dividend per share (\$)	Price Return (\$)	Total Return (\$)
	(1)	(2)	(3)	(4)		$= (3)/(2) - 1$	$= [(3) + (4)]/(2) - 1$
A	600	40	37	2		-7.50%	-2.50%
B	500	50	52	1.5		4.00%	7.00%
Total return = $[(-2.5 + 7)/2]$						2.25%	

12. C.

Behavioural biases in which investors tend to avoid realizing losses but, rather, seek to realize gains is the disposition effect.

Questions

- For portfolio managers of active funds, market indices are least useful as:
 - Proxies to measure risk-adjusted performance.
 - Model portfolios.
 - Benchmarks.
- An internal evaluation of the trading behavior of three fund managers of a mutual fund company during the past year has revealed the following:

Manager X	Was slower than peers when reacting to changes in information
Manager Y	Rarely realized investment losses but realized most of the investment gains
Manager Z	Tended to overreact by disliking losses more than liking comparable gains

Which of the three managers most likely displayed the disposition effect bias?

- Manager Z
 - Manager Y
 - Manager X
- Which of the following statements regarding a commodity index is most accurate?
 - Commodity index returns differ from the changes in the prices of their underlying commodities.
 - Commodity indices in the same markets will share similar risk and return profiles.
 - Commodity indices commonly use an equal weighting method.
 - An informed trader buys a company's stock. His close friends, who lack information or expertise, imitate his action and buy the stock. Which of the following statements concerning this behavioral bias is most accurate? It:
 - Is inconsistent with rational behavior.
 - Is identical to representativeness.
 - Improves market efficiency.
 - The type of voting in board elections that is most beneficial to shareholders with a small number of shares is best described as:
 - Cumulative voting.
 - Statutory voting.
 - Voting by proxy.
 - An industry experiencing intense competitive rivalry among incumbent companies is best characterized by:
 - A small number of competitors and low fixed costs.
 - Customers basing purchase decisions largely on price.
 - Differentiated products and low exit barriers.
 - Which of the following decisions most likely occurs as part of the index management phase of constructing and managing an index?
 - When to rebalance
 - How to identify which securities should be selected from the target market
 - Which weighting method to use.
 - An industry characterized by rising volumes, improving profitability, falling prices, and relatively low competition among companies is most likely in which of the following life-cycle stages?
 - Embryonic
 - Growth
 - Mature
 - Which of the following statements is most accurate?
 - Putable common shares provide benefits to both the issuing company and investors.
 - Investors owning a small number of common shares would prefer statutory voting to cumulative voting.

- c) Convertible preference shares are more volatile and riskier than the underlying common shares.
10. Unlike commercial industry classification systems, industry classification systems developed by governments most likely:
- a) are updated more frequently.
 - b) are more transparent.
 - c) Include private companies.
11. After the public announcement of the merger of two firms, an investor makes abnormal returns by going long on the target firm and short on the acquiring firm. This most likely violates which form of market efficiency?
- a) Semi-strong and strong forms
 - b) Semi-strong form only
 - c) Weak and semi-strong forms
12. The financial systems that are operationally efficient are most likely characterized by:
- a) Liquid markets with low commissions and order price impacts.
 - b) The use of resources where they are most valuable.
 - c) Security prices that reflect fundamental values

Answers

1. B.

Market indices are used as model portfolios for index funds and exchange-traded funds, but they are not useful as model portfolios for active funds.

CFA Level I

"Security Market Indices,"

2. B.

The disposition effect relates to the behavioral bias in which investors tend to avoid realizing losses but, rather, seek to realize gains. Manager Y has displayed this bias.

CFA Level I

"Market Efficiency,"

3. A.

The performance of commodity indices can be different from their underlying commodities because the indices consist of futures contracts on the commodities rather than the actual commodities. Commodity index returns reflect the risk-free interest rate, the changes in future prices, and the roll yield.

CFA Level I

"Security Market Indices,"

4. C.

This behavioral bias is an example of an information cascade wherein the transmission of information is from those participants who act first and whose decisions influence the decisions of others. The behavior of informed traders acting first and uninformed traders imitating the informed traders is consistent with rationality. The imitation trading by the uninformed traders helps the market incorporate relevant information and improves market efficiency.

CFA Level I

"Market Efficiency,"

5. A.

Cumulative voting allows shareholders to direct their total voting rights to specific candidates, as opposed to having to allocate their voting rights evenly among all candidates. Thus, applying all of the votes to one candidate provides the opportunity for a higher level of representation on the board than would be allowed under statutory voting.

CFA Level I

"The Corporate Governance of Listed Companies",

6. B.

The factor that most influences customer purchase decisions is likely to also be the focus of competitive rivalry in the industry. In general, industries in which price is a large factor in customer purchase decisions tend to be more competitive than industries in which customers value other attributes more highly.

CFA Level I

"Introduction to Industry and Company Analysis"

7. A.

One of the decisions that must be made during the management phase of constructing and managing an index is when to rebalance the index. Rebalancing is necessary to maintain the weight of each security consistent with the index's weighting method.

CFA Level I

"Security Market Indices,"

8. B.

An industry in growth stage is characterized by rising volumes, improving profitability, falling prices, and relatively low competition among companies.

CFA Level I

"Introduction to Industry and Company Analysis,"

9. A.

The put option feature facilitates raising capital because the shares are more appealing to investors. As such, it provides a benefit to the issuing company. It also helps investors limit their potential losses because they can sell the shares back to the issuing company if the market price falls below the pre-specified put price. Therefore, puttable common shares are beneficial to both the issuing company and the investors.

CFA Level I

"Overview of Equity Securities,"

10. C.

Industry classification systems developed by governments do not distinguish between public and private companies, whereas commercial classification systems include only publicly traded organizations.

CFA Level I

"Introduction to Industry and Company Analysis,

11. A.

In a semi-strong efficient market, prices adjust quickly and accurately to new information. In this case, prices would quickly adjust to the merger announcement, and if the market is a semi-strong efficient market, investors acting after the merger announcement would not be able to earn abnormal returns. Therefore, the market is not semi-strong-form efficient. A market that is not semi-strong-form efficient is also not strong-form efficient. Thus, violating the semi-strong-form efficiency also implies violating the strong-form efficiency. However, the market could still be weak form efficient because past prices are not being used to make abnormal profits. Thus, we cannot say that the weak-form market efficiency has been violated.

CFA Level I

"Market Efficiency,"

12. A.

Operationally efficient markets are liquid markets in which the costs of arranging trades, commissions, bid-ask spreads, and order price impacts, are low.

CFA Level I

"Market Organization and Structure,"

Questions

- Information-motivated traders are *most likely* to differ from pure investors in that they:
 - pay lower transaction fees.
 - hold well-diversified portfolios.
 - expect to earn excess returns.
- The index weighting that results in portfolio weights shifting away from securities that have increased in relative value toward securities that have fallen in relative value whenever the portfolio is rebalanced is *most* accurately described as:
 - equal weighting.
 - fundamental weighting.
 - float-adjusted market-capitalization weighting.
- The behavioral bias in which investors tend to avoid realizing losses but rather seek to realize gains is *best* described as:
 - the gambler's fallacy.
 - the disposition effect.
 - mental accounting.
- A fund manager compiles the following data on two companies:

	Company A	Company B
Return on assets (ROA)	10.9%	9.0%
Return on equity (ROE)	15.4%	14.3%
Dividend payout ratio	0.35	0.30
Required rate of return	13.0%	12.4%
Weighted average cost of capital	11.8%	11.7%

The *best* conclusion the fund manager can make is that Company A's stock is more attractive than Company B's stock because of its:

- smaller price-to-earnings ratio (P/E).
 - higher dividend growth rate.
 - greater financial leverage.
- A portfolio manager analyzes a market and discovers that it is not possible to achieve consistent and superior risk-adjusted returns, net of all expenses. This market is *most likely* characterized by:
 - restrictions on short selling.
 - persistent anomalies.
 - informational efficiency.
 - An investor writes a put option on FTSE 100 Index futures. Which of the following *best* describes the investor's position with respect to the put contract and her exposure to the underlying index future, respectively?
 - Short, long
 - Long, short
 - Short, short
 - If the number of financial analysts who follow or analyze a company increases substantially, then the market for this company's shares will *most likely* become:
 - overvalued.

- B. more efficient.
- C. more attractive for active investors.
8. Which of the following financial intermediaries is *most likely* to provide liquidity service to its clients?
- A. Exchanges
- B. Brokers
- C. Dealers
9. Which of the following statements is *least* accurate with respect to fixed-income indices?
- A. The indices are susceptible to turnover of the underlying securities.
- B. Compared with equity indices, it is easier and less expensive to replicate fixed-income indices.
- C. Many of the underlying securities in the index tend to be illiquid.
10. Which of the following statements concerning a security market index is *most* accurate?
- A. The divisor will be adjusted to prevent changes not related to prices of constituent securities.
- B. At inception, the total return version of an index will be greater than the price version of an index.
- C. Estimated market prices of constituent securities are not used to calculate the index value.
11. If the following three stocks are held in a portfolio, the portfolio's total return on an equal weighted basis is *closest* to:

Stock	Number of Shares Owned	Beginning of Period Price per Share (\$)	End of Period Price per Share (\$)	Dividend per Share during the Period (\$)
A	500	40	37	2.00
B	320	50	52	1.50
C	800	30	34	0.00

- A. 6.37%.
- B. 5.94%.
- C. 3.28%.
11. An analyst gathers the following information about a company's equity security:
- Trailing price-to-earnings multiple: 10x
 - Last year's EPS: \$5.00
 - Forecasted EPS growth rate: 10%
12. If the analyst estimates that the security is undervalued by \$4, the estimated intrinsic value is *closest* to:
- A. \$54.
- B. \$59.
- C. \$46.

Answers

1. Answer = C

Information-motivated traders expect to earn returns in excess of market returns because they trade on securities they believe the market has over- or undervalued. Unlike pure investors, they expect to earn a return on their information in addition to the normal return expected for bearing risk. Excess returns are generated when the market recognizes and corrects the valuation error on such a security.

CFA Level I

"Market Organization and Structure," Larry Harris Section 2.1.6

2. Answer = B

Fundamentally weighted indices generally will have a contrarian "effect" in that the portfolio weights will shift away from securities that have increased in relative value and toward securities that have fallen in relative value whenever the portfolio is rebalanced.

CFA Level I

"Security Market Indices," Paul D. Kaplan and Dorothy C. Kelly Section 3.2.4

3. Answer = B

Behavioral biases in which investors tend to avoid realizing losses but, rather, seek to realize gains is the disposition effect.

CFA Level I

"Market Efficiency," by W. Sean Cleary, Howard J. Atkinson, and Pamela Peterson Drake Section 5.3

4. Answer = A

From the following computations, Company A's stock is more attractive than Company B's stock because of its smaller P/E.

	Company A	Company B
Dividend growth rate (g)		
$g = \text{ROE} \times (1 - \text{Dividend payout ratio})$	$15.4 \times (1 - 0.35) = 10.0\%$	$14.3 \times (1 - 0.30) = 10.0\%$
$\text{P/E} = \frac{\text{Dividend payout ratio}}{r - g}$		
$r - g$	$0.35 / (0.13 - 0.10) = \mathbf{11.7x}$	$0.30 / (0.124 - 0.10) = \mathbf{12.5x}$
Financial leverage (ROE/ROA)	$15.4 / 10.9 = 1.4x$	$14.3 / 9.0 = 1.6x$

CFA Level I

"Financial Analysis Techniques," Elaine Henry, Thomas R. Robinson, and Jan Hendrik van Greuning Section 4.6.2

"Equity Valuation: Concepts and Basic Tools," John J. Nagorniak and Stephen E. Wilcox Section 5.1

5. Answer = C

In an informationally efficient market, consistent and superior risk-adjusted returns (net of all expenses) are not achievable.

CFA Level I

"Market Efficiency," W. Sean Cleary, Howard J. Atkinson, and Pamela Peterson Drake Section 2.1

6. Answer = A

The investor has written a put contract, which means she is short the option. She, therefore, must satisfy the obligation to purchase the asset if requested to do so by the put owner. The investor has a long exposure to the risk of the underlying index future because she benefits when its quoted price increases—that is, when the put declines in value (or suffers a loss when its quoted price decreases as the put increases in value).

CFA Level I

"Market Organization and Structure," Larry Harris Section 5, Exhibit 1

7. Answer = B

The number of financial analysts who follow or analyze a security or asset should be positively related to market efficiency. Therefore, if more analysts cover a company, the market for this company's shares will most likely become more efficient.

CFA Level I

"Market Efficiency," W. Sean Cleary, Howard J. Atkinson, and Pamela Peterson Drake Section 2.3

8. Answer = C

The service that dealers provide is liquidity. Liquidity is the ability to buy or sell with low transaction costs when investors want to trade. By allowing their clients to trade when they want to trade, dealers provide liquidity to them.

CFA Level I

"Market Organization and Structure," Larry Harris Sections 4.1, 4.2

9. Answer = B

Compared with equity indices, the large number of fixed-income securities—combined with the lack of liquidity of some securities—has made it more costly and difficult for investors to replicate fixed-income indices and duplicate their performance.

CFA Level I

"Security Market Indices," Paul D. Kaplan and Dorothy C. Kelly Section 6.1

10. Answer = A

An index provider will adjust the value of the divisor as necessary to avoid changes in the index value that are unrelated to changes in the prices of constituent securities.

CFA Level I

"Security Market Indices," Paul D. Kaplan and Dorothy C. Kelly Section 2

11. Answer = B

Equal weighting assigns an equal weight to each constituent security at inception. Therefore, it is the sum of the total return from each security divided by the number of securities in the portfolios.

Stock	$(P_1 - P_0 + D)/P_0$	Total Return (%)
A	$(37 - 40 + 2.00)/40 =$	-2.5
B	$(52 - 50 + 1.50)/50 =$	7.00
C	$(34 - 30 + 0)/30 =$	13.33
Portfolio return with equal weighting:		
$(-2.50 + 7.00 + 13.33)/3 =$		5.94

CFA Level I

"Security Market Indices," Paul D. Kaplan and Dorothy C. Kelly Section 3.2.2

12. Answer = A

The current market value, or price, of a security is calculated as follows:

$$P = \frac{P}{E} \times E,$$

Where

P Is the current market value, or price, of the security

$\frac{P}{E}$ Is the trailing price-to-earnings multiple (10x)

E Is last year's EPS (\$5)

The current market value of the security is $P = \frac{P}{E} \times E = 10 \times \$5 = \$50$.

The security is undervalued by \$4. Therefore, the estimated intrinsic value is $\$50 + \$4 = \$54$.

CFA Level I

"Market Efficiency," W. Sean Cleary, Howard J. Atkinson, and Pamela Peterson Drake Section 2.2

"Equity Valuation: Concepts and Basic Tools," John J. Nagorniak and Stephen E. Wilcox Sections 3 and 5

Questions

1. The following data pertain to a company that can be appropriately valued using the Gordon growth model. The dividend is expected to grow indefinitely at the existing sustainable growth rate.

EPS growth rate (three-year average)	7.50%
Current dividend per share	\$3.00
Return on equity	15%
Dividend payout ratio	45%
Investors' required rate of return	16%

The stock's intrinsic value is *closest* to:

- A. \$41.90.
 B. \$37.94.
 C. \$34.62.
2. Which of the following statements about the forms of market efficiency is *least* accurate? If the form of market efficiency is:
- A. semi-strong, then security prices fully reflect all past market data.
 B. weak, then investment strategies based on fundamental analysis could achieve abnormal returns.
 C. strong, then prices reflect only private information.
3. The following table shows information on three different investment strategies with equivalent systematic risk:

Strategy	Type of Strategy	Annualized Data	
		Fees and Expenses	Net Return
1	Passive	0%	15%
2	Exploits price patterns	1%	14%
3	Uses fundamental analysis	2%	

The return, gross of fees and expenses that causes Strategy 3 to be *most* consistent with the strong form of market efficiency is:

- A. 16%.
 B. 18%.
 C. 17%.
4. The voting rights of an unsponsored depository receipt (DR) belong to the:
- A. direct owners of the foreign common shares.
 B. foreign company whose shares are held by the depository.
 C. depository bank.
5. A trader is able to obtain persistent abnormal returns by adopting an investment strategy that purchases stocks that have recently experienced high returns. This strategy exploits a marketpricing anomaly *best* described as:
- A. the overreaction effect.
 B. data mining.
 C. momentum.
6. Security market indices can be used to calculate alphas, which are *best* described as:
- A. the systematic risk of a security, using the index as a proxy for the entire market.
 B. the difference between the return of the actively managed portfolio and the return of the passive portfolio.
 C. a measure of market sentiment.

7. If a test rejects the hypothesis that market prices reflect private information but does not reject the hypothesis that they reflect past market data and public information, then the form of market efficiency is *best* described as:
- strong.
 - weak.
 - semi-strong.
8. A trader buys 500 shares of a stock on margin at \$36 a share using an initial leverage ratio of 1.66. The maintenance margin requirement for the position is 30%. The stock price at which the margin call will occur is *closest* to:
- \$20.57.
 - \$25.20.
 - \$30.86.
9. Compared with unregulated markets, regulated markets are *best* characterized by:
- reduced arbitrage opportunities.
 - higher transaction costs.
 - lower trading volumes.
10. An investor gathers the following data to estimate the intrinsic value of a company's stock using the justified forward price-to-earnings ratio (P/E) approach.

Next year's earnings per share	\$3.00
Return on equity	12.5%
Dividend payout ratio	60%
Required return on shares	10%

The intrinsic value per share is *closest* to:

- \$72.
 - \$48.
 - \$36.
11. An analyst will *most likely* put a "sell" recommendation on a stock when its:
- market value is lower than fundamental value.
 - market value is higher than intrinsic value.
 - intrinsic value is positive.
12. Compared with public equity markets, which of the following statements is *most* accurate about private equity markets? Operating in the private market:
- offers stronger incentives to improve corporate governance.
 - allows management to better adopt a long-term focus.
 - allows more opportunities to raise capital.

Answers

1. Answer = A

$V_0 = D_0 (1 + g) / (r - g)$, where

Sustainable growth rate = $g = b \times \text{ROE}$; $b = (1 - \text{Payout ratio})$ $g = (1 - 0.45) \times 15\% = 8.25\%$;

$V_0 = (\$3 \times 1.0825) \div (0.16 - 0.0825) = \mathbf{\$41.90}$.

CFA Level I

"Equity Valuation: Concepts and Basic Tools," John J. Nagorniak and Stephen E. Wilcox Section 4.2

2. Answer = C

If markets are strong-form efficient, prices reflect not only private information but also past market data and public information. If markets are weak-form efficient, investment strategies based on fundamental analysis of public information and past market data could achieve abnormal returns. The semi-strong form of market efficiency also encompasses the weak form. Therefore, security prices reflect not only publicly known and available information but also all past market data.

CFA Level I

"Market Efficiency," W. Sean Cleary, Howard J. Atkinson, and Pamela Peterson Drake Section 3

3. Answer = C

For a violation of the strong form of market efficiency to occur, the strategy based on fundamental analysis must achieve a net return higher than the net return of the passive strategy, on a risk adjusted basis. This threshold corresponds to 15% because both strategies had the same systematic risk and the passive strategy has no fees or expenses. To find the gross return on the strategy that uses fundamental analysis, the fees and expenses must be added to the net return: Gross return = Net return + Fees and expenses = 15% + 2% = 17%. Anything in excess of 17% would violate the strong form of market efficiency for the fundamental analysis strategy.

CFA Level I

"Market Efficiency," W. Sean Cleary, Howard J. Atkinson, and Pamela Peterson Drake Sections 3.4 and 2.1

4. Answer = C

In the case of unsponsored DRs, the depository bank, not the investors in the DR, retains the voting rights.

CFA Level I

"Overview of Equity Securities," Ryan C. Fuhrmann, and Asjeet S. Lamba Section 5.2

5. Answer = C

A momentum anomaly occurs when securities that have experienced high short-term returns continue to generate higher returns in subsequent periods. Therefore, if a trader can obtain persistent abnormal returns by adopting an investment strategy that purchases stocks that have recently experienced high returns, then he or she is exploiting a momentum anomaly.

CFA Level I

"Market Efficiency," W. Sean Cleary, Howard J. Atkinson, and Pamela Peterson Drake Section 4.1

6. Answer = B

Security market indices serve as market proxies when measuring risk-adjusted performance. Alpha, the difference between the return of the actively managed portfolio and the return of the passive portfolio, is a measure of risk-adjusted return.

CFA Level I

"Security Market Indices," Paul D. Kaplan and Dorothy C. Kelly Section 4.2

7. Answer = C

The forms of market efficiency are as follows:

Forms of Market Efficiency	Market Prices Reflect:		
	Past Market Data	Public Information	Private Information
Weak	√		
Semi-strong	√	√	
Strong	√	√	√

If a test rejects the hypothesis that market prices reflect private information but does not reject the hypothesis that they reflect past market data and public information, then there is evidence that the form of market efficiency is semi-strong (because only past market data and public information are reflected in market prices).

CFA Level I

"Market Efficiency," W. Sean Cleary, Howard J. Atkinson, and Pamela Peterson Drake Section 3

8. Answer = A

Initial equity (%) in the margin transaction = $1/\text{Leverage ratio} = 1/1.66 = 0.60$;

Initial equity per share at the time of purchase = $\$36 \times 0.60 = \21.60 ;

Price (P) at which margin call occurs:

Equity per share/Price per share = Maintenance margin (%)

$= (\$21.60 + P - \$36)/P = 0.30$;

$0.7P = \$14.40$;

P = \$20.57.

CFA Level I

"Market Organization and Structure," Larry Harris Section 5.2

9. Answer = A

Because regulated markets are more informationally efficient, there are fewer arbitrage opportunities.

CFA Level I

"Market Organization and Structure," Larry Harris Section 10

10. Answer = C

Given that the Intrinsic value is $P_0 = P_0/E_1 \times E_1$ and

Justified forward P/E is $P_0/E_1 = p/(r - g)$,

where: p = payout ratio,

Dividend growth rate = $(1 - \text{Payout ratio}) \times \text{ROE} = (1 - 0.6) \times 12.5 = 5\%$,

Justified forward P/E = $P_0/E_1: 0.60 / (0.10 - 0.05) = 12x$, so Intrinsic value = $12 \times \$3 = \36 .

CFA Level I

"Equity Valuation: Concepts and Basic Tools," John J. Nagorniak and Stephen E. Wilcox Section 5.1

11. Answer = B

Intrinsic value is the true value so an analyst will put a "sell" recommendation on a stock when its market value, the price at which a stock is traded, is higher than intrinsic value.

CFA Level I

"Market Efficiency," W. Sean Cleary, Howard J. Atkinson and Pamela Peterson Drake Section 2.2

12. Answer = B

The management of a public firm is under pressures to meet shorter-term demands, such as meeting quarterly sales and earnings projections from analysts. Private owners are thus better able to focus on longer-term value creation opportunities.

CFA Level I

"Overview of Equity Securities," Ryan C. Fuhrmann, and Asjeet S. Lamba Section 4

Chapter 7

Derivatives Investments

Mock A

Questions

1. For a forward contract with a value of zero, a situation where the spot price is above the forward price is best explained by high:
 - a. Interest rates.
 - b. Convenience yield.
 - c. Storage costs.
2. Convenience yield is best described as a nonmonetary benefit of holding a(n):
 - a. Forward contract.
 - b. Option contract.
 - c. Asset.
3. According to put-call-forward parity, if the put in a protective put with forward contract expires out of the money, the payoff is most likely equal to:
 - a. The face value of a risk-free bond.
 - b. Zero.
 - c. The market value of the underlying asset.
4. Which of the following statements best describes changes in the value of a long forward position during its life?
 - a. As the time to maturity goes down, the value of the position goes up.
 - b. As the price of the underlying goes up, the value of the position goes up.
 - c. As interest rates go down, the value of the position goes up.
5. A derivative can best be described as a financial instrument that:
 - a. Duplicates the underlying asset's performance.
 - b. Passes through the underlying asset's returns.
 - c. transforms the underlying asset's performance
6. Which of the following derivatives is least likely to be classified as a contingent claim?
 - a. A call option contract
 - b. A credit default swap
 - c. A futures contract

Answers

1. B.

If the convenience yield is high, holding the underlying confers large benefits, thus the spot price can exceed the forward price for a forward contract with a value of zero. Based on the formula:

$$V_t(T) = S_t - (y - \theta)(1+r)^t - F_0(T)(1+r)^{-(T-t)}$$

and an initial value $V_t(0)$ of zero, large benefits θ explains why the spot price can exceed the forward price.

2015 CFA Level 1

"Basics of Derivative Pricing and Valuation,"

2. C.

Convenience yield represents the nonmonetary advantage of holding the asset.

CFA Level I

"Basics of Derivative Pricing and Valuation,"

3. C.

A protective put with forward contract is defined as a long position in (1) a bond which has the face value equal to the forward contract, (2) a forward contract and (3) a long position in a put. If the put expires out of the money, the value of the overall position is equal to the market value of the asset.

+ $F_0(t)$ (payoff of bond)

+ $S_T - F_0(t)$ (payoff of forward)

+ 0 (payoff of option)

= S_T (payoff of strategy)

CFA Level I

"Basics of Derivative Pricing and Valuation,"

4. B.

Given the formula for the value of a forward contract:

$$V_t(T) = S_t - F_0(T)(1+r)^{-(T-t)}$$

It follows that the value of the contract goes up as the price of the underlying goes up.

2015 CFA Level 1

"Basics of Derivative Pricing and Valuation,"

5. C.

The best characterization of a derivative is that it typically transforms the underlying asset's performance.

CFA Level I

"Derivative Markets and Instruments,"

6. C.

A futures contract is classified as a forward commitment in which the buyer undertakes to purchase the underlying asset from the seller at a later date and at a price agreed on by the two parties when the contract is initiated.

CFA Level I

"Derivative Markets and Instruments,"

Questions

1. In contrast to over-the-counter options, futures contracts most likely:
 - a) Are not exposed to default risk.
 - b) Represent a right rather than a commitment.
 - c) Are private, customized transactions.
2. An investor has purchased a share of stock for \$190. A call option on this stock, expiring in seven months and with an exercise price of \$200, is priced at \$11.40. If the investor enters into a covered call now, the profit on this strategy if the stock price at expiration is \$215 is closest to:
 - a) \$28.60.
 - b) -\$3.60.
 - c) \$21.40.
3. Which of the following is least likely one of the main benefits of derivative markets? Derivative markets:
 - a) Enable companies to more easily practice risk management.
 - b) Reveal prices and volatility of the underlying assets.
 - c) Exhibit lower volatility compared with the spot market.
4. The pricing of forwards and futures will most likely differ if:
 - a) Futures prices and interest rates are uncorrelated.
 - b) Futures prices and interest rates are negatively correlated.
 - c) Interest rates exhibit zero volatility.
5. A forward rate agreement most likely differs from most other forward contracts, because:
 - a) It involves an option component.
 - b) Positions cannot be closed out prior to maturity.
 - c) Its underlying is not an asset.
6. An investor notices that the price of an American call option is above the price of a European call option with otherwise identical features. What is the most likely reason for this difference?
 - a) The underlying will go ex-dividend.
 - b) The options are close to expiration.
 - c) The options are deep-in-the-money.

Answers

1. A.

Over-the-counter options are exposed to default risk, but futures contracts are standardized transactions that take place on futures exchanges and are not exposed to default risk.

CFA Level I

"Derivative Markets and Instruments,"

2. C.

$$n = S_T - S_0 - \max(0, \$215 - \$200) + \$11.40 = \$21.40$$

$$\Pi = \$215 - \$190 - \max(0, \$215 - \$200) + \$11.40 = \$21.40.$$

CFA Level I

"Risk Management Applications of Option Strategies,"

3. C.

Derivative markets are not necessarily more or less volatile than spot markets. Derivative markets reveal prices and volatilities of the underlying assets and facilitate risk management.

CFA Level I

"Derivative Markets and Instruments,"

4. B.

The pricing of forwards and futures will differ if futures prices and interest rates are negatively correlated. A negative correlation between futures prices and interest rates makes forwards more desirable than futures in the long position.

CFA Level I

"Basics of Derivative Pricing and Valuation,"

5. C.

Forward rate agreements, unlike most other forward contracts, do not have an asset as an underlying. Instead, the underlying is an interest rate.

CFA Level I

"Basics of Derivative Pricing and Valuation,"

6. A.

American call prices can differ from European call prices only if there are cash flows on the underlying.

CFA Level I

"Basics of Derivative Pricing and Valuation,"

Questions

1. Using put-call parity, a long call can *best* be replicated by going:
 - A. long the put, short the asset and long the bond.
 - B. long the put, long the asset and short the bond.
 - C. short the put, long the asset and short the bond.
2. According to put-call-forward parity, if the put in a protective put with forward contract expires out of the money, the payoff is *most likely* equal to:
 - A. zero.
 - B. the face value of a risk-free bond.
 - C. the market value of the underlying asset.
3. According to put-call parity, if a fiduciary call expires in the money, the payoff is *most likely* equal to the:
 - A. market value of the asset.
 - B. difference between the market value of the asset and the face value of the risk-free bond.
 - C. face value of the risk-free bond.
4. A high convenience yield is *most likely* associated with holding:
 - A. commodities.
 - B. equities.
 - C. bonds.
5. According to put-call-forward parity, the difference between the price of a put and the price of a call is *most likely* equal to the difference between:
 - A. forward price and spot price discounted at the risk-free rate.
 - B. exercise price and forward price discounted at the risk-free rate.
 - C. spot price and exercise price discounted at the risk-free rate.
6. In the binomial model, the difference between the up and down factors *best* represents the:
 - A. moneyness of an option.
 - B. pseudo probability.
 - C. volatility of the underlying.

Answers

1. Answer = B

According to put-call parity, a long call is equal to long put, long asset, short bond.

CFA Level I

"Basics of Derivative Pricing and Valuation," Don M. Chance Section 4.1.9

2. Answer = C

A protective put with forward contract is defined as a long position in (1) a bond which has the face value equal to the forward contract, (2) a forward contract and (3) a long position in a put. If the put expires out of the money, the value of the overall position is equal to the market value of the asset. + $F_0(t)$ (payoff of bond)

+ $S_T - F_0(t)$ (payoff of forward)

+ 0 (payoff of option)

= S_T (payoff of strategy)

CFA Level I

"Basics of Derivative Pricing and Valuation," Don M. Chance Section 4.1.10

3. Answer = A

A fiduciary call, defined as a long position in a call and in a risk-free bond, generates a payoff that is equal to the market value of the asset if it expires in the money.

CFA Level I

"Basics of Derivative Pricing and Valuation," Don M. Chance Section 4.1.9

4. Answer = A

Convenience yield is primarily associated with commodities and generally exists as a result of difficulty in shorting the commodity or unusually tight supplies.

CFA Level I

"Basics of Derivative Pricing and Valuation," Don M. Chance Section 2.2.5

5. Answer = B

Put-call-forward parity can be written as:

$$p_0 - c_0 = [X - F_0(T)] / (1 + r)^T$$

This means that the difference between the price of a put and the price of a call is equal to the difference between exercise price and forward price discounted at the risk-free rate. .

CFA Level I

"Basics of Derivative Pricing and Valuation," Don M. Chance Section 4.1.9

6. Answer = C

The volatility of the underlying is captured in the binomial model by the difference between the up and down factors.

CFA Level I

"Basics of Derivative Pricing and Valuation," Don M. Chance Section 4.2

Questions

1. Conceptually, a FRA *most likely* allows a company that wants to invest money in the future to lock in a rate by making a:
 - A. variable payment and receiving a fixed payment.
 - B. fixed payment and receiving a different fixed payment.
 - C. fixed payment and receiving a variable payment.
2. During its life, the value of a forward contract is *most likely* equal to the price of the underlying minus the price of the:
 - A. forward, discounted over the remaining term of the contract.
 - B. forward.
 - C. forward, discounted over the original term of the contract.
3. Forward rate agreements are *most likely* used to hedge an exposure in the:
 - A. foreign exchange market.
 - B. money market.
 - C. equity market.
4. A corporation issues five-year fixed-rate bonds. Its treasurer expects interest rates to decline for all maturities for at least the next year. She enters into a one-year agreement with a bank to receive quarterly fixed-rate payments and to make payments based on floating rates benchmarked on three-month LIBOR. This agreement is *best* described as a:
 - A. futures contract.
 - B. swap.
 - C. forward contract.
5. Which of the following statements is *least* accurate concerning differences in the pricing of forwards and futures?
 - A. Interest rate volatility can explain pricing differences.
 - B. Pricing differences can arise if futures prices and interest rates are uncorrelated.
 - C. Differences in the pattern of cash flows of forwards and futures can explain pricing differences.
6. A forward rate agreement *most likely* differs from most other forward contracts, because:
 - A. its underlying is not an asset.
 - B. positions cannot be closed out prior to maturity.
 - C. it involves an option component.

Answers

1. Answer = A

FRAs are forward contracts that conceptually allow lenders to lock in a fixed payment on a future investment by receiving a known payment and making an unknown payment which offsets the unknown future interest payment.

CFA Level I

"Basics of Derivative Pricing and Valuation," Don M. Chance Section 3.1.4

2. Answer = A

The value of a forward contract is the spot price of the underlying minus the present value of the forward contract. Calculating the present value requires adjusting the time period to account for the remaining term of the contract.

CFA Level I

"Basics of Derivative Pricing and Valuation," Don M. Chance Section 3.1.3

3. Answer = B

Forward rate agreements are used to hedge interest rate exposure present in the money market.

CFA Level I

"Basics of Derivative Pricing and Valuation," Don M. Chance Section 3.1.4

4. Answer = B

A swap is a series of forward payments. Specifically, a swap is an agreement between two parties to exchange a series of future cash flows. The corporation receives fixed interest rate payments and makes variable interest rate payments. Given that the contract is for one year and the floating rate is based on three-month LIBOR, at least four payments will be made during the year.

CFA Level I

"Derivative Markets and Instruments," Don M. Chance Section 4.1

5. Answer = B

If futures prices and interest rates are uncorrelated, the prices of forwards and futures will be identical.

CFA Level I

"Basics of Derivative Pricing and Valuation," Don M. Chance Section 3.2

6. Answer = A

Forward rate agreements, unlike most other forward contracts, do not have an asset as an underlying. Instead, the underlying is an interest rate.

CFA Level I

"Basics of Derivative Pricing and Valuation," Don M. Chance Section 3.1.4

Chapter 8

Fixed Income

Mock A

Questions

1. An analyst uses a valuation model to estimate the value of an option-free bond at 92.733 to yield 11%. If the value is 94.474 for a 60 bp decrease in yield and 91.041 for a 60 bp increase in yield, the approximate modified duration of the bond is closest to:
 - a. 1.85.
 - b. 6.17.
 - c. 3.09.
2. In a securitization structure, credit tranching allows investors to choose between:
 - a. Subordinated bonds and senior bonds.
 - b. Partially amortizing loans and fully amortizing loans.
 - c. Extension risk and contraction risk.
3. Which of the following is least likely a component of the "Four Cs of Credit Analysis" framework?
 - a. Competition
 - b. Collateral
 - c. Covenants
4. The type of residential mortgage least likely to contain a "balloon" payment is a(n)
 - a. Interest-only mortgage.
 - b. Partially amortizing mortgage.
 - c. Fully amortizing mortgage.
5. If an issuer is required to retire a specified portion of the bond's principal each year, the bond most likely:
 - a. Is callable.
 - b. Has a sinking fund provision.
 - c. Is a step-up note.
6. Consider a five-year option-free bond that is priced at a discount to par value. Assuming the discount rate does not change, one year from now the value of the bond will most likely:
 - a. Stay the same.
 - b. Decrease.
 - c. Increase.
7. An investor purchases a 5% coupon bond maturing in 15 years for par value. Immediately after purchase, the yield required by the market increases. The investor would then most likely have to sell the bond at:
 - a. A premium.
 - b. par.
 - c. A discount.
8. Which of the following bonds is most likely to trade at a lower price relative to an otherwise identical option-free bond?
 - a. Callable bond
 - b. Convertible bond
 - c. Puttable bond
9. Which of the following most likely exhibits negative convexity?
 - a. An option-free bond
 - b. A callable bond
 - c. A puttable bond

10. Consider two bonds that are identical except for their coupon rates. The bond that will have the highest interest rate risk most likely has the:
 - a. Coupon rate closest to its market yield.
 - b. Highest coupon rate.
 - c. Lowest coupon rate.
11. If a bank wants the ability to retire debt prior to maturity in order to take advantage of lower borrowing rates, it most likely issues a:
 - a. Callable bond.
 - b. Convertible bond.
 - c. putable bond.
12. Ted Nguyen is an investor domiciled in a country with an original issue discount tax provision. He purchases a zero-coupon bond at a deep discount to par value with the intention of holding the bond until maturity. At maturity, he will most likely face:
 - a. A capital gain.
 - b. Neither a capital loss nor gain.
 - c. a capital loss

Answers

1. C.

The approximate modified duration of a bond is, $ApproxModDur = \frac{(PV_-) - (PV_+)}{2 \times (\Delta Yield) \times (PV_0)}$ where PV_- , PV_0 , and PV_+ are the values of the bond when the yield falls, under the current yield, and when the yield rises, respectively, and $\Delta Yield$ is the size of the yield change.

Therefore,

$$ApproxModDur = \frac{94.474 - 91.041}{2 \times 0.0060 \times 92.733} = 3.09$$

CFA Level 1

"Understanding Fixed Income Risk and Return,"

2. A.

Credit tranching allows investors to choose between subordinate and senior bond classes as a means of credit enhancement. The purpose of this structure is to redistribute the credit risk associated with the collateral.

CFA Level I

"Introduction to Asset-Backed Securities,"

3. A.

The "Four Cs of Credit Analysis" framework includes capacity, collateral, covenants, and character. Competition is not one of the components.

CFA Level I

"Fundamentals of Credit Analysis,"

4. C.

A fully amortizing mortgage is least likely to contain a balloon payment because the sum of all the scheduled principal repayments during the mortgage's life is such that when the last mortgage payment is made the loan is paid in full.

CFA Level I

"Introduction to Asset-Backed Securities"

5. B.

A sinking fund provision requires retirement of a portion of the bond's principal every year, rather than retirement of the entire issue at maturity.

CFA Level 1

"Fixed-Income Securities: Defining Elements,"

6. C.

The bond is priced below its par value but will be worth exactly par value at maturity. Over time, assuming a stable discount rate, the value of the bond must rise so that it is equal to par at maturity. That is, the price is "pulled to par."

CFA Level 1

"Introduction to Fixed-Income Valuation,"

7. C.

The bond would sell below par or at a discount if the yield required by the market rises above the coupon rate. Because the bond initially was purchased at par, the coupon rate equals the yield required by the market. Subsequently, if yields rise above the coupon, the bond's market price would fall below par.

CFA Level 1

"Introduction to Fixed-Income Valuation,"

8. A.

A callable bond benefits the issuer because it gives the issuer the right to redeem all (or part) of the bonds before the maturity date.

Thus, the price of a callable bond will typically be lower than the price of an otherwise identical non-callable bond.

CFA Level I

"Fixed-Income Securities: Defining Elements,"

9. B.

A callable bond exhibits negative convexity at low yield levels and positive convexity at high yield levels.

CFA Level 1

"Understanding Fixed-Income Risk and Return,"

10. C.

A lower coupon rate means that more of the bond's value comes from repayment of face value, which occurs at the end of the bond's life.

CFA Level I

"Introduction to Fixed-Income Valuation,"

11. A.

Callable bonds give issuers the ability to retire debt prior to maturity. The most compelling reason for them to do so is to take advantage of lower borrowing rates.

CFA Level 1

"Fixed-Income Markets: Issuance, Trading, and Funding,"

12. B.

An original issue discount tax provision allows the investor to increase the cost basis of the bond, so when the bond matures, the investor faces no capital gain or loss.

CFA Level 1

"Fixed-Income Securities: Defining Elements,"

Questions

1. One limitation as to why using the average duration of the bonds in a portfolio does not properly reflect that portfolio's yield curve risk is that the approach assumes:
 - a) A parallel shift in the yield curve.
 - b) All the bonds have the same discount rate.
 - c) A non-parallel shift in the yield curve.
2. Centro Corp. recently issued a floating-rate note (FRN) that includes a feature that prevents its coupon rate from falling below a prespecified minimum rate. This feature in an FRN is most likely referred to as a:
 - a) Cap.
 - b) Collar.
 - c) Floor.
3. Which of the following is least likely a component of yield spread?
 - a) Expected inflation rate
 - b) Credit risk
 - c) Taxation
4. If the yield-to-maturity on an annual-pay bond is 7.75%, the bond-equivalent yield is closest to:
 - a) 8.05%.
 - b) 7.61%.
 - c) 7.90%.
5. Consider two 10-year bonds, one that contains no embedded options and the other that gives its owner the right to convert the bond to a fixed number of shares of the issuer's common stock. The convertibility option in the second bond cannot be exercised for five years. The bonds are otherwise identical. Compared with the yield on the convertible bond, the yield on the option-free bond is most likely:
 - a) Higher.
 - b) The same.
 - c) Lower.
6. Consider bonds that have the same yield to maturity and maturity. The bond with the greatest reinvestment risk is most likely the one selling at:
 - a) A premium.
 - b) A discount.
 - c) par.
7. Compared with investment-grade bonds, the spread movements on high-yield bonds are influenced:
 - a) Less by interest rate changes and exhibits a greater correlation with movements in equity markets.
 - b) More by interest rate changes and exhibit a greater correlation with movements in equity markets.
 - c) Less by interest rate changes and exhibits a lower correlation with movements in equity markets.
8. Which of the following statements is least accurate regarding the factors that affect the interest rate risk characteristics of an option-free bond?
 - a) The higher the yield, the greater the bond's price sensitivity to changes in interest rates.
 - b) The longer the bond's maturity, the greater the bond's price sensitivity to changes in interest rates.
 - c) The lower the coupon rate, the greater the bond's price sensitivity to changes in interest rates.
9. A South Korean electronics company issued bonds denominated in US dollars in the United States and registered with the SEC. These bonds are most likely known as a:
 - a) Global bond.
 - b) Foreign bond.
 - c) Eurobond.
10. The Delfain Corporation reported a significant improvement in profitability which was followed by a material upgrade in its credit rating. The market responded by immediately requiring a 100 basis point narrower spread

to Gilts on Delfain's 8-year bond. If the bond's modified duration is 6.0 and its convexity is 55.0 the return impact of this change is closest to:

- a) 7.10%.
- b) -5.73%.
- c) 6.28%.

11. On 15 December 2013, Alpha Corp. issued a 10-year callable bond paying an annual coupon of 8%. The bond is callable in whole or in part at any time after 15 December 2018. This type of callable bond is most likely referred to as:

- a) Bermuda style.
- b) American style.
- c) European style.

12. The duration and convexity of an option-free bond priced at \$90.25 are 10.34 and 151.60, respectively. If yields increase by 200 bps, the percentage change of the price is closest to:

- a) -23.71%.
- b) -17.65%.
- c) -20.68%.

Answers

1. A.

A limitation to using the average duration approach in calculating portfolio duration is that it assumes all interest rates across the yield curve change by the same amount and, therefore, each bond's price changes by the same percentage.

CFA Level I

"Understanding Fixed Income Risk and Return,"

2. C.

An FRN with a floor on the coupon rate prevents the coupon rate from falling below a prespecified minimum rate.

CFA Level I

"Fixed-Income Securities: Defining Elements,"

3. A.

Building blocks of the yield curve are spread (risk premium) and a benchmark (risk-free rate of return). Expected inflation rate and expected real rate are components of the risk-free rate of return (i.e., the benchmark).

CFA Level 1

"Introduction to Fixed-Income Valuation,"

4. B.

The bond-equivalent yield $2 \times (1.0775^{0.5} - 1) = 0.07605$ or 7.61%

CFA Level 1 "Introduction to Fixed-Income Valuation,"

5. A.

The convertibility option provides a benefit to the investor, who will accept a lower yield on the convertible bond compared with the option-free bond.

CFA Level 1

"Fixed-Income Securities: Defining Elements,"

6. A.

Yield to maturity is based on the assumption that a bond is held to maturity, does not default, and has its coupon payments reinvested at the yield to maturity. The bond selling at a premium has the highest coupon rate and is expected to earn the most reinvestment income from reinvesting those coupon payments at the yield to maturity. If the reinvestment rate falls, this bond will suffer the greatest loss.

CFA Level 1

"Understanding Fixed-Income Risk and Return,"

7. A.

High-yield bonds can be thought of as a hybrid between investment-grade bonds and equity securities. Their spread movements are less influenced by interest rate changes than are investment-grade bonds and they exhibit greater correlation with movements in equity markets.

CFA Level I

"Fundamentals of Credit Analysis",

8. B.

Option-free bonds have positive convexity. The higher the yield to maturity, the lower the duration (and thus the lower the interest rate risk).

CFA Level I

"Understanding Fixed-Income Risk and Return,"

9. B.

Bonds issued by entities that are incorporated in another country are called foreign bonds. Therefore, the bonds issued by a South Korean company in the United States are known as foreign bonds.

CFA Level 1

"Fixed-Income Securities: Defining Elements,"

10. C.

The return impact of a 60 basis points fall in the bond's yield can be computed as:

Return impact $\approx - (MDur \times \Delta Spread) + \frac{1}{2} Cv \times (\Delta Spread)^2$.

Return impact $\approx - (6.0 \times -0.01) + \frac{1}{2}(55.0) \times (-0.01)^2 = 6.28\%$.

CFA Level I "Fundamentals of Credit Analysis",

11. B.

An American-style callable bond is a bond in which the issuer has the right to call the bonds at any time starting on the first call date.

CFA Level 1

"Fixed-Income Securities: Defining Elements,"

12. B

The percentage change in price is calculated as follows: Duration effect:

$$-10.34 \times (0.02) = -20.68\%$$

$$\text{And convexity effect: } = \frac{1}{2} \times 151.60 \times (0.02)^2 = 3.03\%$$

Total percentage change is the sum of duration effect and convexity effect:

$$-20.68\% + 3.03\% = -17.65\%$$

CFA Level 1

"Understanding Fixed-Income Risk and Return,"

Questions

1. To obtain the spot yield curve, a bond analyst would *most likely* use the most:
 - A. recently issued and actively traded government bonds.
 - B. seasoned and actively traded government bonds.
 - C. recently issued and actively traded corporate bonds.
2. What type of risk *most likely* affects an investor's ability to buy and sell bonds in the desired amounts and at the desired time?
 - A. Default
 - B. Market liquidity
 - C. Spread
3. The option-free bonds issued by ALS Corp. are currently priced at 108.50. Based on a portfolio manager's valuation model, a 1bp increase in interest rates will result in the bond price falling to 108.40, whereas a 1bp decrease in interest rates will result in the bond price rising to 108.59. The price value of a basis point (PVBP) for the bonds is *closest* to:
 - A. 0.095.
 - B. 0.088.
 - C. 0.190.
4. The value of a 10-year, 6% coupon, \$100 par value bond with semiannual payments, assuming an annual discount rate of 7%, is *closest* to:
 - A. \$99.07.
 - B. \$92.89.
 - C. \$107.44.
5. An investor sells a bond at the quoted price of \$98.00. In addition, she receives accrued interest of \$4.40. The flat price of the bond is equal to the:
 - A. par value plus accrued interest.
 - B. agreed on bond price excluding accrued interest.
 - C. accrued interest plus the agreed on bond price.
6. A two-year spot rate of 5% is *most likely* the:
 - A. yield to maturity on a coupon-paying bond maturing at the end of Year 2.
 - B. yield to maturity on a zero-coupon bond maturing at the end of Year 2.
 - C. coupon rate in Year 2 on a coupon-paying bond maturing at the end of Year 4.
7. The process of securitization is *least likely* to allow banks to:
 - A. originate loans.
 - B. reduce the layers between borrowers and ultimate investors.
 - C. repackage loans into simpler structures.
8. Consider a five-year option-free bond that is priced at a discount to par value. Assuming the discount rate does not change, one year from now the value of the bond will *most likely*:
 - A. decrease.

- B. stay the same.
 - C. increase.
9. An investor who owns a mortgage pass-through security is exposed to contraction risk which is the risk that when interest rates:
- A. decline, the security will effectively have a longer maturity than was anticipated at the time of purchase.
 - B. decline, the security will effectively have a shorter maturity than was anticipated at the time of purchase.
 - C. rise, the security will effectively have a shorter maturity than was anticipated at the time of purchase.
10. Which of the following *most likely* exhibits negative convexity?
- A. An option-free bond
 - B. A puttable bond
 - C. A callable bond
11. Consider two bonds that are identical except for their coupon rates. The bond that will have the highest interest rate risk *most likely* has the:
- A. lowest coupon rate.
 - B. highest coupon rate.
 - C. coupon rate closest to its market yield.
12. In a securitization structure, credit tranching allows investors to choose between:
- A. extension risk and contraction risk.
 - B. subordinated bonds and senior bonds.
 - C. partially amortizing loans and fully amortizing loans.

Answers

1. Answer = A

To obtain the spot yield curve a bond analyst would prefer to use the most recently issued and actively traded government bonds. Such bonds will have similar liquidity as well as fewer tax effects because they will be priced closer to par value.

CFA Level I

"Introduction to Fixed-Income Valuation", James F. Adams and Donald J. Smith Section 4

2. Answer = B

The size of the spread between the bid price and the ask price is the primary measure of market liquidity of the issue. Market liquidity risk is the risk that the investor will have to sell a bond below its indicated value. The wider the bid-ask spread, the greater the market liquidity risk.

CFA Level I

"Fundamentals of Credit Analysis," Christopher L. Gootkind Section 2

3. Answer = A

The bond's PVBP is computed using
$$PVBP = \frac{(PV_{\downarrow}) - (PV_{\uparrow})}{2}.$$

$$\text{so, } \frac{108.59 - 108.40}{2} = 0.095.$$

CFA Level 1

"Understanding Fixed-Income Risk and Return," James F. Adams and Donald J. Smith Section 3.5

4. Answer = B

A security with 19 semiannual payments of \$3 interest and a 20th payment of \$103 (interest plus return of face value) with a semiannual discount rate of 3.5% is computed as:

$$P = \frac{3}{1.035^1} + \frac{3}{1.035^2} + \frac{3}{1.035^3} + \dots + \frac{3}{1.035^{19}} + \frac{103}{1.035^{20}}$$

$$P = 92.89.$$

CFA Level 1

"Introduction to Fixed-Income Valuation," James F. Adams and Donald J. Smith Section 2.1

5. Answer = B

The agreed on bond price without accrued interest is referred to as the flat price.

CFA Level 1

"Introduction to Fixed-Income Valuation," James F. Adams and Donald J. Smith Section 3.1

6. Answer = B

A spot rate is defined as the yield to maturity on a zero-coupon bond maturing at the date of that cash flow.

CFA Level I

"Introduction to Fixed-Income Valuation," James F. Adams and Donald J. Smith Section 2.4

7. Answer = C

Securitization allows banks to originate (or create) loans and the process results in a reduction in the layers between borrowers and ultimate investors. The loans are repackaged into more complex, not simpler, structures.

CFA Level I

"Introduction to Asset-Backed Securities", Frank J. Fabozzi Section 2

8. Answer = C

The bond is priced below its par value but will be worth exactly par value at maturity. Over time, assuming a stable discount rate, the value of the bond must rise so that it is equal to par at maturity. That is, the price is "pulled to par."

CFA Level 1

"Introduction to Fixed-Income Valuation," James F. Adams and Donald J. Smith Section 2.3

9. Answer = B

Contraction risk is the risk faced by investors when interest rates fall in that the security will effectively have a shorter maturity than was anticipated at the time of purchase because homeowners can refinance at new, lower interest rates.

CFA Level I

"Introduction to Asset-Backed Securities", Frank J. Fabozzi Section 5.1.6

10. Answer = C

A callable bond exhibits negative convexity at low yield levels and positive convexity at high yield levels.

CFA Level 1

"Understanding Fixed-Income Risk and Return," James F. Adams and Donald J. Smith Section 3.6

11. Answer = A

A lower coupon rate means that more of the bond's value comes from repayment of face value, which occurs at the end of the bond's life.

CFA Level I

"Introduction to Fixed-Income Valuation," James F. Adams and Donald J. Smith Section 2.3

12. Answer = B

Credit tranching allows investors to choose between subordinate and senior bond classes as a means of credit enhancement. The purpose of this structure is to redistribute the credit risk associated with the collateral.

CFA Level I

"Introduction to Asset-Backed Securities," Frank J. Fabozzi Section 3.3

Questions

1. Which of the following statements is *least* accurate regarding the factors that affect the interest rate risk characteristics of an option-free bond?
 - A. The longer the bond's maturity, the greater the bond's price sensitivity to changes in interest rates.
 - B. The lower the coupon rate, the greater the bond's price sensitivity to changes in interest rates.
 - C. The higher the yield, the greater the bond's price sensitivity to changes in interest rates.
2. Centro Corp. recently issued a floating-rate note (FRN) that includes a feature that prevents its coupon rate from falling below a prespecified minimum rate. This feature in an FRN is *most likely* referred to as a:
 - A. cap.
 - B. floor.
 - C. collar.
3. If the yield-to-maturity on an annual-pay bond is 7.75%, the bond-equivalent yield is *closest* to:
 - A. 7.90%.
 - B. 8.05%.
 - C. 7.61%.
4. A South Korean electronics company issued bonds denominated in US dollars in the United States and registered with the SEC. These bonds are *most likely* known as a:
 - A. foreign bond.
 - B. global bond.
 - C. eurobond.
5. Consider a \$100 par value bond, with an 8% coupon paid annually, maturing in 20 years. If the bond currently sells for \$96.47, the yield to maturity is *closest* to:
 - A. 8.37%.
 - B. 8.29%.
 - C. 7.41%.
6. Consider bonds that have the same yield to maturity and maturity. The bond with the greatest reinvestment risk is *most likely* the one selling at:
 - A. par.
 - B. a discount.
 - C. a premium.
7. A 90-day commercial paper issue is quoted at a discount rate of 4.75% for a 360-day year. The bond equivalent yield for this instrument is *closest* to:
 - A. 4.87%.
 - B. 4.75%.
 - C. 4.81%.
8. In a rising interest rate environment, the effective duration of a puttable bond relative to an otherwise identical non-puttable bond, will *most likely* be:
 - A. higher.
 - B. lower.
 - C. the same.

9. A credit analyst observes the following information for Zeta Corp. and its industry.

	Zeta Corp.	Industry Median
Return on capital (%)	19.0%	20.0%
Total debt/Total capital (%)	42.0%	15.5%
FFO/Total debt (%)	45.3%	40.0%
Total debt/EBITDA (x)	3.5x	1.2x
EBITDA interest coverage (x)	4.0x	7.5x

Based on this information, it is *most likely* that the credit risk of Zeta Corp. is:

- A. below its industry peers.
 - B. above its industry peers.
 - C. similar to its industry peers.
10. Which of the following is *least likely* a component of yield spread?
- A. Expected inflation rate
 - B. Taxation
 - C. Credit risk
11. Which of the following 90-day money market instruments *most likely* offers the investor the highest rate of return?

Money Market Instrument	Quoted Rate	Quotation Basis	Day Convention
Instrument A	5.78%	360	Discount rate
Instrument B	5.80%	365	Discount rate
Instrument C	5.96%	365	Add-on rate

- A. Instrument C
 - B. Instrument A
 - C. Instrument B
12. Using the following US Treasury forward rates, the value of a 2½-year \$100 par value Treasury bond with a 5% coupon rate is *closest* to:

Period	Years	Forward Rate
1	0.5	1.20%
2	1	1.80%
3	1.5	2.30%
4	2	2.70%
5	2.5	3.00%

- A. \$104.87.
- B. \$106.83.
- C. \$101.52.

Answers

1. Answer = C

Option-free bonds have positive convexity. The higher the yield to maturity, the lower the duration (and thus the lower the interest rate risk).

CFA Level I

"Understanding Fixed-Income Risk and Return," by James F. Adams and Donald J. Smith Section 3.3

2. Answer = B

An FRN with a floor on the coupon rate prevents the coupon rate from falling below a prespecified minimum rate.

CFA Level I

"Fixed-Income Securities: Defining Elements," Moorad Choudhry and Stephen E. Wilcox Section 4.2

3. Answer = C

The bond-equivalent yield =

$$2 \times (1.0775^{0.5} - 1) = 0.07605 \text{ or } 7.61\%$$

CFA Level 1

"Introduction to Fixed-Income Valuation," James F. Adams and Donald J. Smith Section 3.3

4. Answer = A

Bonds issued by entities that are incorporated in another country are called foreign bonds. Therefore, the bonds issued by a South Korean company in the United States are known as foreign bonds.

CFA Level 1

"Fixed-Income Securities: Defining Elements," Moorad Choudhry and Stephen E. Wilcox Section 3.2

5. Answer = A

A security with a present value of 96.47, 19 interest payments of 8, and a 20th payment of principal plus interest (108) has a yield to maturity of **8.37%**.

CFA Level 1

"Introduction to Fixed-Income Valuation," James F. Adams and Donald J. Smith Section 3

6. Answer = C

Yield to maturity is based on the assumption that a bond is held to maturity, does not default, and has its coupon payments reinvested at the yield to maturity. The bond selling at a premium has the highest coupon rate and is expected to earn the most reinvestment income from reinvesting those coupon payments at the yield to maturity. If the reinvestment rate falls, this bond will suffer the greatest loss.

CFA Level 1

"Understanding Fixed-Income Risk and Return," James F. Adams and Donald J. Smith Section 2

7. Answer = A

The price of the commercial paper per 100 of par value is: $PV = FV \times \left(1 - \frac{\text{Days}}{\text{Year}} \times DR\right)$ where PV

and FV are the price and face value of the money market instrument, Days is the number of days between settlement and maturity, Year is number of days in the year, and DR is the discount rate stated as an annual percentage.

So, $PV = 100 \times \left(1 - \frac{90}{360} \times 0.0475\right) = 98.8125$. The bond equivalent yield is,

$$AOR = \frac{\text{Year}}{\text{Days}} \times \left(\frac{FV - PV}{PV}\right). \text{ So, } AOR = \frac{365}{90} \times \left(\frac{100 - 98.8125}{98.8125}\right) = 4.874\%.$$

CFA Level 1

"Introduction to Fixed-Income Valuation," James F. Adams and Donald J. Smith Section 3.5

8. Answer = B

When interest rates are rising, the put option becomes more valuable to the investor. The ability to sell the bond at par value limits the price depreciation as rates rise. So, the presence of an embedded put option reduces the sensitivity of the bond price to changes in interest rates, resulting in a lower effective duration.

CFA Level 1

"Understanding Fixed-Income Risk and Return," James F. Adams and Donald J. Smith Section 3.3

9. Answer = B

While the company has a similar return on capital it has significantly higher leverage as well as lower EBITDA interest coverage ratio than its industry peers. It is likely that the company's credit risk will be above its industry peers.

CFA Level I

"Fundamentals of Credit Analysis", Christopher L. Gootkind Section 5.2.1

10. Answer = A

Building blocks of the yield curve are spread (risk premium) and a benchmark (risk-free rate of return). Expected inflation rate and expected real rate are components of the risk-free rate of return (i.e., the benchmark).

CFA Level 1

"Introduction to Fixed-Income Valuation," James F. Adams and Donald J. Smith Section 5.1

11. Answer = A

Instrument C provides a bond equivalent yield of 5.96%, compared with 5.946% for Instrument A and 5.883% for Instrument B.

CFA Level I

"Introduction to Fixed-Income Valuation," by James F. Adams and Donald J. Smith Section 3.5

12. Answer = B

The value of the bond is

$$\begin{aligned} & \frac{2.5}{(1 + .012/2)} + \frac{2.5}{(1 + .012/2) \times (1 + .018/2)} + \frac{2.5}{(1 + .012/2) \times (1 + .018/2) \times (1 + .023/2)} \\ & + \frac{2.5}{(1 + .012/2) \times (1 + .018/2) \times (1 + .023/2) \times (1 + .027/2)} \\ & + \frac{102.5}{(1 + .012/2) \times (1 + .018/2) \times (1 + .023/2) \times (1 + .027/2) \times (1 + .030/2)} = \$106.83 \end{aligned}$$

CFA Level I

"Introduction to Fixed-Income Valuation," James F. Adams and Donald J. Smith Section 4

Chapter 9

Alternative Investments

Mock A

Questions

1. For a hedge fund investor, a benefit of investing in a fund of funds is least likely the:
 - a. Ability to negotiate better redemption terms.
 - b. Multilayered fee structure.
 - c. Higher level of due diligence expertise.
2. The most likely impact of adding commodities to a portfolio of equities and bonds is to:
 - a. Increase risk.
 - b. Provide higher current income.
 - c. Reduce exposure to inflation.
3. Based on the historical record, adding alternative investments to a traditional investment portfolio consisting of publicly traded debt and equity will most likely decrease the portfolio's:
 - a. Liquidity.
 - b. Risk-adjusted return.
 - c. Downside risk.
4. The return on a commodity index is likely to be different from returns on the underlying commodities because:
 - a. Indices are constructed using futures contracts.
 - b. Assets are not marked to market.
 - c. Data are subject to survivorship bias.

Answers

1. A.

Funds of funds have a multilayered fee structure that will reduce the returns to the investor.

CFA Level I

"Introduction to Alternative Investments,"

2. C.

Over the long term, commodity prices are closely related to inflation and thus including commodities in a portfolio of equities and bonds will reduce its exposure to inflation.

CFA Level I

"Introduction to Alternative Investments,"

3. A.

Many categories of alternative assets have low liquidity because of the fund structures used (e.g., limited partnerships for hedge funds and private equity) or high transactions costs for underlying assets (e.g., real estate). Alternative assets have generally had high downside risks. However, low correlations with traditional asset classes suggest strong diversifying potential and high returns result in relatively strong Sharpe ratios (high risk-adjusted returns).

CFA Level I

"Introduction to Alternative Investments,"

4. A.

Because commodity indices are constructed using commodity futures and not the underlying commodities, there can be differences between commodity index returns and the returns of the underlying commodities.

CFA Level I

Questions

1. Compared to traditional investments, alternative investments least likely demonstrate which of the following characteristics?
 - a) Underlying investments that are illiquid.
 - b) A high degree of regulation.
 - c) Narrow manager specialization.
2. Concentrated portfolio strategies are attractive because of their:
 - a) Ability to track market indices.
 - b) Low risk.
 - c) Potential to generate alpha.
3. An investor who has positions in multiple long-short equity hedge funds and is concerned about whether these positions are sufficiently diversified will mostly likely be concerned about the lack of:
 - a) Frequent independent valuations.
 - b) Liquidity in the underlying assets.
 - c) Transparency in reported positions.
4. If the price of a commodity futures contract is below the spot price, it is most likely that the:
 - a) Cost of carry exceeds the convenience yield.
 - b) Convenience yield exceeds storage costs.
 - c) Roll yield is negative.

Answers

1. B.

Alternative investments are less regulated and transparent than traditional investments such as equity and debt securities.

CFA Level I

"Introduction to Alternative Investments,"

2. C.

Concentrated portfolio strategies focus on only a few securities, strategies, or managers. This focus reduces diversification but may enable investors to achieve alpha.

CFA Level I

"Introduction to Alternative Investments,"

3. C.

Long-short hedge funds invest in liquid, publicly traded equity (taking long and short positions); therefore, the underlying positions can be reversed easily and there is no need for independent valuations because current market prices are available. The investor will have difficulty in determining if the different funds are holding diverse or concentrated positions (both within each fund and between funds) because hedge funds generally do not reveal their holdings.

CFA Level I

"Introduction to Alternative Investments,"

4. B.

The convenience yield must exceed the cost of carry to arrive at a futures price below the spot price because the futures price is approximately equal to the spot price $[(1 + r) + \text{Storage cost} - \text{Convenience yield}]$ and the cost of carry is defined as interest cost plus storage cost. Given that interest cost is always positive, the convenience yield must also exceed storage costs to arrive at a futures price below the spot price.

CFA Level I

"Introduction to Alternative Investments,"

Questions

1. The *most likely* impact of adding commodities to a portfolio of equities and bonds is to:
 - A. increase risk.
 - B. reduce exposure to inflation.
 - C. provide higher current income.
2. Alternative investments that rely on estimates rather than observable market prices for valuation purposes are *most likely* to report:
 - A. returns that are understated.
 - B. volatility of returns that is understated.
 - C. correlations of returns with the returns of traditional assets that are overstated.
3. Relative to traditional investments, alternative investments are *best* characterized as having:
 - A. higher correlations with other asset classes.
 - B. unique legal and tax considerations.
 - C. greater liquidity.
4. The real estate index *most likely* to suffer from sample selection bias is a(n):
 - A. REIT index.
 - B. appraisal index.
 - C. repeat sales index.

Answers

1. Answer = B

Over the long term, commodity prices are closely related to inflation and thus including commodities in a portfolio of equities and bonds will reduce its exposure to inflation.

CFA Level I

"Introduction to Alternative Investments," Terri Duhon, George Spentzos, and Scott D. Stewart Section 6.3

2. Answer = B

The use of estimates tends to smooth the return series. As a consequence, the volatility of returns will be understated.

CFA Level I

"Introduction to Alternative Investments," Terri Duhon, George Spentzos, and Scott D. Stewart Section 8.2

3. Answer = B

Alternative investments are more likely characterized as having unique legal and tax considerations because of the broad range and complexity of the investments.

CFA Level I

"Introduction to Alternative Investments," Terri Duhon, George Spentzos, and Scott D. Stewart Section 2

4. Answer = C

Only properties that sell in each period and are included in the index and vary over time which may not be representative of the whole market.

CFA Level I

"Introduction to Alternative Investments," by Terri Duhon, George Spentzos, and Scott D. Stewart Section 5.3

Questions

1. Illiquidity is *most likely* a major concern when investing in:
 - A. private equity.
 - B. real estate investment trusts.
 - C. commodities.
2. Do management fees *most likely* get paid to the manager of a hedge fund, regardless of the fund's performance?
 - A. No, only when the fund's net asset value exceeds the previous high-water mark
 - B. No, only when the fund's gross return is positive
 - C. Yes
3. Concentrated portfolio strategies are attractive because of their:
 - A. potential to generate alpha.
 - B. ability to track market indices.
 - C. low risk.
4. Compared with other investment asset classes, an investment in real estate is *least likely* to be characterized by:
 - A. basic indivisibility.
 - B. homogeneity.
 - C. fixed location.

Answers

1. Answer = A

Once a commitment in a private equity fund has been made, the investor has very limited liquidity options.

CFA Level I

"Introduction to Alternative Investments," Terri Duhon, George Spentzos, and Scott D. Stewart Section 4.5

2. Answer = C

Regardless of performance, the management fee is always paid to the fund manager.

CFA Level I

"Introduction to Alternative Investments," Terri Duhon, George Spentzos, and Scott D. Stewart Section 3.3.1

3. Answer = A

Concentrated portfolio strategies focus on only a few securities, strategies, or managers. This focus reduces diversification but may enable investors to achieve alpha.

CFA Level I

"Introduction to Alternative Investments," Terri Duhon, George Spentzos, and Scott D. Stewart Section 2.2

4. Answer = B

Because no two properties are identical, homogeneity is not a feature of an investment in real estate.

CFA Level I

"Introduction to Alternative Investments," Terri Duhon, George Spentzos, and Scott D. Stewart Section 5

Chapter 10

Portfolio Management

Mock A

Questions

1. Which of the following is least likely a part of the execution step of the portfolio management process?
 - a. Performance measurement
 - b. Portfolio construction
 - c. Security analysis
2. The risk-free rate is 5% and the market risk premium is 8%. If the beta of TRL Corp. is 1.5, based on the capital asset pricing model (CAPM), the expected return of TRL's stock is closest to:
 - a. 9.5%.
 - b. 17.0%.
 - c. 15.5%.
3. Following its decision to divest its non-core assets, analysts expect HCL Corp's standard deviation of returns to rise to 30% and its correlation with the market portfolio to remain unchanged at 0.8. The risk-free rate and the market risk premium are expected to remain unchanged at 6% and 8%, respectively. However, the market portfolio's standard deviation of returns is expected to decrease to 15%. The firm's expected return after the restructure is closest to:
 - a. 18.8%.
 - b. 9.2%.
 - c. 17.6%.
4. The point of tangency between the capital allocation line (CAL) and the efficient frontier of risky assets most likely identifies the:
 - a. Global minimum-variance portfolio.
 - b. Optimal risky portfolio.
 - c. optimal investor portfolio
5. Which of the following institutional investors is most likely to have a low tolerance for investment risk and relatively high liquidity needs?
 - a. Insurance company
 - b. Defined benefit pension plan
 - c. Charitable foundation
6. The top level of a risk management system most likely is:
 - a. Risk governance.
 - b. Defined policies or procedures.
 - c. Strategic analysis or integration.
7. The execution step of the portfolio management process includes:
 - a. Preparing the investment policy statement.
 - b. Monitoring the portfolio performance.
 - c. Finalizing the asset allocation.
8. An investment policy statement's risk objective states that over a 12-month period, with a probability of 95%, the client's portfolio must not lose more than 5% of its value. This statement is most likely a(n):
 - a. Total risk objective.
 - b. Relative risk objective.
 - c. Absolute risk objective.

9. A 35-year old financial analyst expects to retire at the age of 65 and is interested in investing to fund her retirement. She describes herself as being financially astute with average risk tolerance. Which asset class is least likely to be suitable for a majority allocation in her portfolio to meet her retirement needs?
 - a. Exchange-listed equities
 - b. Long-term bonds
 - c. US Treasury bills
10. Stock X and Stock Y have the same level of total risk. Stock X has twice the systematic risk of Stock Y and half its non-systematic risk. Stock X's expected return will most likely be:
 - a. The same as the expected return of Stock Y.
 - b. Higher than the expected return of Stock Y.
 - c. Lower than the expected return of Stock Y.

Answers

1. A.

Performance measurement is a part of the feedback step of the portfolio management process. The execution step includes asset allocation, security analysis, and portfolio construction.

CFA Level I

"Portfolio Management: An Overview,"

2. B.

Using the CAPM relationship of $E(R_i) = R_f + [E(R_m) - R_f]\beta_i$, we can estimate the expected return as: $E(R_i) = 0.05 + (0.08)(1.5) = 17.0\%$.

CFA Level I

"Portfolio Risk and Return: Part II,"

3. A.

We first compute the firm's beta using: $\beta_i = \frac{\rho_{im}\sigma_i}{\sigma_m}$ The beta is: $\beta_i = \frac{0.8(0.30)}{0.15} = 1.6$ The expected return is computed using:

$$E(R_i) = R_f + [E(R_m) - R_f]\beta_i. \text{ So, } E(R_i) = 0.06 + (0.08)1.6 = 18.8\%$$

CFA Level I

"Portfolio Risk and Return: Part II,"

4. B.

The optimal risky portfolio lies at the point of tangency between the capital allocation line and the efficient frontier of risky assets.

CFA Level I

"Portfolio Risk and Return: Part I,"

5. A.

Insurance companies need to be relatively conservative and liquid, given the necessity of paying claims when due.

CFA Level I

"Portfolio Management: An Overview,"

6. A.

Normally a role of the board of directors of a company, risk governance is where goals and responsibilities are defined and top-level decisions (such as determining the company's risk tolerance).

CFA Level I

"Risk Management: An Introduction,"

7. C.

Asset allocation occurs in the execution step.

CFA Level I

"Portfolio Management: An Overview,"

8. C.

The statement is an absolute risk objective because it expresses a maximum loss in value with an associated probability of loss.

CFA Level I

"Basics of Portfolio Planning and Construction,"

9. C.

The investor has a 30-year time horizon and average risk tolerance, so she is able to accept the additional risk associated with exchange-listed equities and long-term bonds compared with US Treasury bills.

CFA Level I

"Basics of Portfolio Planning and Construction,"

10. B.

Because Stock X has a higher systematic risk level compared with Stock Y, its expected return will be higher than that of Stock Y.

CFA Level I

"Portfolio Risk and Return: Part II,"

Questions

- A major benefit of employing a risk budgeting process is that it most likely:
 - Eliminates the need for hedging within the organization.
 - Allows the organization to determine its enterprise risk tolerance.
 - forces risk tradeoffs across the organization
- In a strategic asset allocation, assets within a specific asset class are least likely to have:
 - Similar risk and return expectations.
 - Low correlations with other asset classes.
 - Low paired correlations.
- Information about three stocks is provided in the following table:

Stock	Expected Return	Beta
Booraem Inc	12.85%	1.5
Heisen Inc	11.27%	1.1
Gutmann Inc	9.51%	0.8

If the expected market return is 9.5% and the average risk-free rate is 1.2%, according to the capital asset pricing model (CAPM) and the security market line (SML), which of the three stocks is most likely overvalued?

- Booraem Inc.
 - Gutmann Inc.
 - Heisen Inc.
- Last year, a portfolio manager earned a return of 12%. The portfolio's beta was 1.5. For the same period, the market return was 7.5%, and the average risk-free rate was 2.7%. Jensen's alpha for this portfolio is closest to:
 - 4.50%.
 - 0.75%.
 - 2.10%.
 - Among other things, an organization's risk tolerance should most likely reflect its:
 - Size.
 - Perception of market stability.
 - Competitive position.
 - A portfolio contains equal weights of two securities having the same standard deviation. If the correlation between the returns of the two securities was to decrease, the portfolio risk would most likely:
 - Decrease.
 - Remain the same.
 - Increase.
 - You are preparing an investment policy statement for a client who manages her own successful marketing consultancy. Her annual income is approximately \$500,000. She describes herself as a finance novice. Most of her savings are invested in bank term deposits and short-term government securities. In her responses to the standard risk assessment questionnaire, she strongly agrees with the statements that she "feels more comfortable putting money in a bank account than in the stock market." Also, she "thinks of the word 'risk' as being a 'loss'". Based on this information, your client's ability and willingness to take risk can best be described as:
 - High ability and willingness.
 - Low ability and high willingness.
 - High ability and low willingness.
 - The covariance of the assets in the following portfolio is closest to:

	Asset 1	Asset 2	Asset 1 vs. Asset 2
Correlation			0.8
Portfolio weight	0.6	0.4	
Variance	3.5%	1.5%	

- a) 0.4%
- b) 2.3%
- c) 1.8%
9. Based on the following historical data, which is closest to the standard deviation for the two-asset portfolio shown in the table?

	Asset A	Asset B	Asset A and Asset B
Standard Deviation	4.7%	7.7%	
Portfolio Weight	0.4	0.6	
Correlation			0.3

- a)
- b) 6.5%
- c) 5.0%
- d) 5.5%
10. The following table shows data for the stock of JKU and a market index.

Expected return of JKU	15%
Expected return of market index	12%
Risk-free rate	5%
Standard deviation of JKU returns	20%
Standard deviation of market index return	15%
Correlation of JKU and market index returns	0.75

Based on the capital asset pricing model (CAPM), JKU is most likely:

- a) Fairly valued.
- b) Overvalued.
- c) Undervalued.

Answers

1. B.

Adding a risk budgeting process causes the organization to consider how its total risk tolerance will be allocated across its subsidiaries. Either the total current risks the subsidiaries are engaging in will exceed the risk tolerance and subsidiaries will have to compete for risk by demonstrating highest returns per unit of risk or the total current risks will be less than the risk tolerance and a search will be underway for the subsidiaries that can best utilize the remaining risk budget. The risk tolerance is determined and then sets the risk budget, rather than being determined by it. Hedging can be a part of risk budgeting if hedging produces the superior risk adjusted returns.

CFA Level I

"Risk Management: An Introduction,"

2. C.

In a strategic asset allocation, assets within a specific asset class have high paired correlations and low correlations with other asset classes.

CFA Level I

"Basics of Portfolio Management and Construction,"

3. A.

Booraem Inc. is overvalued because it lies below the SML. The expected return, 12.85%, is less than the required return. According to the CAPM, the required return for Booraem Inc. is 0.1365, or 13.65%: $0.1365 = 0.012 + 1.5(0.095 - 0.012)$.

CFA Level I

"Portfolio Risk and Return: Part II,"

4. C.

Jensen's alpha = $0.12 - [0.027 + 1.5(0.075 - 0.027)] = 0.021$, or 2.10%.

CFA Level I

"Portfolio Risk and Return: Part II,"

5. C.

An organization with a strong competitive position can recover from losses more easily than one with a weaker competitive position. Therefore an organization's risk tolerance should reflect its competitive position. An organization's size does not define the risks sources it faces or the relative losses it can absorb; therefore it should not be reflected in its risk tolerance. Neither the risk sources affecting an organization nor the size of the losses an organization can absorb are a function of its perception of market stability.

CFA Level 1

"Risk Management: An Introduction,"

6. A.

The formula for the return standard deviation (risk) of a two asset portfolio is:

$$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1w_2 \text{Cov}(R_1, R_2)}$$

The formula for portfolio risk shows that portfolio risk decreases as the correlation decreases.

CFA Level I

"Portfolio Risk and Return: Part I,"

7. C.

Although the client owns a successful business and has a high income, she exhibits above-average risk aversion, indicating that her ability to take risk is high but her willingness to take risk is low.

CFA Level I

"Basics of Portfolio Planning and Construction,"

8. C.

The covariance is calculated from the standard deviations of the two assets and their correlation. The portfolio weights are not relevant.

$$\text{Cov} = \rho_{12} \times \sigma_1 \times \sigma_2$$

$$= \rho_{12} \times \sqrt{\text{var}_1} \times \sqrt{\text{var}_2}$$

$$= 0.8 \times \sqrt{3.5\%} \times \sqrt{1.5\%} = 1.8\%$$

CFA Level I

"Portfolio Risk and Return: Part I,"

9. A.

The standard deviation of a two asset portfolio is calculated as follows:

$$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1w_2 \rho_{12} \sigma_1\sigma_2}$$

CFA Level I

"Portfolio Risk and Return: Part I,"

10. C.

$\beta_{JKU} = \rho_{JKU,M} \times \sigma_{JKU}/\sigma_M = 0.75 \times 0.2/0.15 = 1.0$ and $E(R_{JKU}) = RFR + \beta_{JKU} \times (R_M - RFR) = 0.05 + 1 \times (0.12 - 0.05) = 0.12$. The required rate of return of JKU is 12%, and the expected return of JKU is 15%. Therefore, JKU is undervalued relative to the security market line (SML); the risk–return relationship lies above the SML.

CFA Level I

"Portfolio Risk and Return: Part II,"

Questions

- An investor whose portfolio lies to the right of the market portfolio on the capital market line (CML) has *most likely*:
 - loaned some funds at the risk-free rate and invested the remaining funds in the market portfolio.
 - borrowed funds at the risk-free rate and invested all available funds in the market portfolio.
 - invested all available funds in the risk-free asset.
- A correlation matrix of the returns for securities A, B, and C is reported below:

Security	A	B	C
A	1		
B	0.5	1	
C	0	-0.5	1

Assuming that the expected return and the standard deviation of each security are the same, a portfolio consisting of an equal allocation of which two securities will be *most effective* for portfolio diversification?

- Securities A and B
 - Securities B and C
 - Securities A and C
- Based on the capital asset pricing model (CAPM), the expected return on FGL Corp's shares is 12%. Using a model independent of the CAPM, an analyst has estimated the returns on the stock at 10%. Based on this information, the analyst is *most likely* to consider the stock to be:
 - overvalued.
 - correctly valued.
 - undervalued.
 - Which of the following performance measures *most likely* relies on systematic risk as opposed to total risk when calculating a risk-adjusted return?
 - Sharpe ratio
 - M-squared
 - Treynor ratio
 - The stock of GBK Corporation has a beta of 0.65. If the risk-free rate of return is 3% and the expected market return is 9%, the expected return for GBK is *closest* to:
 - 3.9%.
 - 6.9%.
 - 10.8%.
 - Stock X and Stock Y have the same level of total risk. Stock X has twice the systematic risk of Stock Y and half its non-systematic risk. Stock X's expected return will *most likely* be:
 - lower than the expected return of Stock Y.
 - the same as the expected return of Stock Y.
 - higher than the expected return of Stock Y.
 - The strategic asset allocation and portfolio rebalancing policy are *most likely* addressed in which section of an investment policy statement?
 - Procedures

- B. Appendices
- C. Investment objectives
8. Following its decision to divest its non-core assets, analysts expect HCL Corp's standard deviation of returns to rise to 30% and its correlation with the market portfolio to remain unchanged at 0.8. The risk-free rate and the market risk premium are expected to remain unchanged at 6% and 8%, respectively. However, the market portfolio's standard deviation of returns is expected to decrease to 15%. The firm's expected return after the restructure is *closest* to:
- A. 9.2%.
- B. 17.6%.
- C. 18.8%.
9. An asset management firm generated the following annual returns in their US large-cap equity portfolio:

Year	Net Return (%)
2008	-34.8
2009	32.2
2010	11.1
2011	-1.4

- The 2012 return needed to achieve a trailing five-year geometric mean annualized return of 5% when calculated at the end of 2012 is *closest* to:
- A. 17.9%.
- B. 27.6%.
- C. 35.2%.
10. The variance of returns of a security and the market portfolio are 0.25 and 0.09, respectively. If the covariance of security returns and market returns is 0.06, the security's beta is *closest* to:
- A. 0.40.
- B. 0.67.
- C. 0.24.

Answers

1. Answer = B

A portfolio lying to the right of the market portfolio on the CML is formed by borrowing funds at the risk-free rate and investing all available funds in the market portfolio.

CFA Level I

"Portfolio Risk and Return: Part II," Vijay Singal Section 2.2.4

2. Answer = B

The negative correlation of -0.5 between investment securities B and C is the lowest and thus is the most effective for portfolio diversification.

CFA Level I

"Portfolio Risk and Return: Part I," Vijay Singal Section 4.3

3. Answer = A

Because the estimated return on the stock is lower than the expected return using the CAPM, the stock does not compensate the investor for the level of risk and so it is most likely overvalued.

CFA Level I

"Portfolio Risk and Return: Part II," Vijay Singal Section 4.3

4. Answer = C

The Treynor ratio measures the return premium of a portfolio versus the risk-free asset relative to the portfolio's beta, which is a measure of systematic risk.

CFA Level I

"Portfolio Risk and Return: Part II," Vijay Singal Section 4.3.2

5. Answer = B

$$E(R_{GBK}) = R_f + \beta_{GBK} \times [E(R_{Mkt}) - R_f] =$$

$$0.03 + 0.65 \times (0.09 - 0.03) = 0.069 \text{ or } 6.9 \% .$$

CFA Level I

"Portfolio Risk and Return: Part II," Vijay Singal Section 3.2.6

6. Answer = C

Because Stock X has a higher systematic risk level compared with Stock Y, its expected return will be higher than that of Stock Y.

CFA Level I

"Portfolio Risk and Return: Part II," Vijay Singal Section 3.1

7. Answer = B

Information related to strategic asset allocation and portfolio rebalancing policy would be placed in the appendices of an investment policy statement.

CFA Level I

"Basics of Portfolio Planning and Construction," Alistair Byrne and Frank E. Smuddle Section 2.2

8. Answer = C

We first compute the firm's return is computed using:

$$\beta_i = \frac{\rho_{i,m} \sigma_i}{\sigma_m}$$

$$\beta_i = \frac{0.8(0.30)}{0.15} = 1.6$$

beta using:

CFA Level I

The beta is: $E(R_i) = R_f + [E(R_m) - R_f]\beta_i$. So, $E(R_i) = 0.06 + (0.08)1.6 = 18.8\%$. The expected

"Portfolio Risk and Return:

Vijay Singal Sections 3.2 and 4.2

Part II,"

9. Answer = C

$$\bar{R}_G = 0.05 = \sqrt[5]{(1 - 0.348)(1 + 0.322)(1 + 0.111)(1 - 0.014)(1 + R_{2012})} - 1$$

Holding period total return (cumulative) factor calculation through 2011:

$$(1 - 0.348) \times (1 + 0.322) \times (1 + 0.111) \times (1 - 0.014) = 0.652 \times 1.322 \times 1.111 \times 0.986 = 0.9442.$$

Compound total return (cumulative) factor at 5% per year of 5% for five years:

$$1.05^5 = 1.2763.$$

Return needed in 2012 to achieve a compound annualized return of 5%:

$$1.2763/0.9442 = 1.3517 = \mathbf{35.2\%}.$$

$$\text{Check: } 0.944 \times 1.352 = 1.276^{(1/5)} = 1.050 = 5\% \text{ annualized.}$$

CFA Level I

"Portfolio Risk and Return: Part I," Vijay Singal

Section 2.1.3

10. Answer = B

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\sigma_m^2} = \frac{0.06}{0.09} = 0.67$$

The security's beta is:

CFA Level I

"Portfolio Risk and Return: Part II," Vijay Singal

Section 3.2

Questions

- In the context of strategic asset allocation, adding asset classes with low correlation will *most likely* improve a portfolio's risk-return trade-off as long as the stand-alone risk of the added asset class:
 - does not exceed its diversification effect.
 - equals its diversification effect.
 - exceeds its diversification effect.
- You are preparing an investment policy statement for a client who manages her own successful marketing consultancy. Her annual income is approximately \$500,000. She describes herself as a finance novice. Most of her savings are invested in bank term deposits and short-term government securities. In her responses to the standard risk assessment questionnaire, she strongly agrees with the statements that she "feels more comfortable putting money in a bank account than in the stock market." Also, she "thinks of the word 'risk' as being a 'loss'". Based on this information, your client's ability and willingness to take risk can best be described as:
 - high ability and low willingness.
 - high ability and willingness.
 - low ability and high willingness.
- A key difference between a wrap account and a mutual fund is that wrap accounts:
 - have assets that are owned directly by the individual.
 - cannot be tailored to the tax needs of a client.
 - have a lower required minimum investment.
- Which of the following is *least likely* an assumption of the capital asset pricing model (CAPM)?
 - Investors are different only with respect to their unique holding periods.
 - An investor can invest as much as he or she desires in any asset.
 - Security prices are not affected by investor trades.
- If Investor A has a lower risk aversion coefficient than Investor B, on the capital allocation line, will Investor B's optimal portfolio *most likely* have a higher expected return?
 - No, because Investor B has a higher risk tolerance
 - No, because Investor B has a lower risk tolerance
 - Yes
- The return measure that *best* allows one to compare asset returns earned over different length time periods is the:
 - annualized return.
 - net portfolio return.
 - holding period return.
- Based on the following historical data, which is *closest* to the standard deviation for the two-asset portfolio shown in the table?

	Asset A	Asset B	Asset A and B
Standard deviation	4.7%	7.7%	
Portfolio weight	0.4	0.6	
Correlation			0.3

- 6.5%
- 5.5%
- 5.0%

8. When considering a portfolio that is optimal for one investor, a second investor with a higher risk aversion would *most likely*:
- A. expect a higher variance for the portfolio.
 - B. derive a lower utility from the portfolio.
 - C. have a lower return expectation for the portfolio.

9. A portfolio has the following returns:

	Portfolio Returns
2006	2.4%
2007	9.6%
2008	-4.0%
2009	5.6%
2010	4.8%
2011	-3.2%

The sample variance of the portfolio is closest to:

- A. 0.23%.
 - B. 0.28%.
 - C. 0.36%.
10. Which of the following types of institutions is *most likely* to have a long investment time horizon and a higher level of risk tolerance?
- A. An endowment
 - B. An insurance company
 - C. A bank

Answers

1. Answer = A

In general, adding assets classes with low correlation improves the risk–return trade-off as long as the stand-alone risk of the added asset class does not exceed its diversification effect.

CFA Level I

"Basics of Portfolio Planning and Construction," Alistair Byrne and Frank E. Smuddle Section 3.2

2. Answer = A

Although the client owns a successful business and has a high income, she exhibits above-average risk aversion, indicating that her ability to take risk is high but her willingness to take risk is low.

CFA Level I

"Basics of Portfolio Planning and Construction," Alistair Byrne and Frank E. Smuddle Section 2.2

3. Answer = A

The key difference between a wrap account and a mutual fund is that in a wrap account, the assets are owned directly by the individual.

CFA Level I

"Portfolio Management: An Overview," Robert M. Conroy and Alistair Byrne Section 5.3.2

4. Answer = A

One of the assumptions of the CAPM is that investors plan for the same single holding period.

CFA Level I

"Portfolio Risk and Return: Part II," Vijay Singal Section 4.1

5. Answer = B

Investor B has a higher risk aversion coefficient, thus a lower risk tolerance and a lower expected return on the capital allocation line.

CFA Level I

"Portfolio Risk and Return: Part I," Vijay Singal Section 3.3

6. Answer = A

The annualized return is an average return measure that can be calculated using return data for a period that is shorter (or longer) than one year. In many cases, it is most convenient to annualize all available returns in order to compare returns when the time periods during which a return is earned or computed vary. It reflects the return that would be earned over a one-year period, assuming that money can be reinvested repeatedly while earning a similar return.

CFA Level I

"Portfolio Risk and Return: Part I," Vijay Singal Section 2.1

7. Answer = B

The standard deviation of a two asset portfolio is calculated as follows:

$$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2}$$

CFA Level I

"Portfolio Risk and Return: Part I," Vijay Singal Section 2.3.3

8. Answer = B

Utility has two terms: the expected return and a negative term based on the portfolio risk weighted by risk aversion. For an identical portfolio, the investor with a higher risk aversion (A) would calculate a lower utility (U).

$$U = E(r) - \frac{1}{2} A \sigma$$

CFA Level I

"Portfolio Risk and Return: Part I," Vijay Singal

Section 3.2

9. Answer = B

The sample variance is calculated as the sum of squared deviations from the arithmetic mean.

$$\text{Mean} = \sum \frac{2.4\% + 9.6\% - 4.0\% + 5.6\% + 4.8\% - 3.2\%}{6} = 2.5\%$$

$$\text{Sample variance} = \frac{\text{sum square derivations}}{N-1}$$

$$= \frac{\sum (2.4\% - 2.5\%)^2 + (9.6\% - 2.5\%)^2 + (-4.0\% - 2.5\%)^2 + (5.6\% - 2.5\%)^2 + (4.8\% - 2.5\%)^2 + (-3.2\% - 2.5\%)^2}{5}$$

$$= 0.28\%$$

CFA Level I

"Portfolio Risk and Return: Part I," Vijay Singal Section 4.1.3

10. Answer = A

Endowments have a long investment time horizon and a high level of risk tolerance.

CFA Level I

"Portfolio Management: An Overview," Robert M. Conroy and Alistair Byrne Section 3.2